

Mapping Recaptures

John D. Robinson

2026-08-24

Contents

Mapping Genotype Recaptures in Central GA	1
Load data	1
Females	2
Males	34
Distances within and between years	62
Digging in...	65

Mapping Genotype Recaptures in Central GA

This markdown will visualize the spatial distribution of bears with *spatial recaptures* from hair samples collected in central Georgia (2023 and 2024). These samples were genotyped via capture-seq (Rapid Genomics) or GT-seq (GTSeek). Capture-seq data were filtered to avoid tightly linked loci (no SNPs within 500kb with $r^2 > 0.8$) and focus on informative markers (MAF > 0.2) and combined with GT-seq data (which have a broader range of MAF), resulting in a total of 1226 SNP loci (see `4_combineGTSeek` for details on combining the data from the two marker panels). Genotypes were subjected to a clone analysis in COLONY (Wang 2016) and KING kinship coefficients (Manichaikul et al. 2010) were calculated in PLINK2 (Chang et al. 2015). Inferences from the two approaches were combined following methods outlined in `10_recapFinal`.

Here, we'll plot the recaptures for bears as a way to assess a couple of things:

1. Confidence in recapture data - are samples spaced too far apart? Are we confident in the combined inferences from COLONY and PLINK?
2. Evidence for moving activity centers - do we have sufficient data to fit a model where activity centers are allowed to move between years? Does this appear to be happening in the dataset?

Load data

We need to load the recapture objects and the locations of the hair snares to visualize the distribution of recaptures...

```
#Load the sampleMetadata file
smd <- read.csv("../sampleinfo/sampleMetaData_cloneID.csv", header = TRUE)

#Load capture history objects
```

```
load("../scr/CGA_CH.Rda")

#Load hair snare location data
load("../sampleinfo/snareLocations.Rda")

#Load the state space
library(raster)

## Loading required package: sp

S360 <- raster("../input/state-space360.tif")

#Source functions to calculate some basic summary stats
source("../input/sumstats_4d.R")
```

Females

In the plots that follow, all detections for individuals with spatial recaptures are plotted on the same map. Primary sampling occasions (years) are shown in different colors, while secondary sampling occasions (weeks) are plotted with different symbols. In cases where the same individual is detected at multiple hair snares *during the same week*, points are connected with a dotted line (colored to indicate the year of the movement). Each plot also gives the total number of detections for a bear, and the number of hair snares J , weeks K , and years T where the individual was detected.

First, for females...

```
#Females
caphist <- Fch$y
Fclones <- Fch$clone

sumstats_4d(caphist)

##          ss.totN      ss.totcaps      ss.avgcaps ss.indwsprecap      ss.avgtraps
##      88.000000      260.000000      2.954545      31.000000      1.602273
##      ss.avgyrs
##      1.284091

nind <- dim(caphist)[1]

yr.col <- c("blue","purple")
wk.symbol <- 0:6

for(i in 1:nind) {
  i.ch <- caphist[i,,]

  nTraps <- sum(rowSums(i.ch) > 0)
  if(nTraps > 1) {
    plot.me <- as.data.frame(which(i.ch == 1, arr.ind = TRUE))
    plot(S360, legend = FALSE, main = paste("Female",i,"- clone", Fclones[i]))
    mtext(paste(length(which(smd$finalClone == Fclones[i])), "samples ->",
                      nrow(plot.me), "dets:"),
```

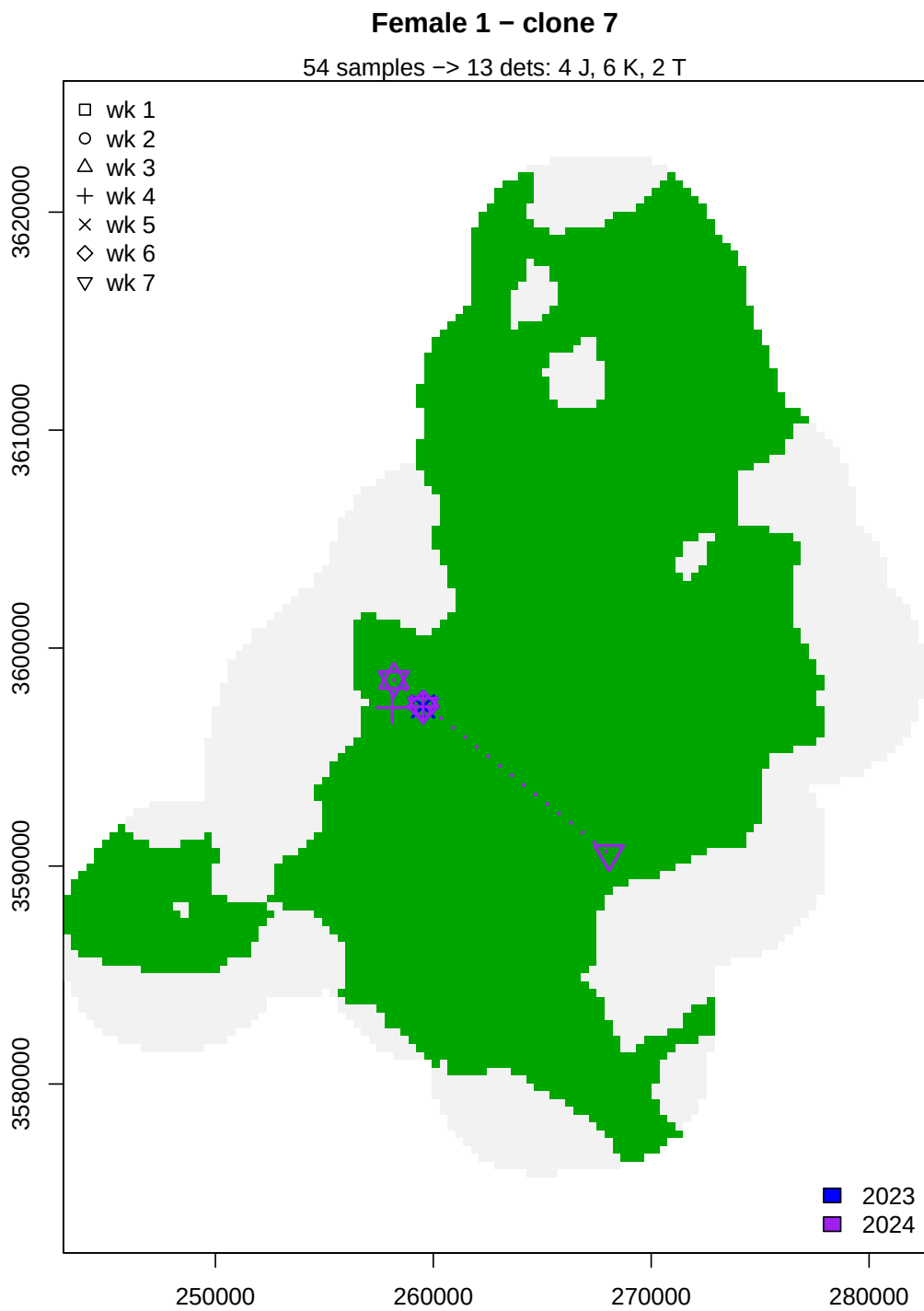
```

        nTraps, "J,",
        length(unique(plot.me$dim2)), "K,",
        length(unique(plot.me$dim3)), "T"))
points(x=locs[plot.me$dim1,1], locs[plot.me$dim1,2],
       pch = wk.symbol[plot.me$dim2], col = yr.col[plot.me$dim3],
       cex = 2, lwd = 2)

for(r in 2:nrow(plot.me)) {
  if(plot.me$dim3[r]==plot.me$dim3[r-1] &
     plot.me$dim2[r]==plot.me$dim2[r-1]) {
    lines(x=c(locs[plot.me$dim1[r-1],1], locs[plot.me$dim1[r],1]),
          y=c(locs[plot.me$dim1[r-1],2], locs[plot.me$dim1[r],2]),
          lty="dotted", col = yr.col[plot.me$dim3[r]],
          lwd = 2.5)
  }
}

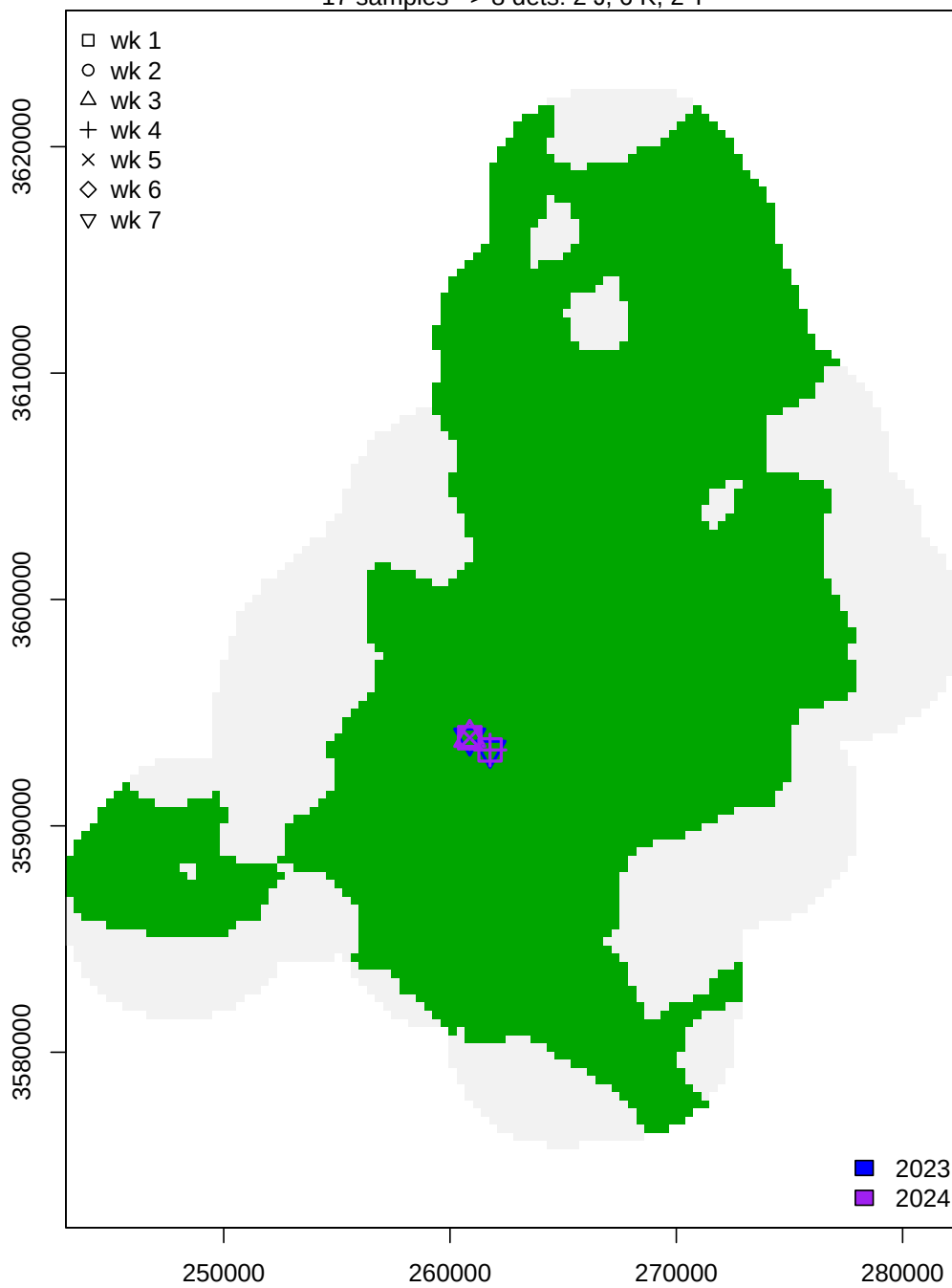
legend("topleft", bty = "n", pch = 0:6, legend = paste("wk",c(1:7)))
legend("bottomright", bty = "n", fill = yr.col, legend = c("2023","2024"))
}
}

```



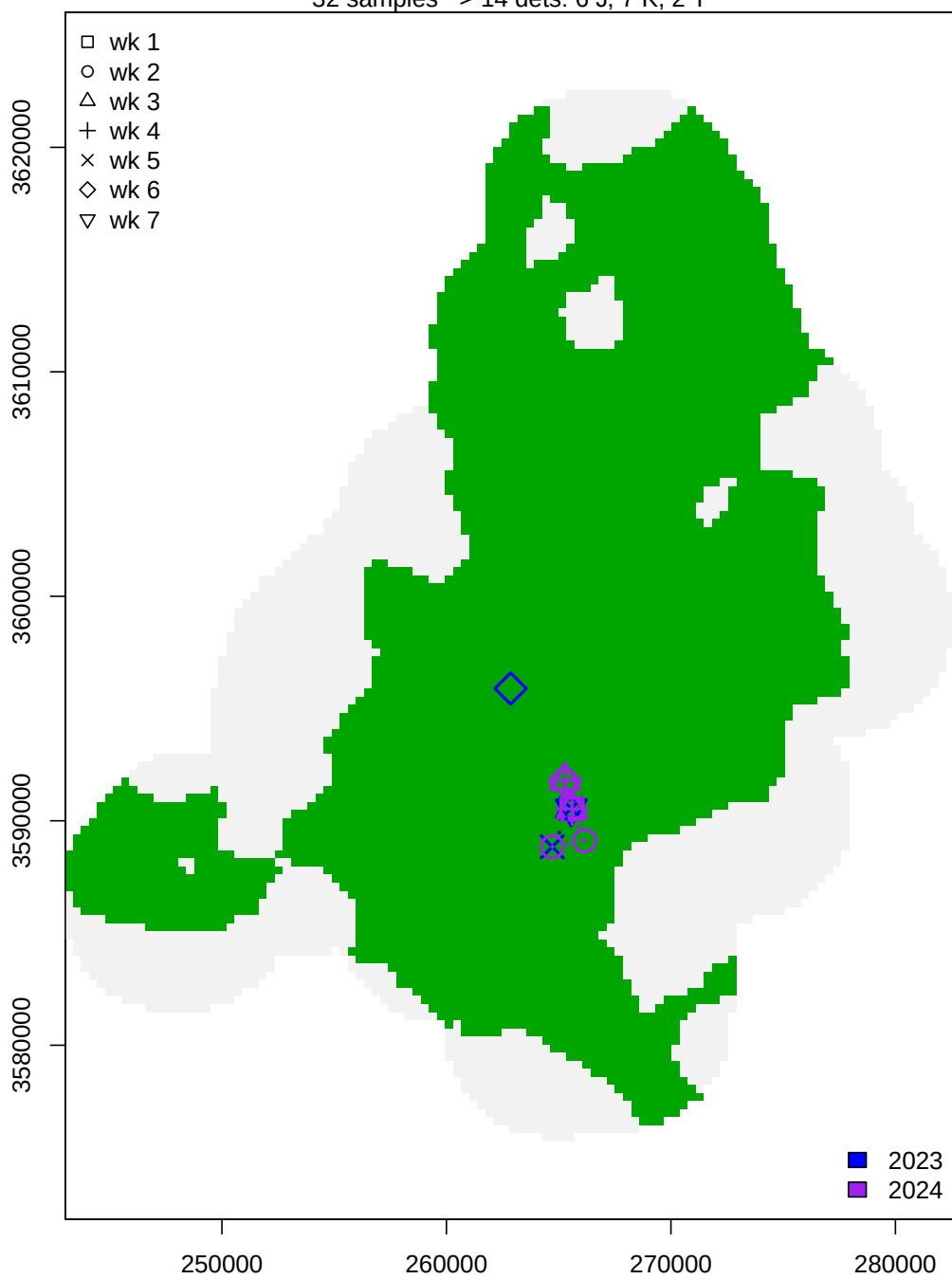
Female 2 – clone 322

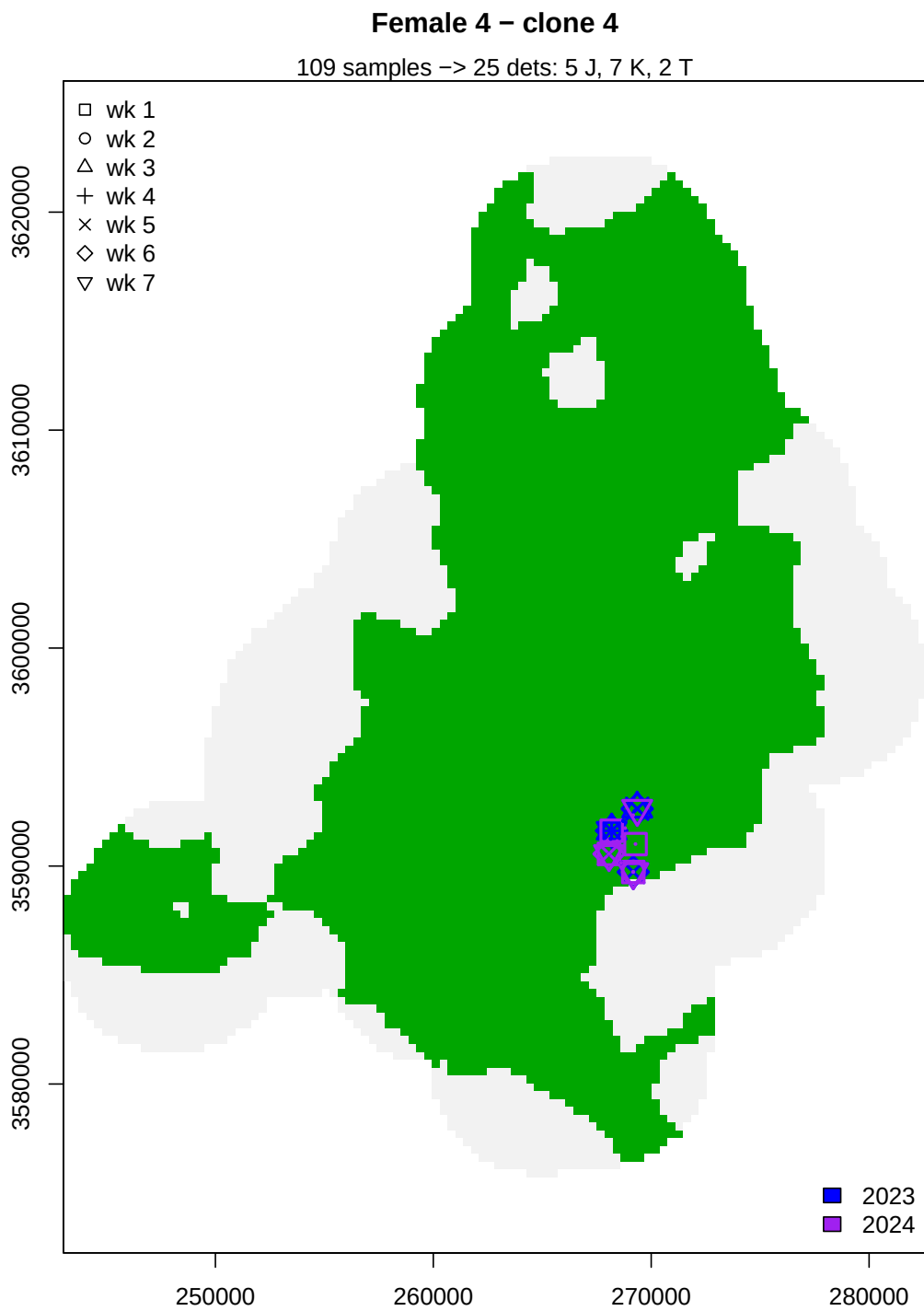
17 samples -> 8 dets: 2 J, 6 K, 2 T



Female 3 – clone 291

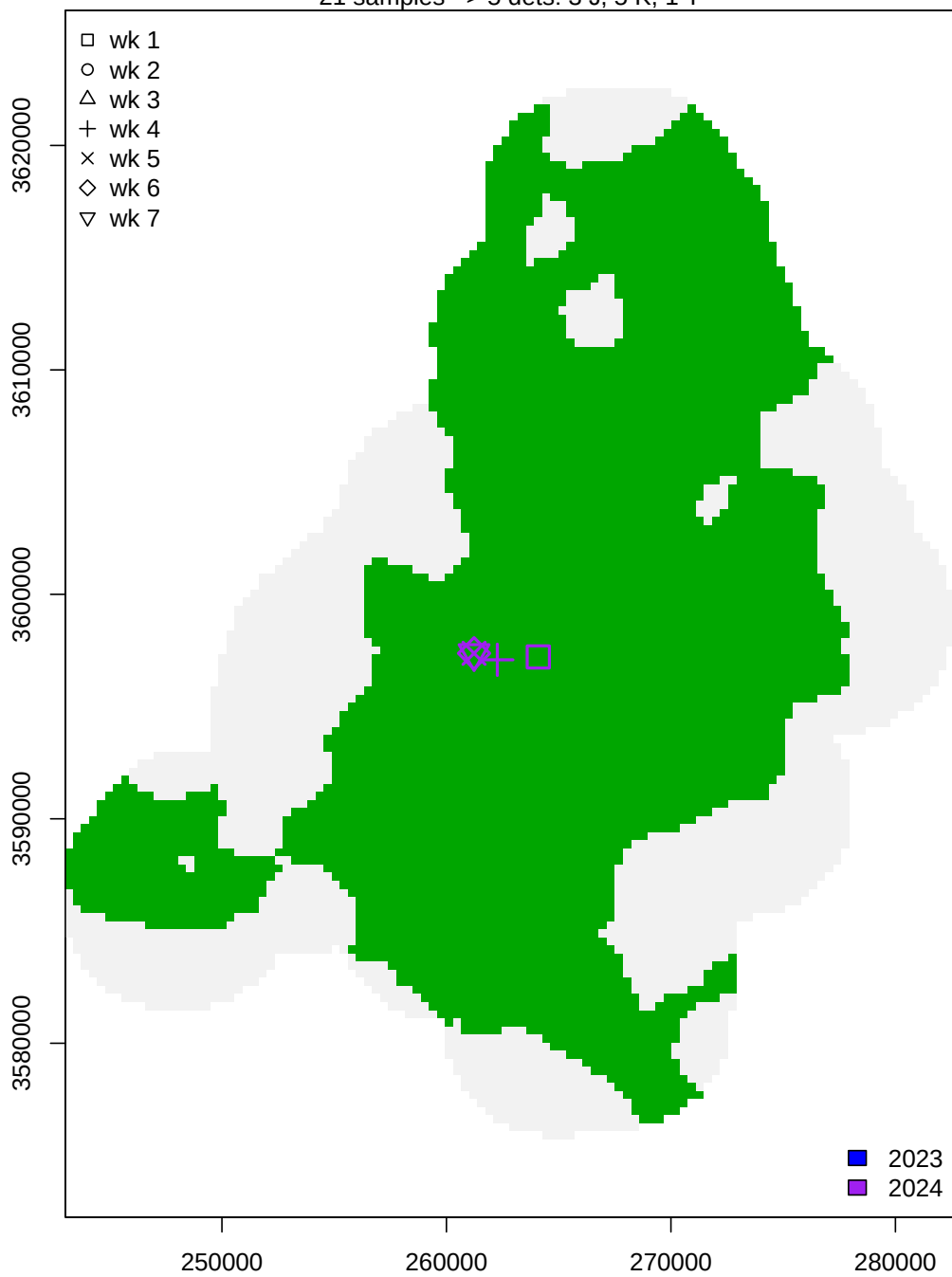
32 samples -> 14 dets: 6 J, 7 K, 2 T

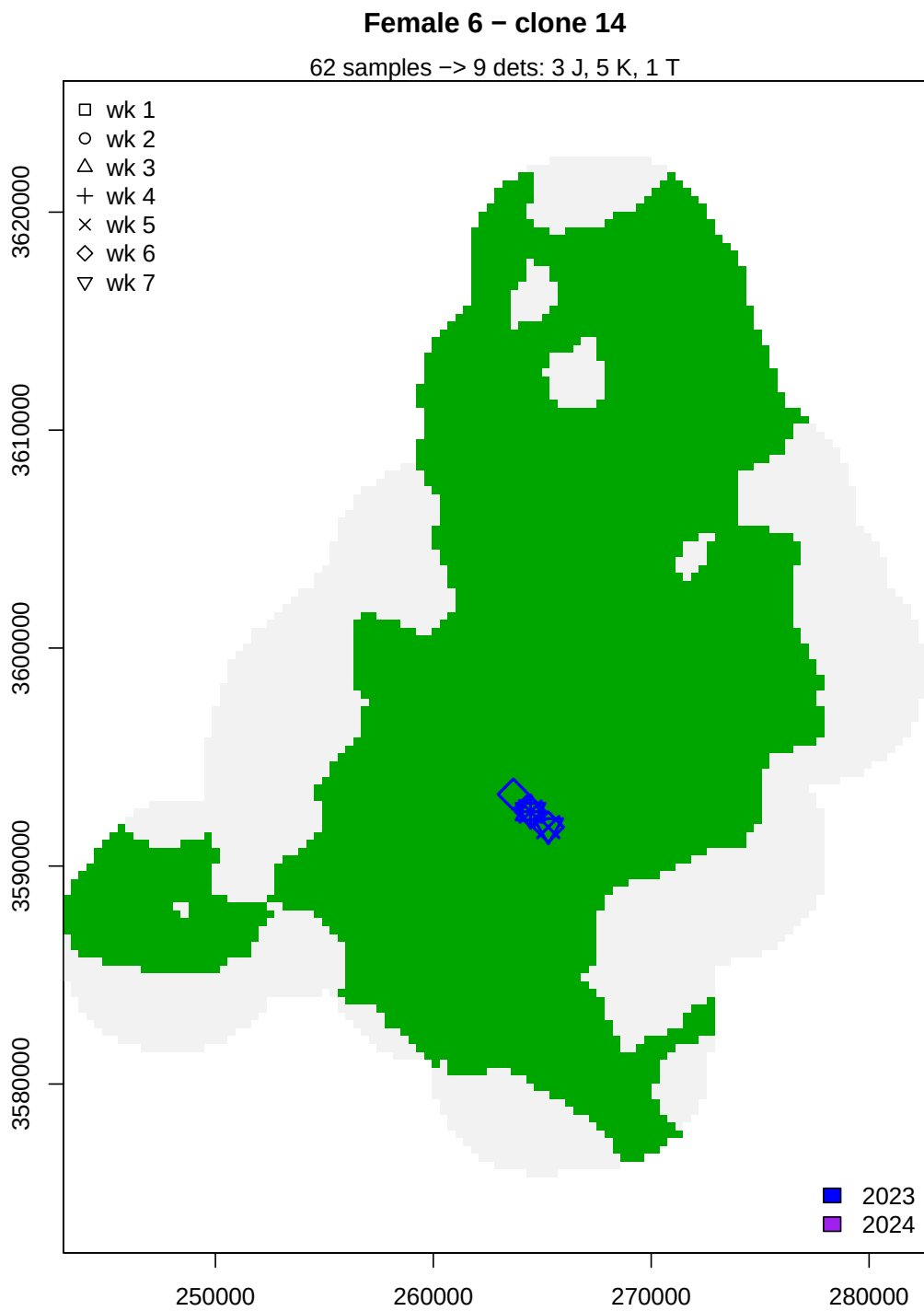


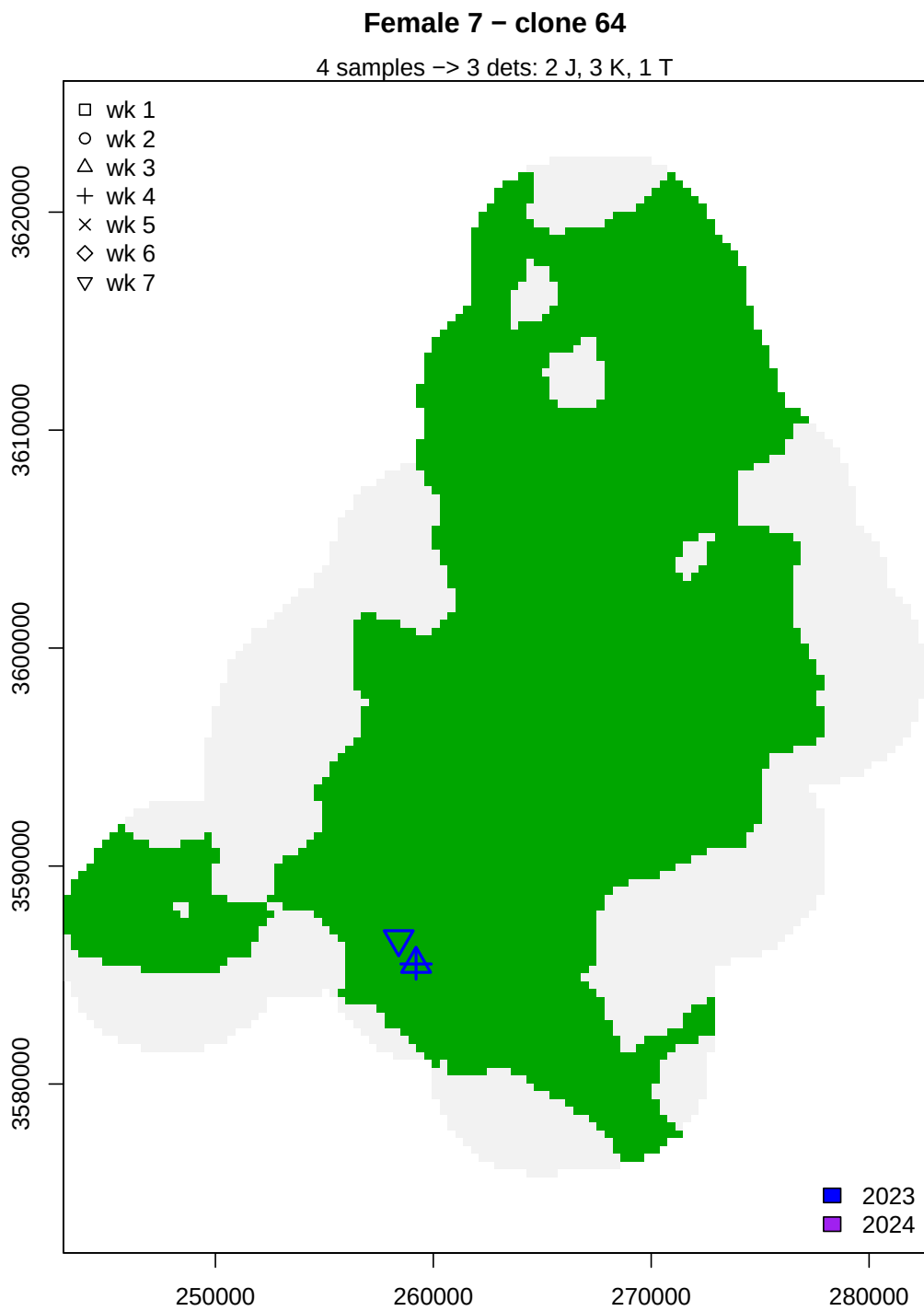


Female 5 – clone 316

21 samples -> 5 dets: 3 J, 5 K, 1 T

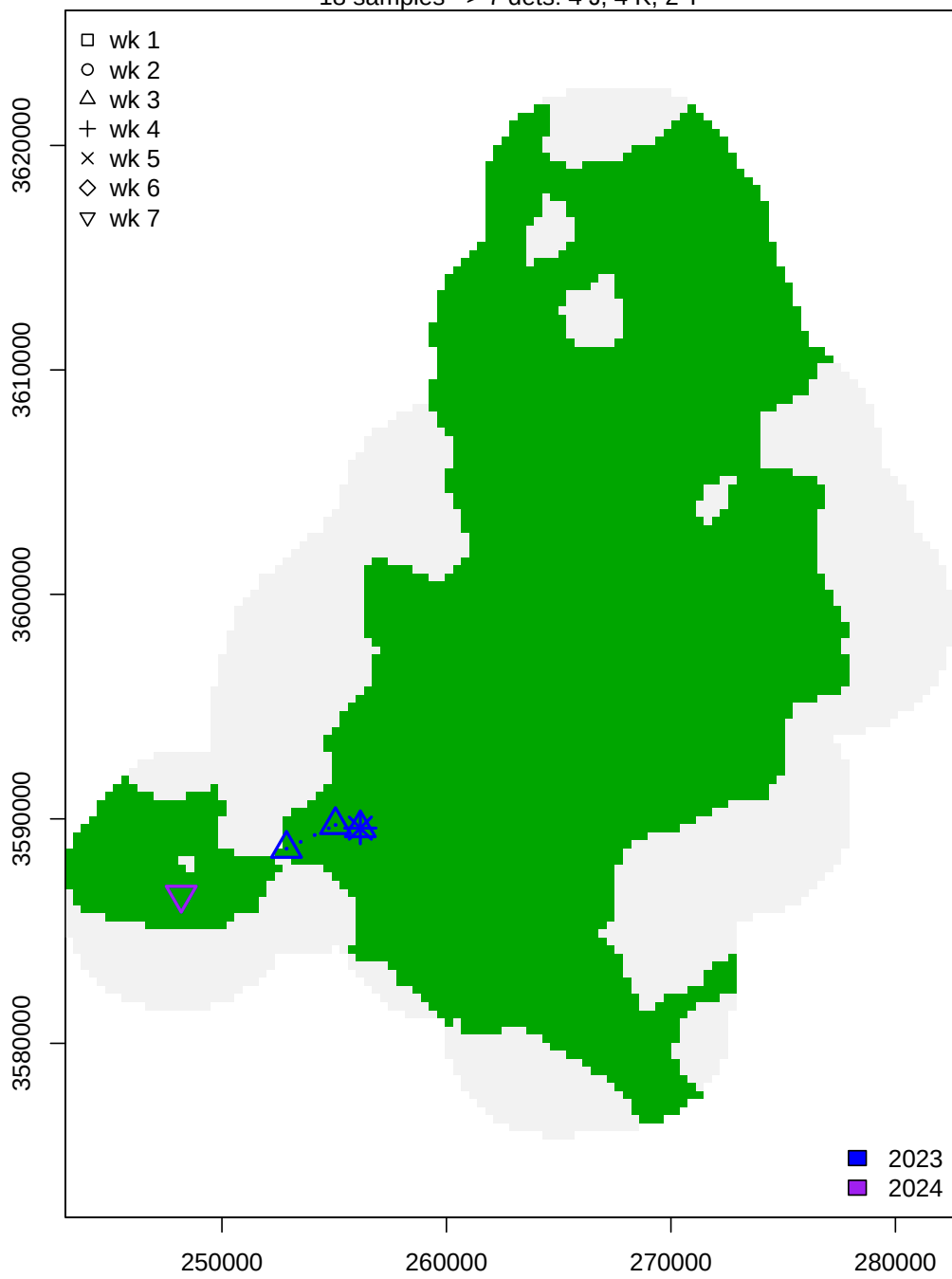


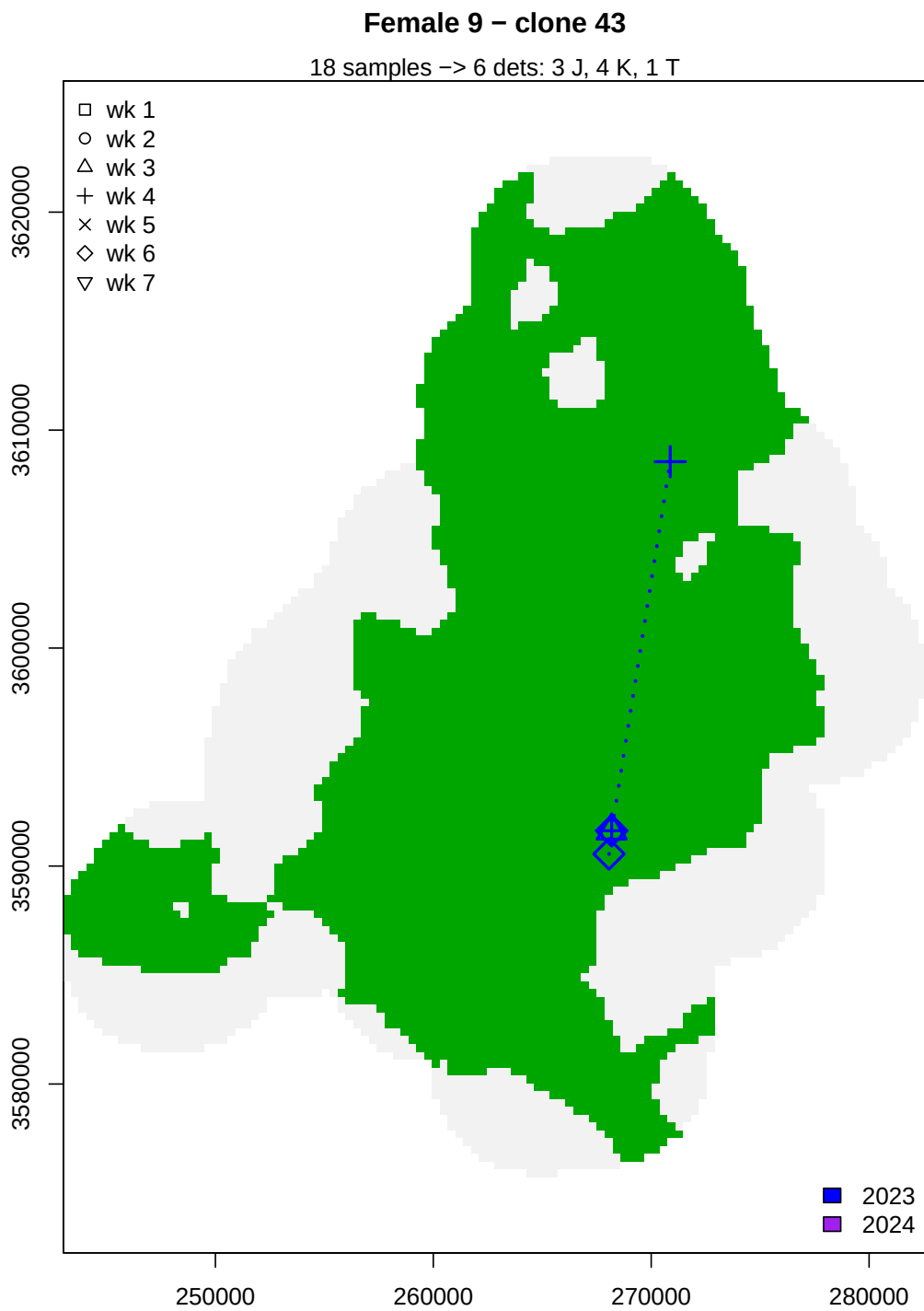


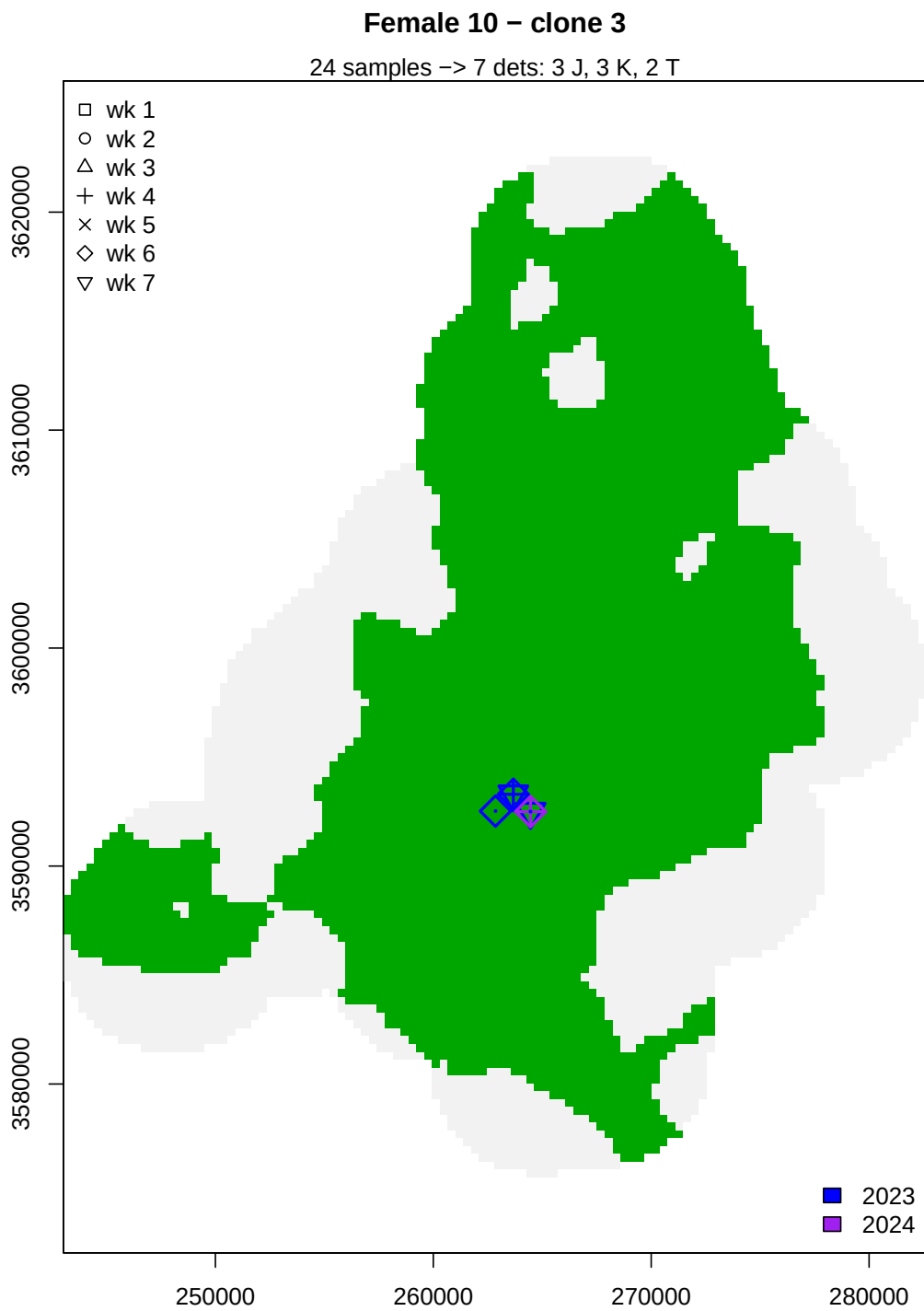


Female 8 – clone 275

18 samples -> 7 dets: 4 J, 4 K, 2 T

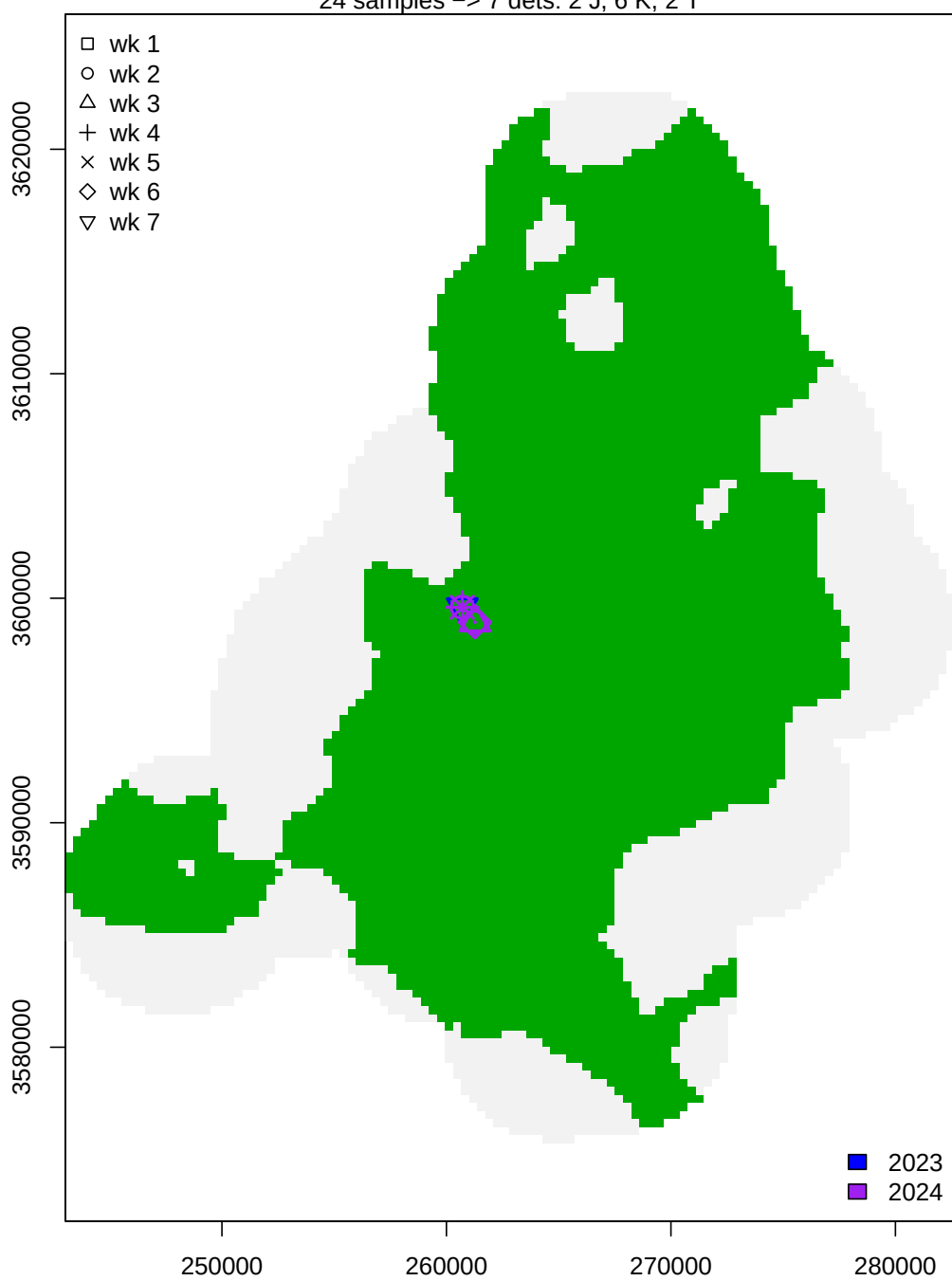






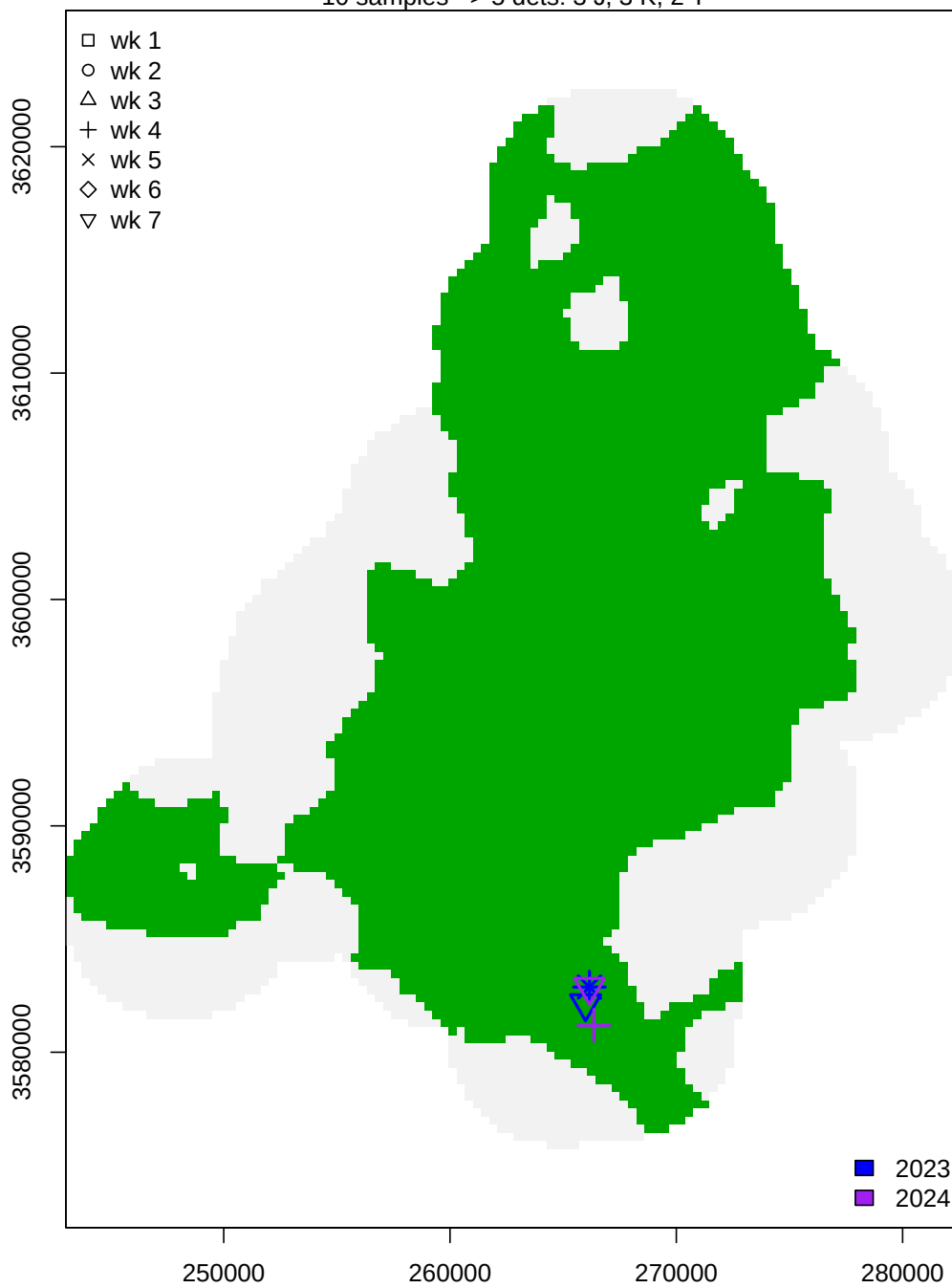
Female 11 – clone 298

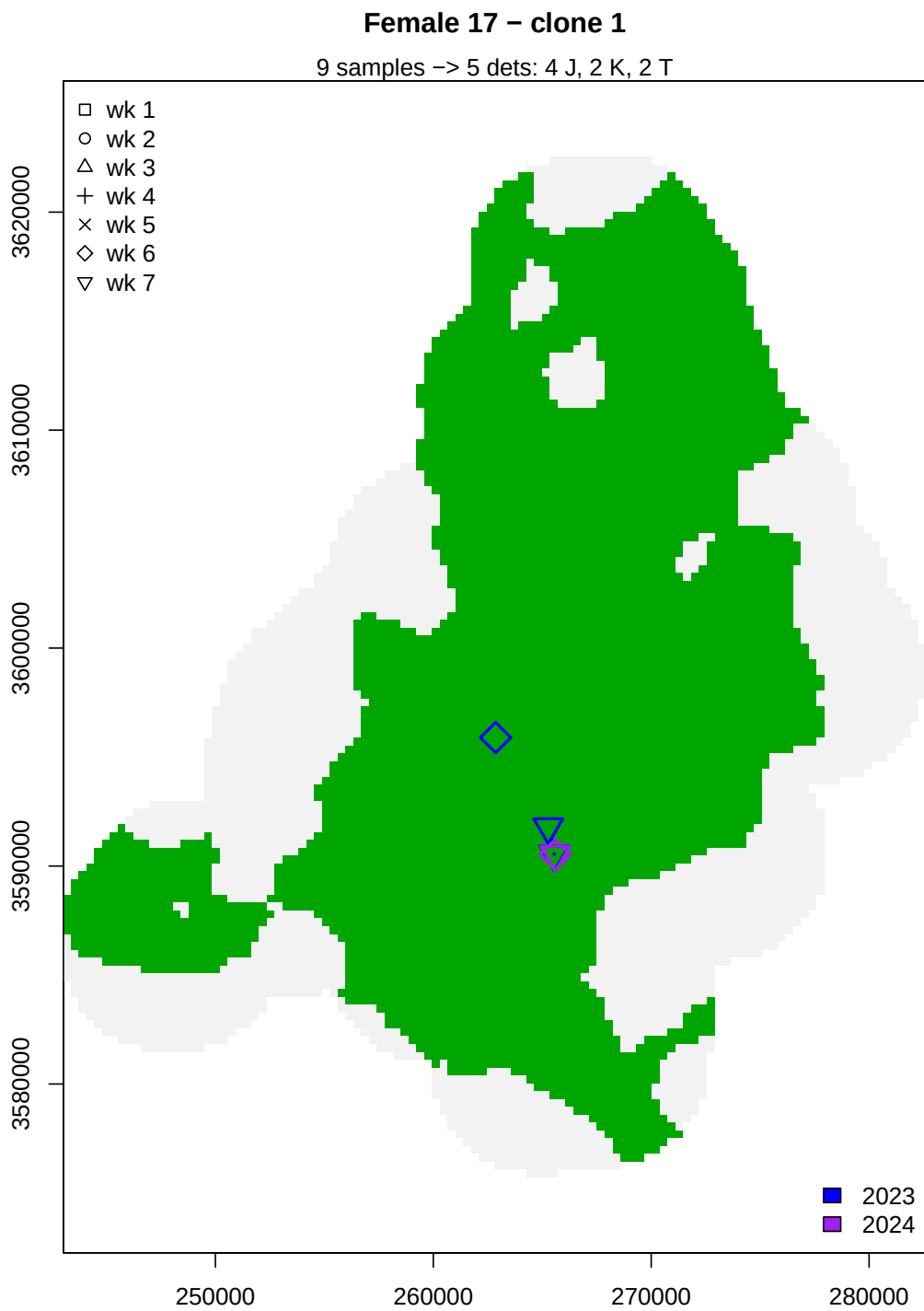
24 samples -> 7 dets: 2 J, 6 K, 2 T



Female 12 – clone 153

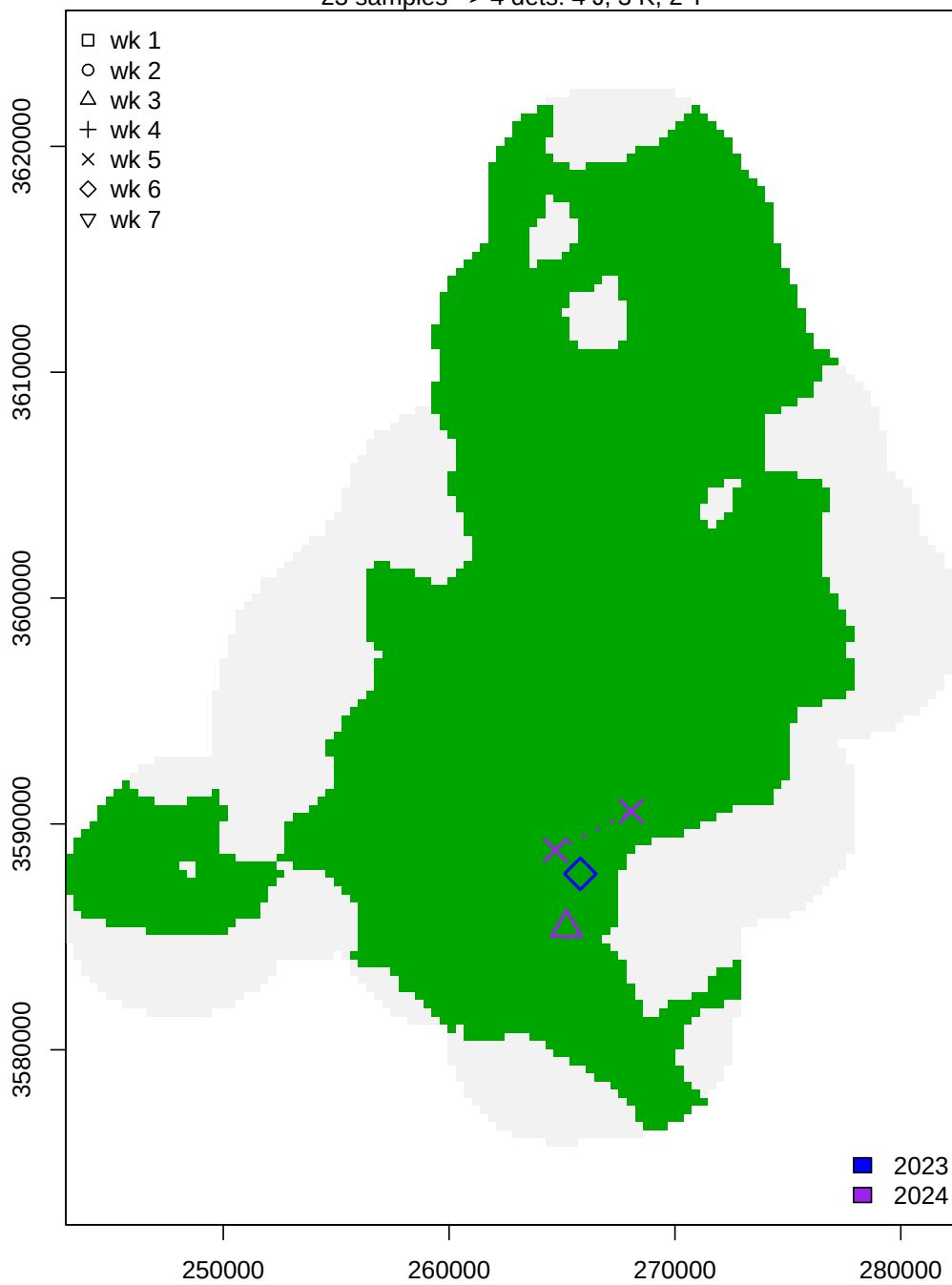
10 samples -> 5 dets: 3 J, 3 K, 2 T

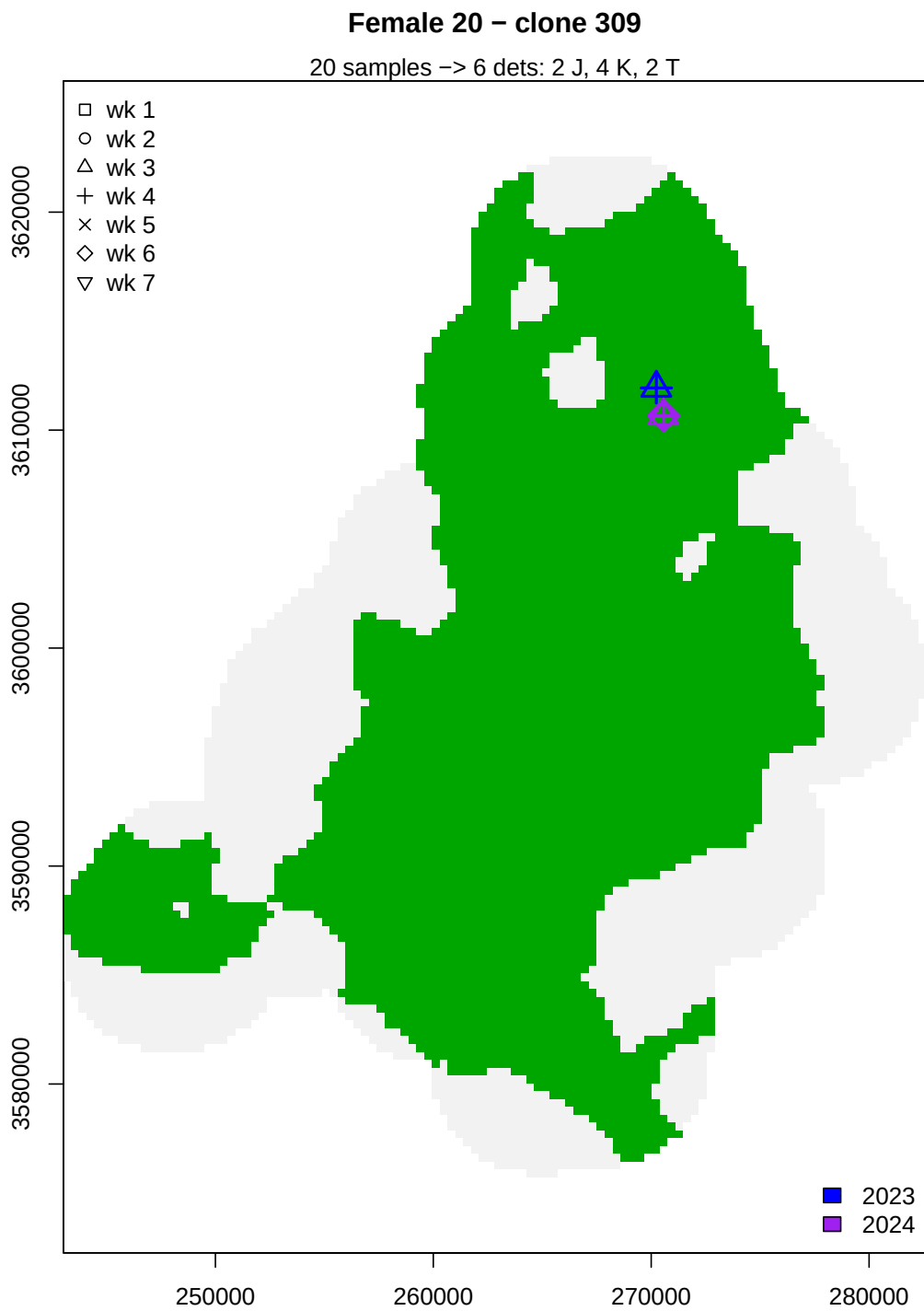




Female 18 – clone 304

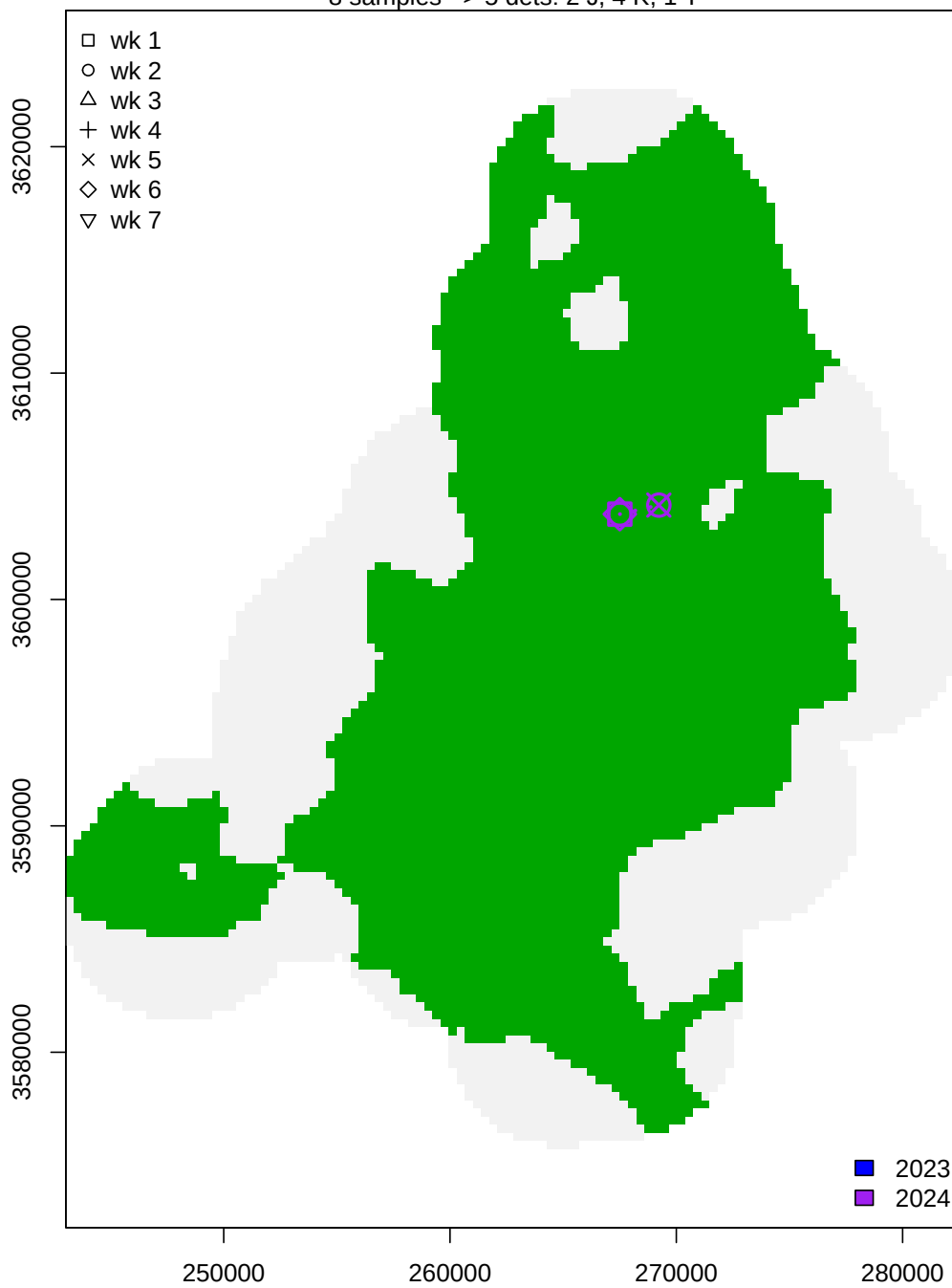
23 samples -> 4 dets: 4 J, 3 K, 2 T





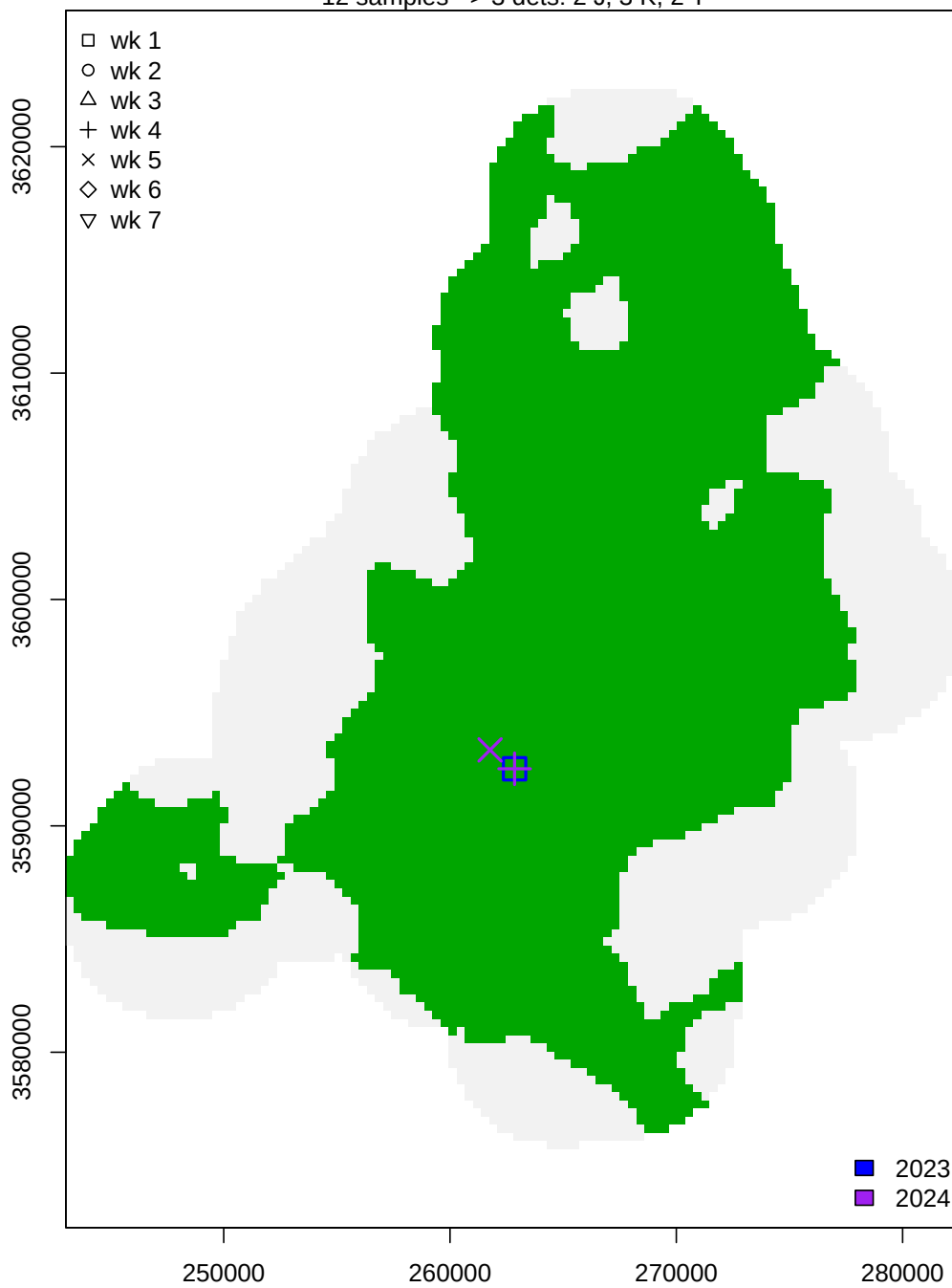
Female 22 – clone 176

8 samples -> 5 dets: 2 J, 4 K, 1 T



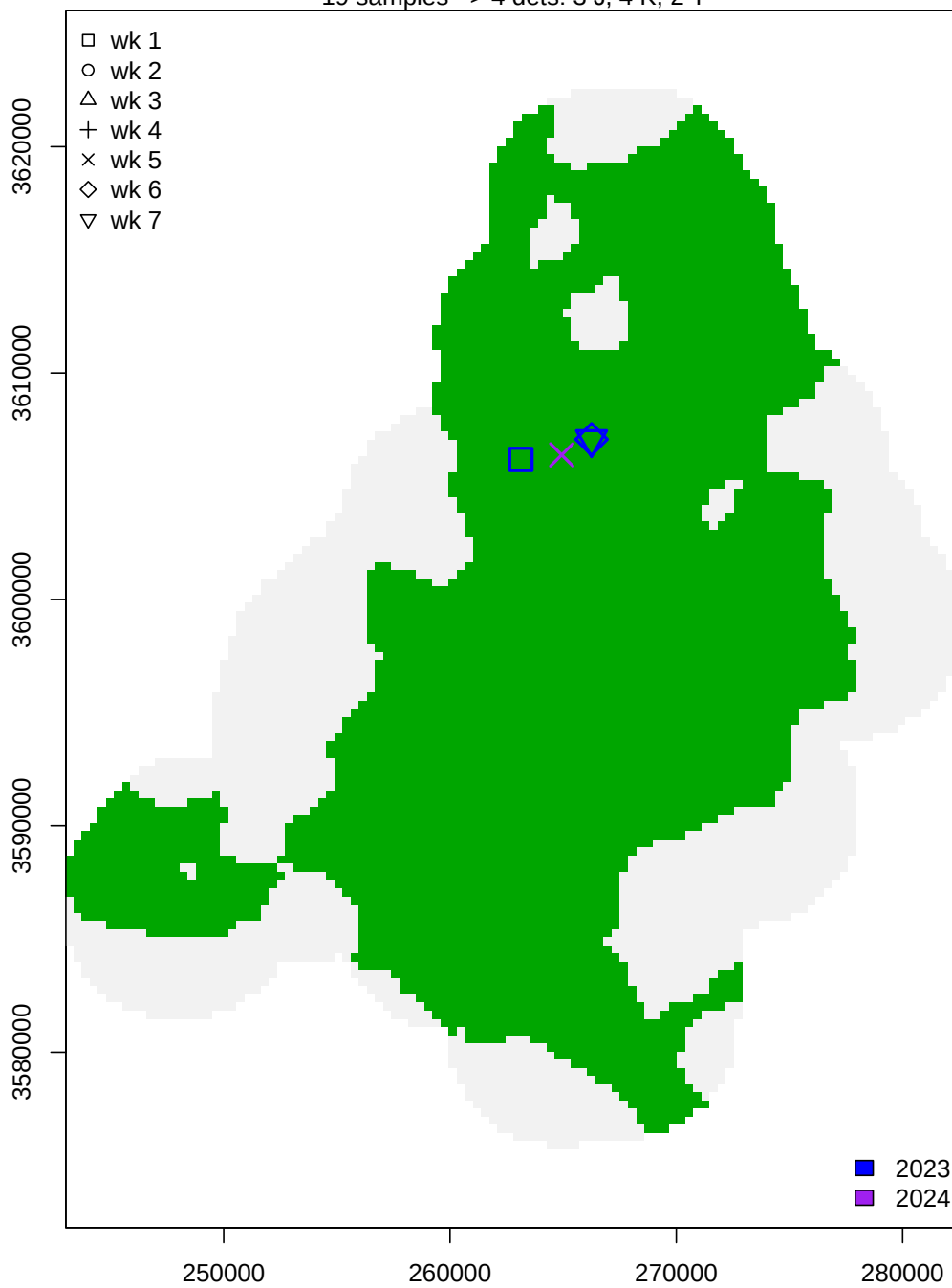
Female 23 – clone 284

12 samples -> 3 dets: 2 J, 3 K, 2 T



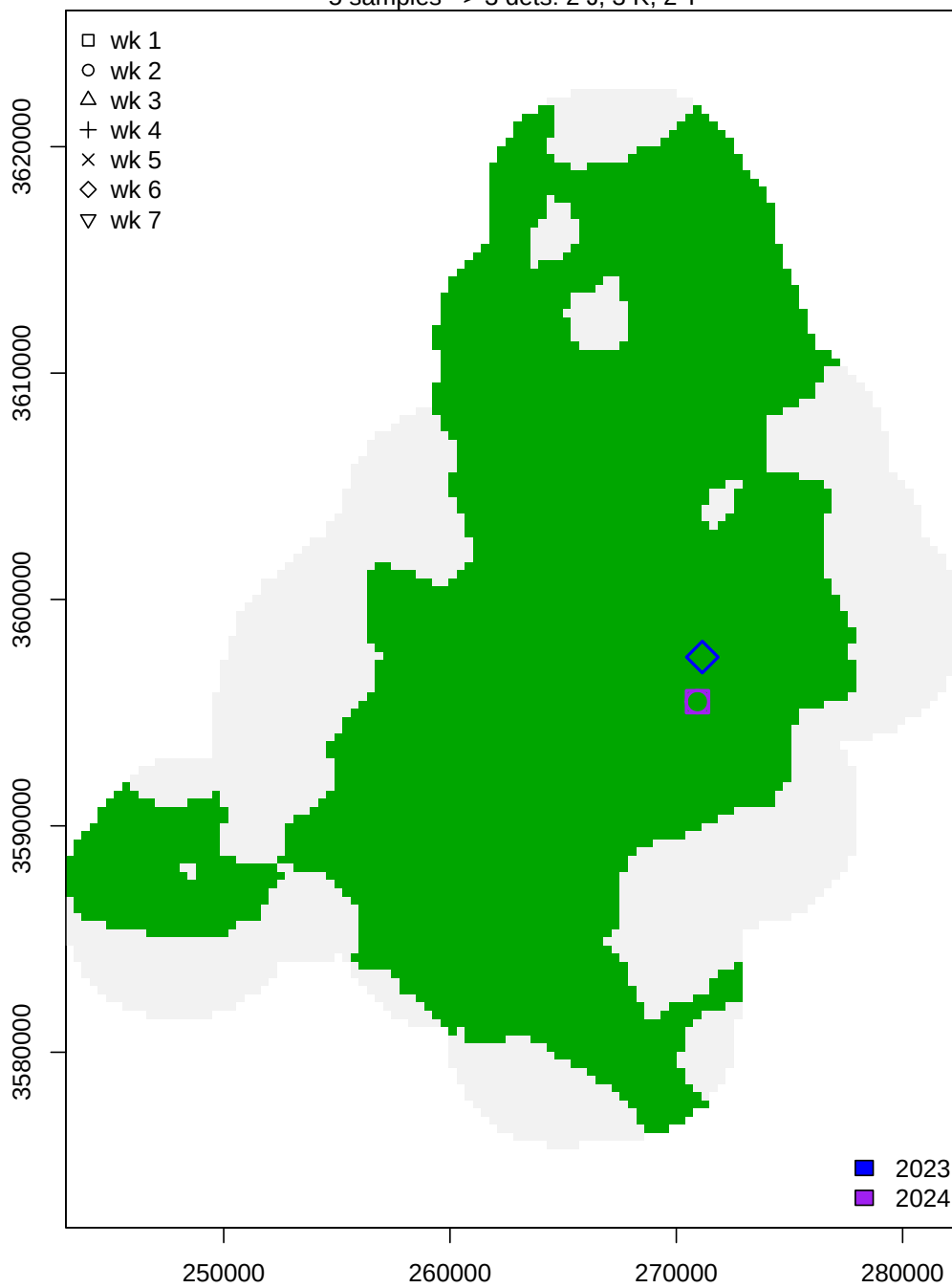
Female 25 – clone 286

19 samples -> 4 dets: 3 J, 4 K, 2 T



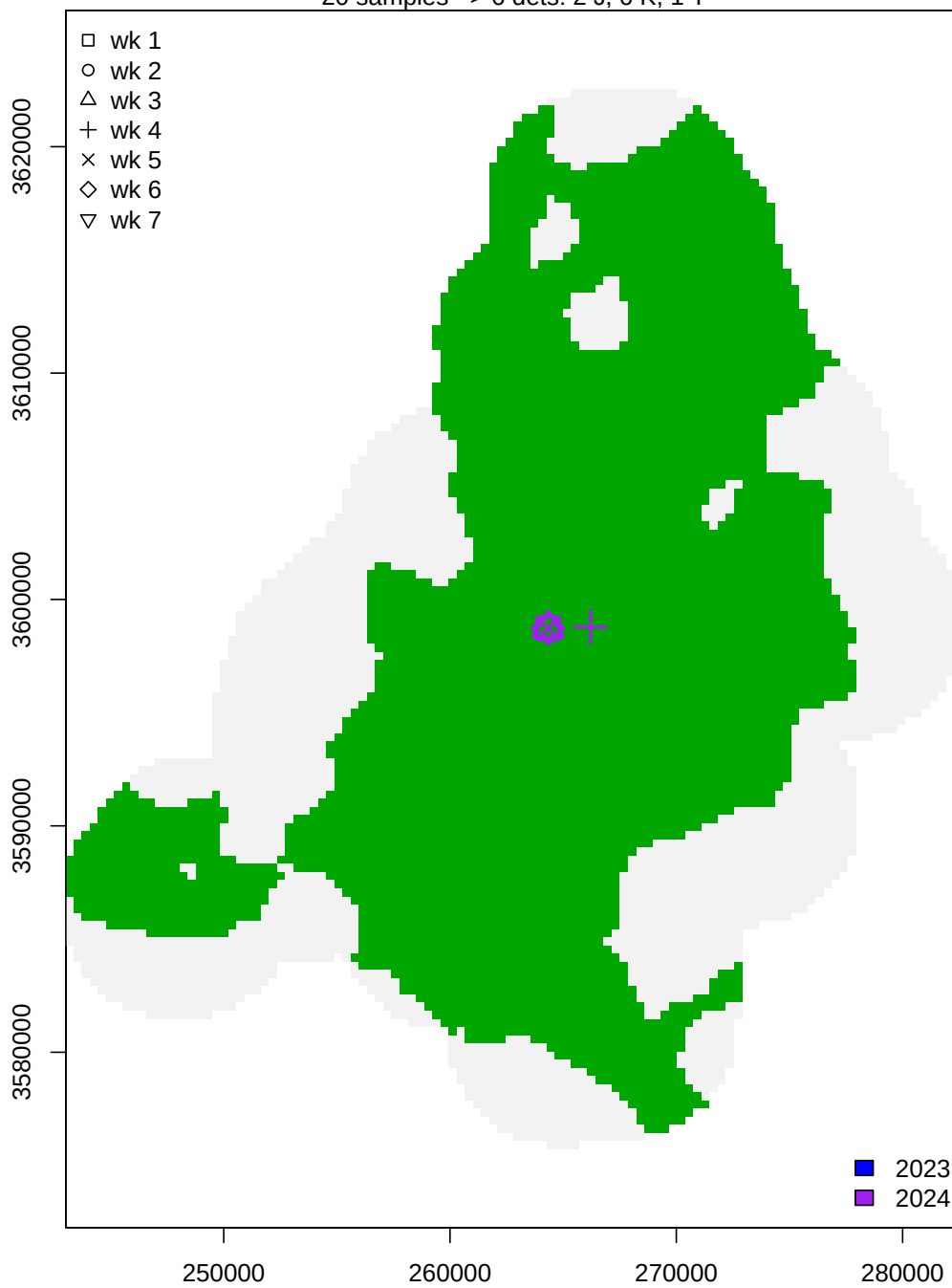
Female 27 – clone 35

5 samples -> 3 dets: 2 J, 3 K, 2 T



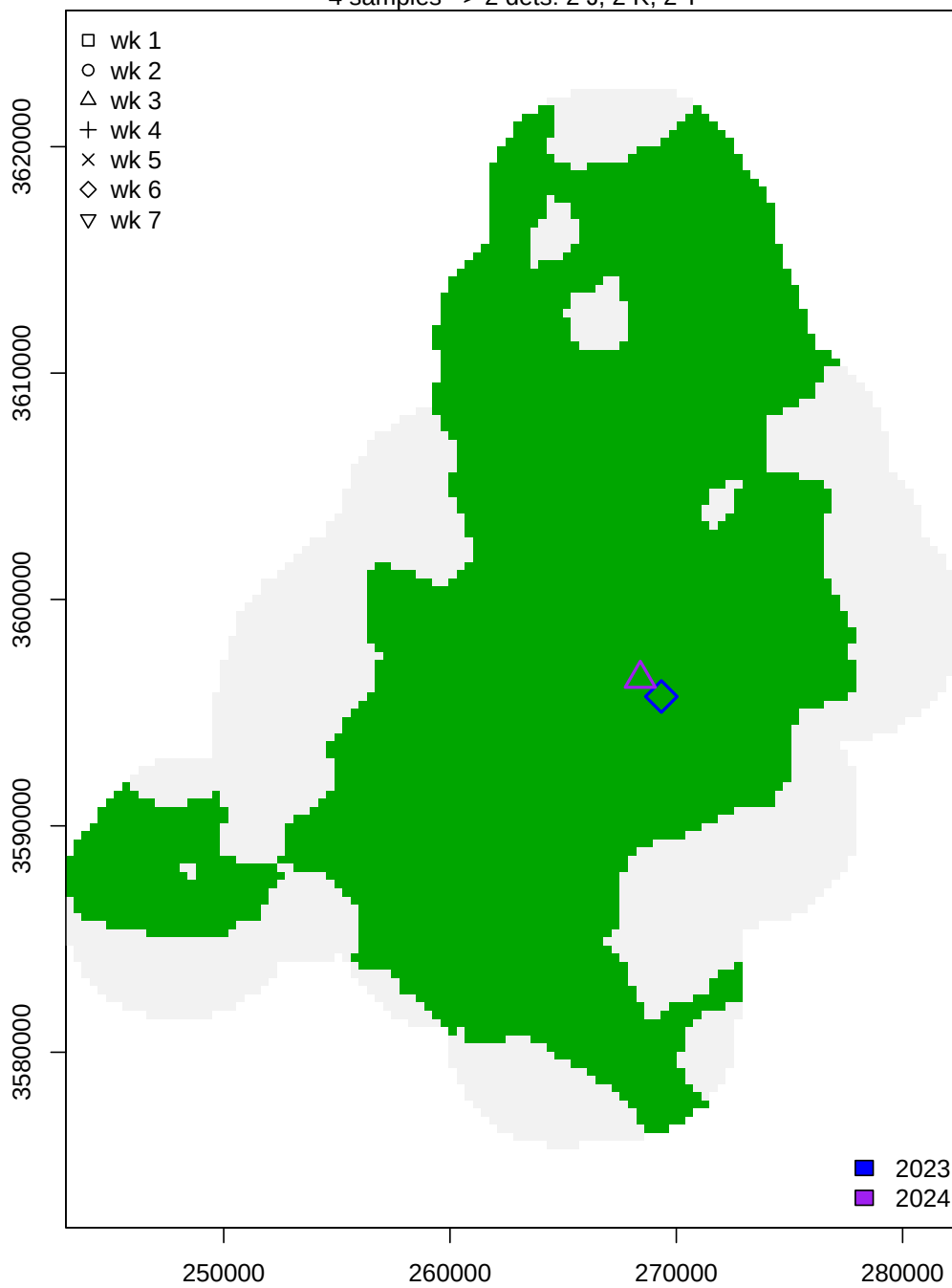
Female 34 – clone 277

20 samples -> 6 dets: 2 J, 6 K, 1 T



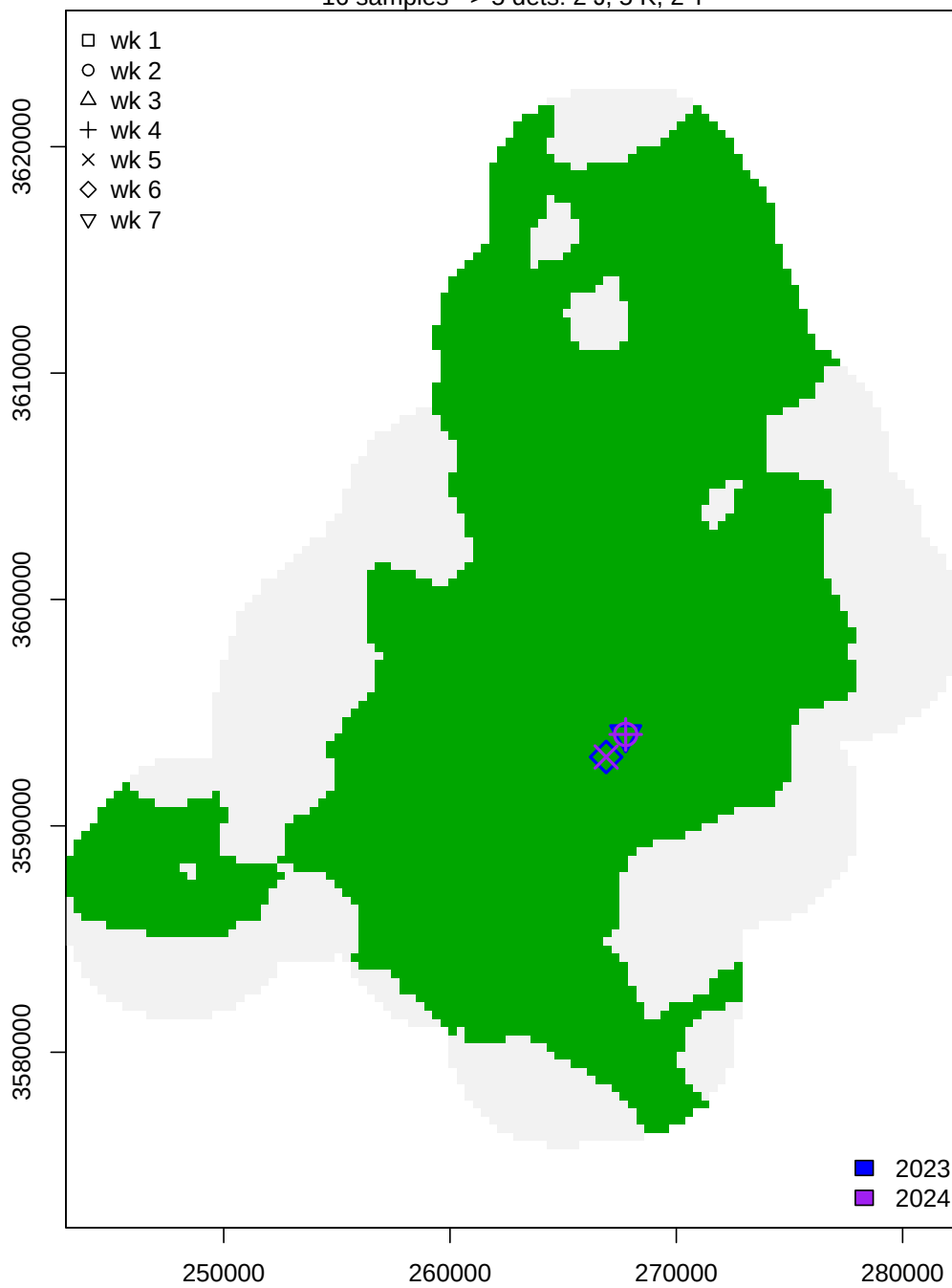
Female 35 – clone 15

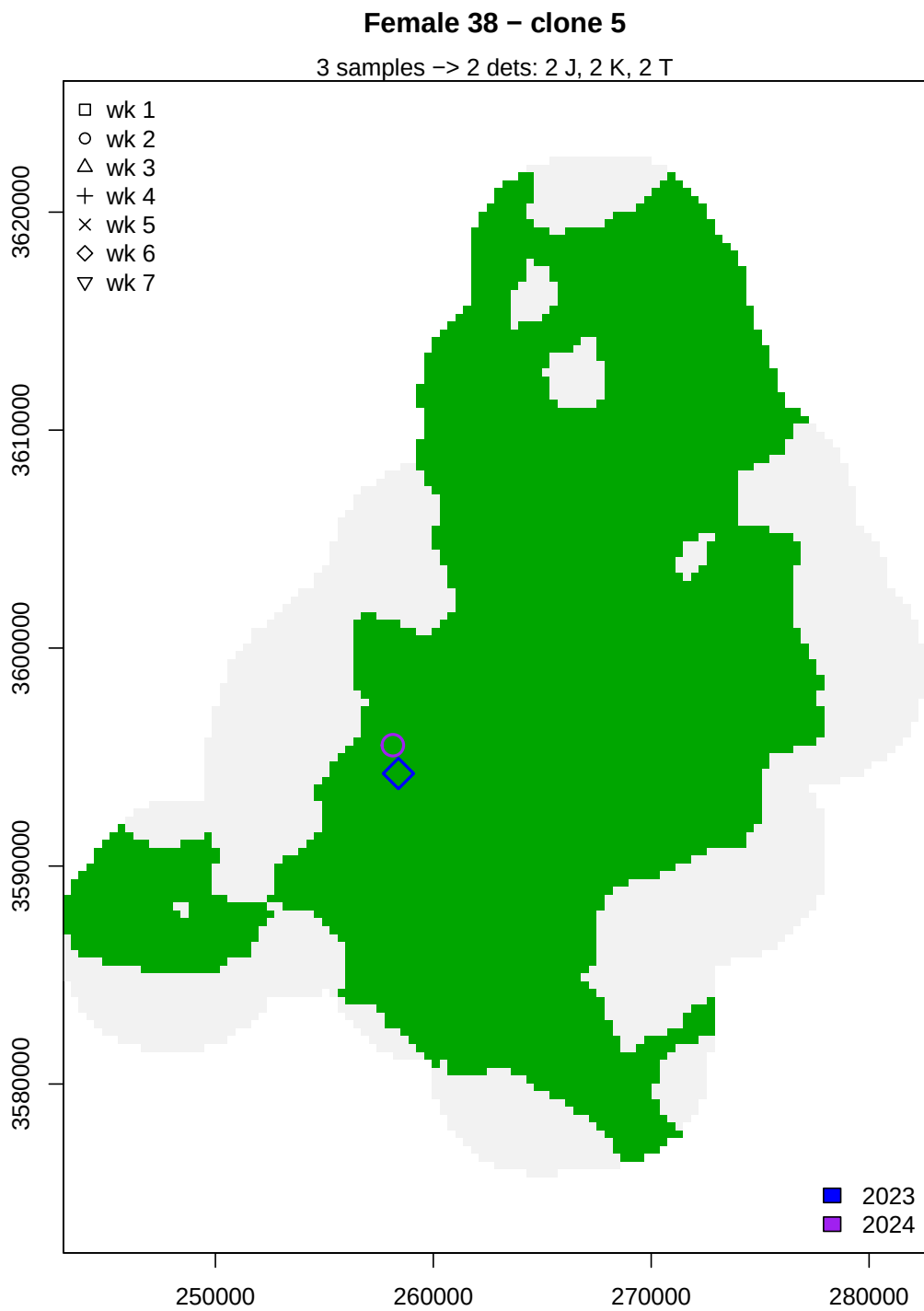
4 samples -> 2 dets: 2 J, 2 K, 2 T



Female 37 – clone 271

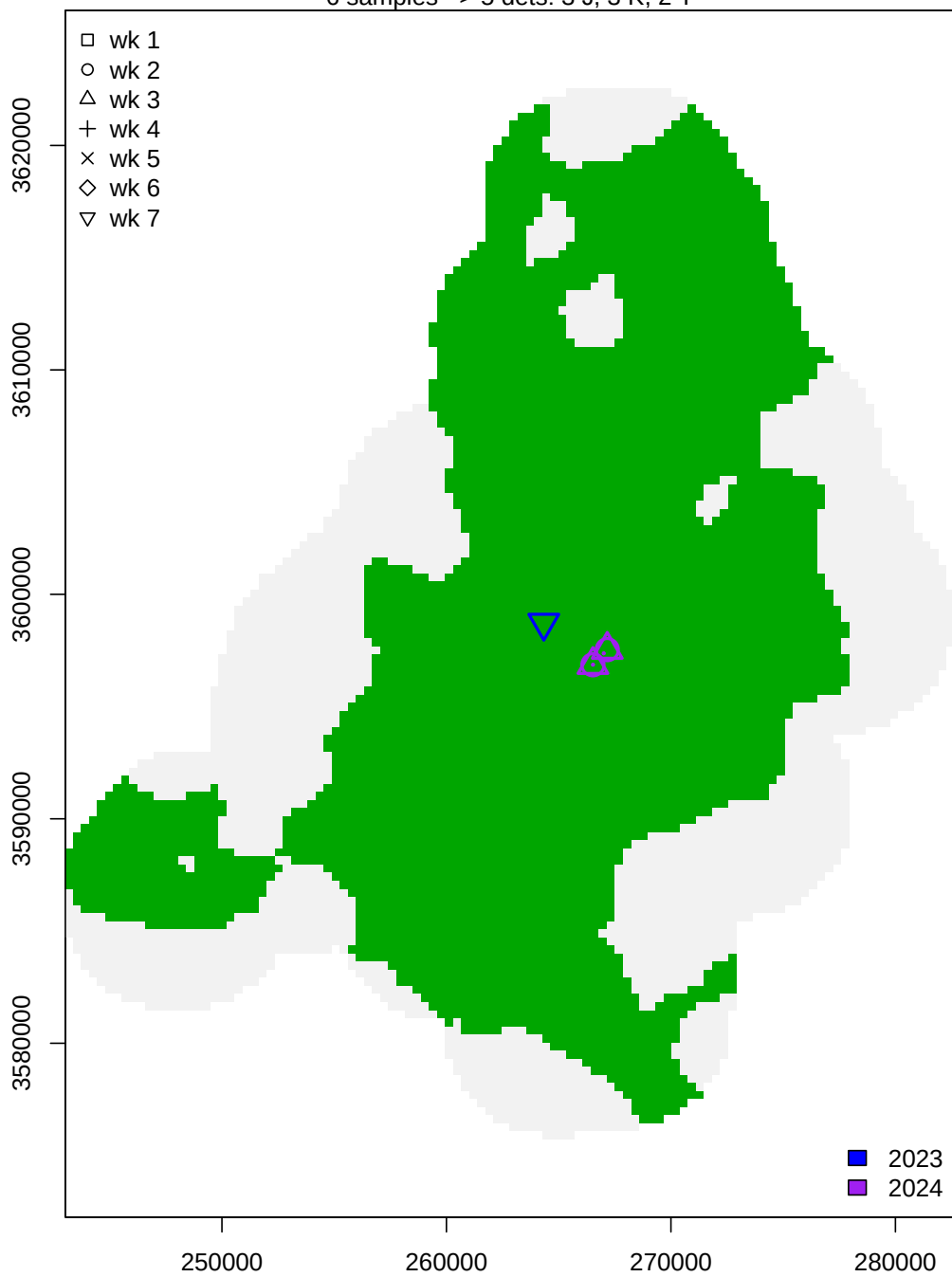
16 samples -> 5 dets: 2 J, 5 K, 2 T





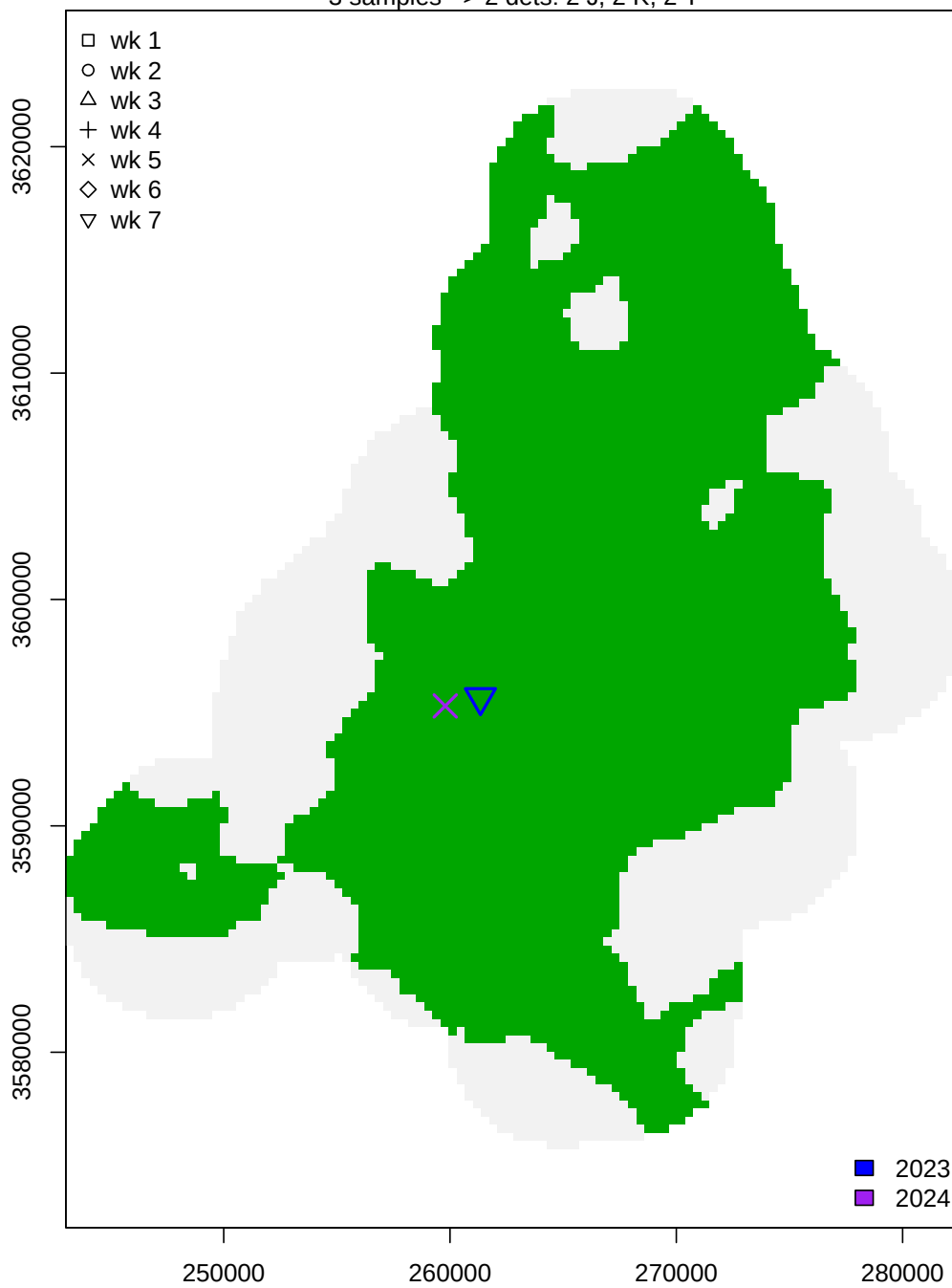
Female 39 – clone 45

6 samples → 5 dets: 3 J, 3 K, 2 T



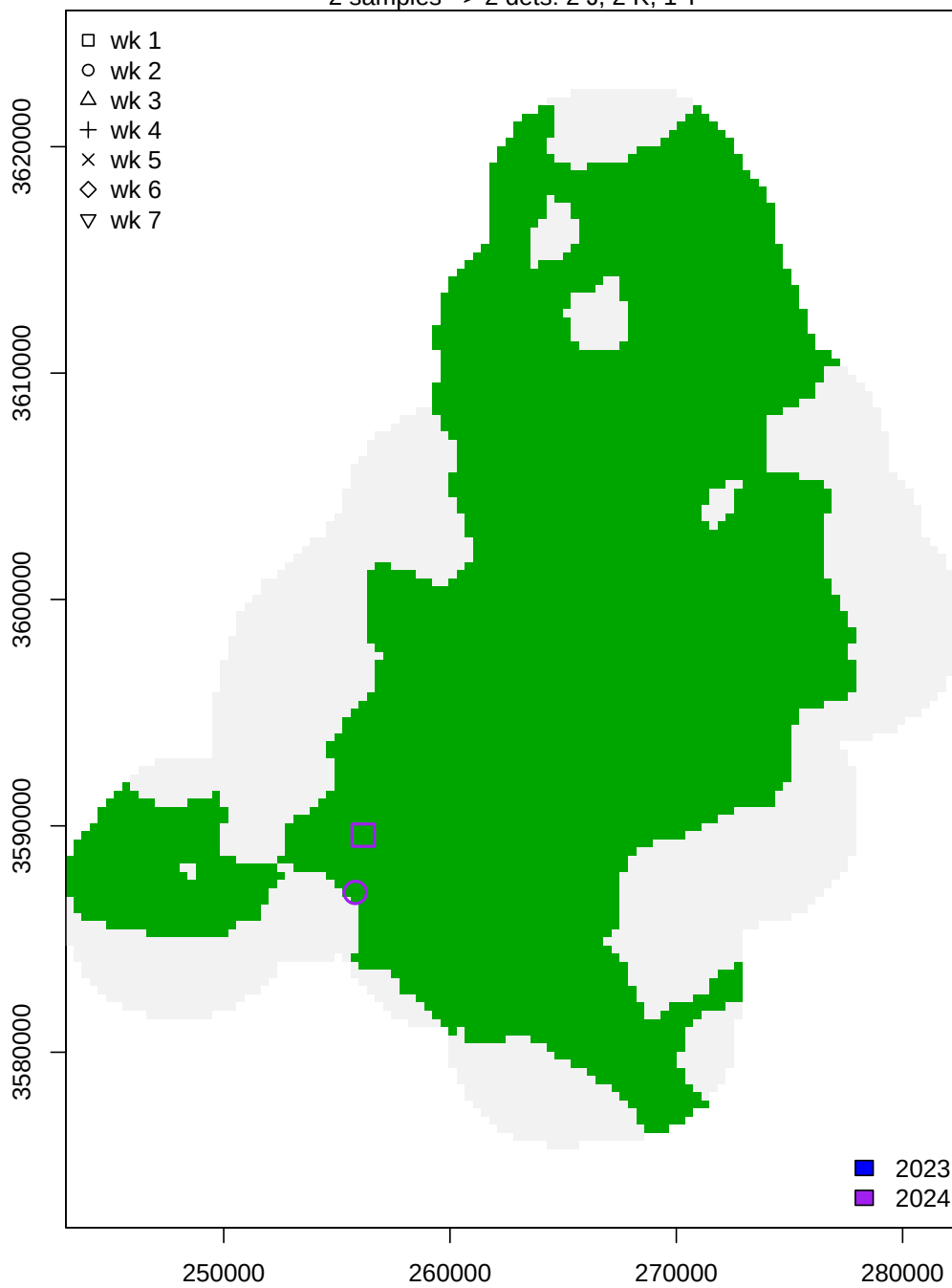
Female 46 – clone 172

3 samples -> 2 dets: 2 J, 2 K, 2 T



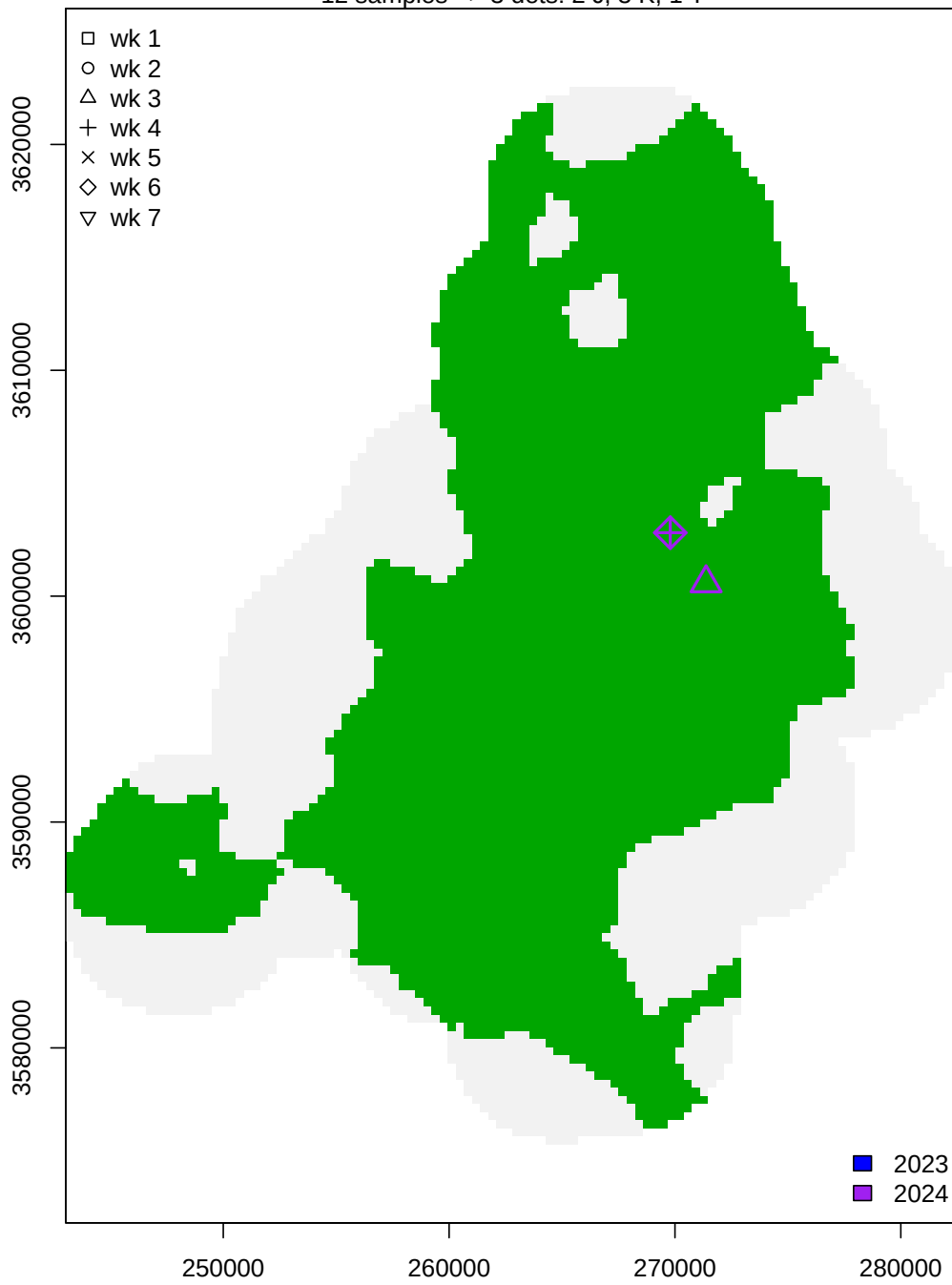
Female 49 – clone 225

2 samples -> 2 dets: 2 J, 2 K, 1 T



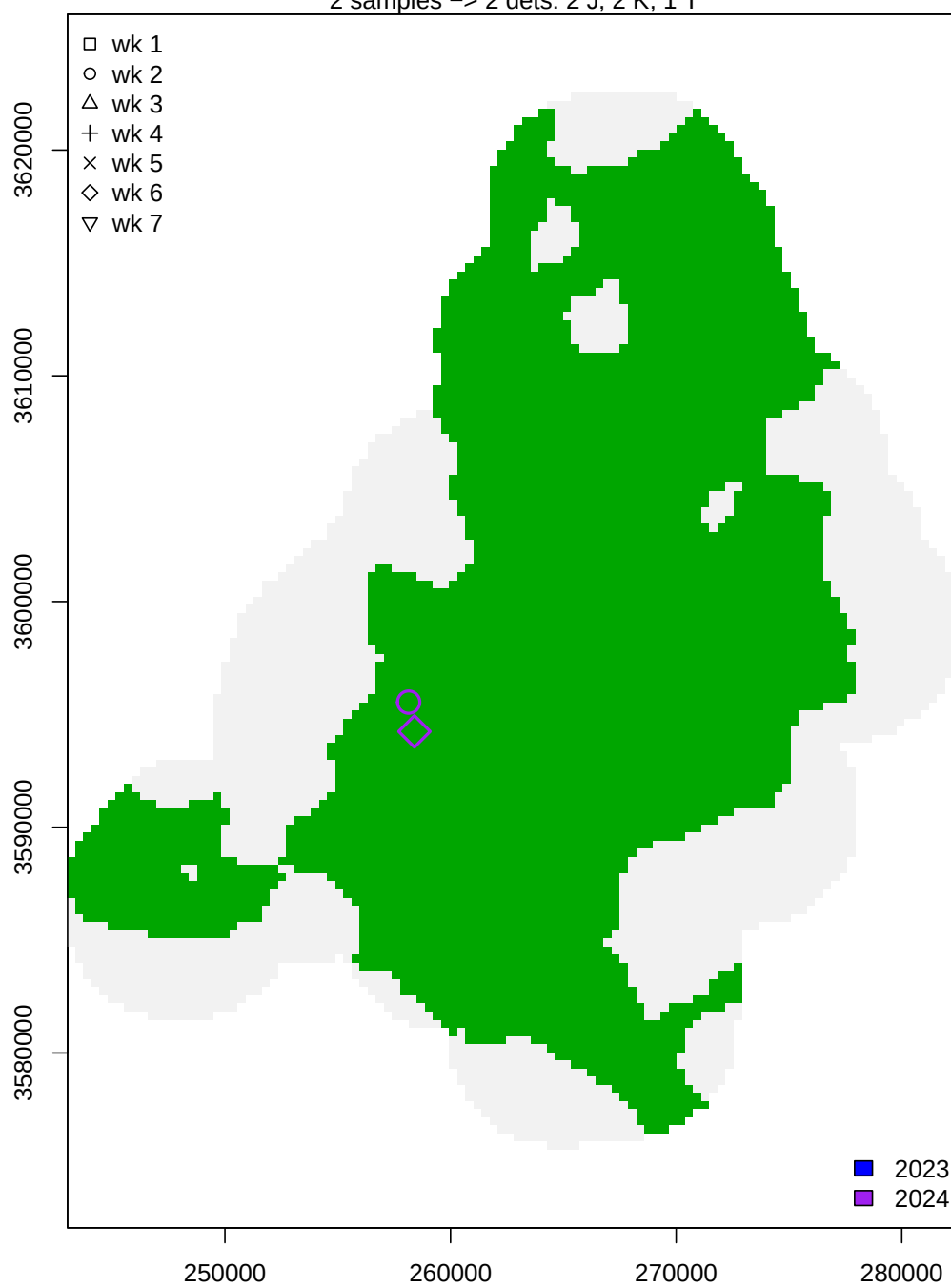
Female 57 – clone 280

12 samples -> 3 dets: 2 J, 3 K, 1 T



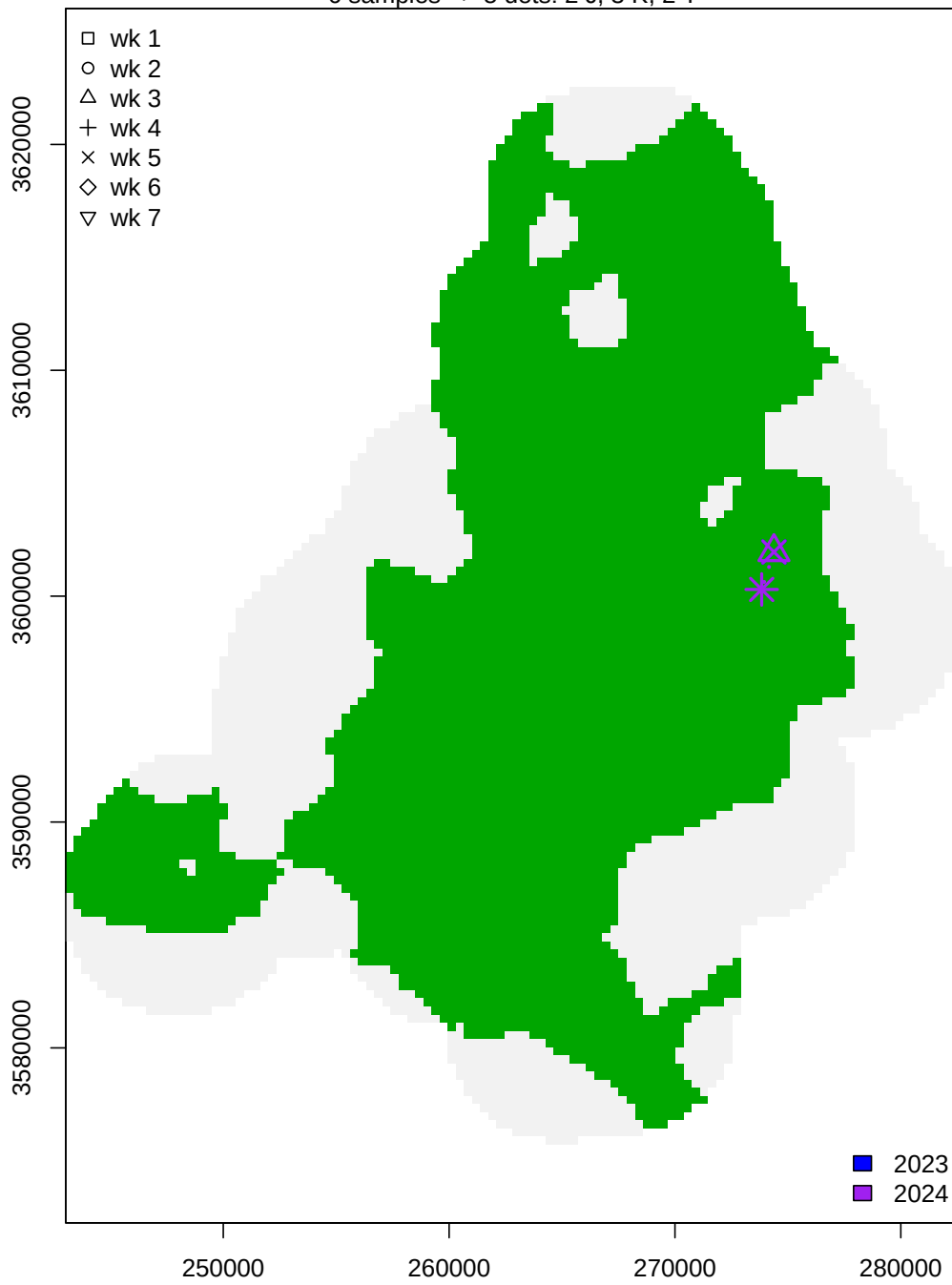
Female 75 – clone 321

2 samples -> 2 dets: 2 J, 2 K, 1 T



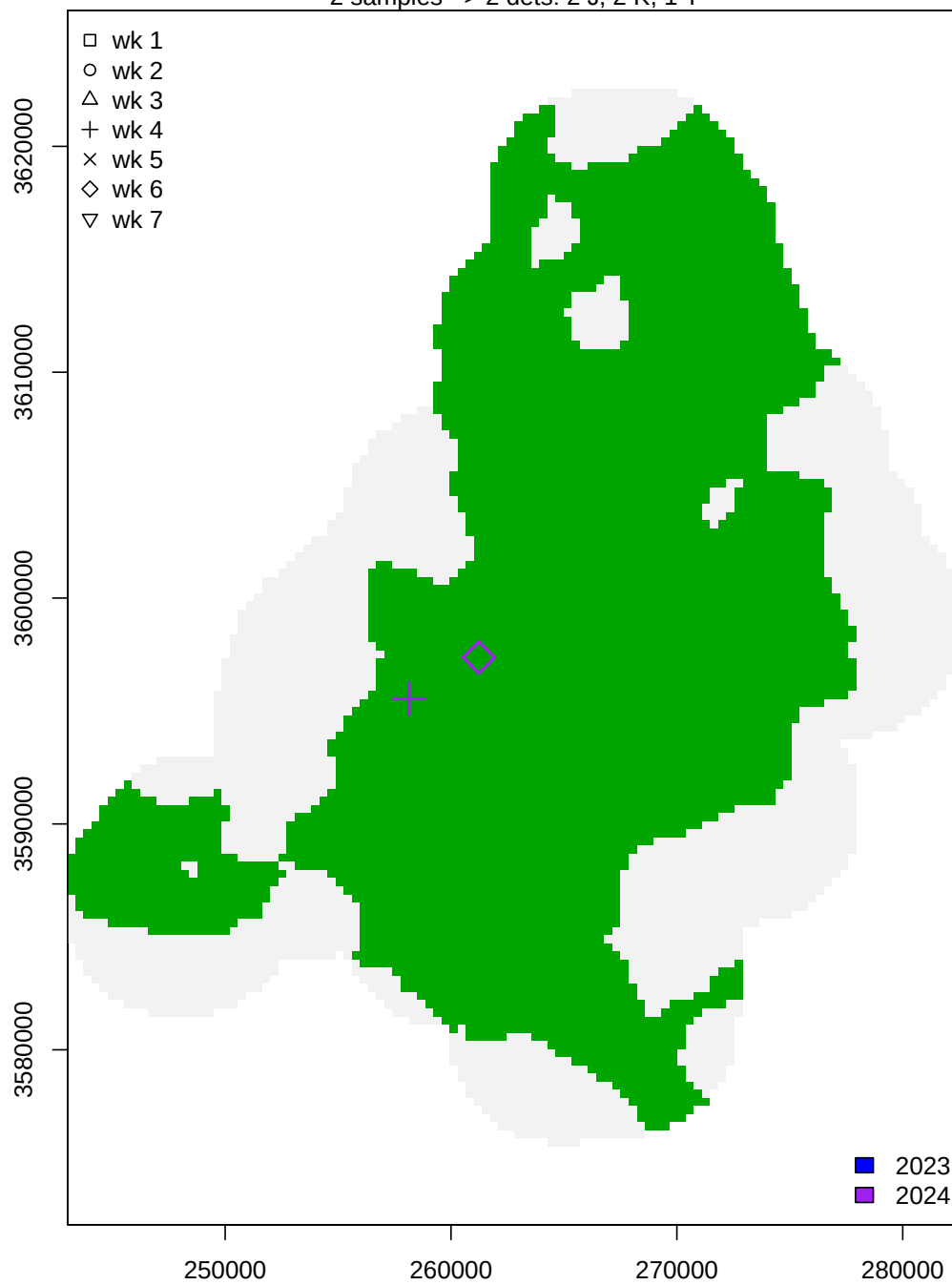
Female 77 – clone 324

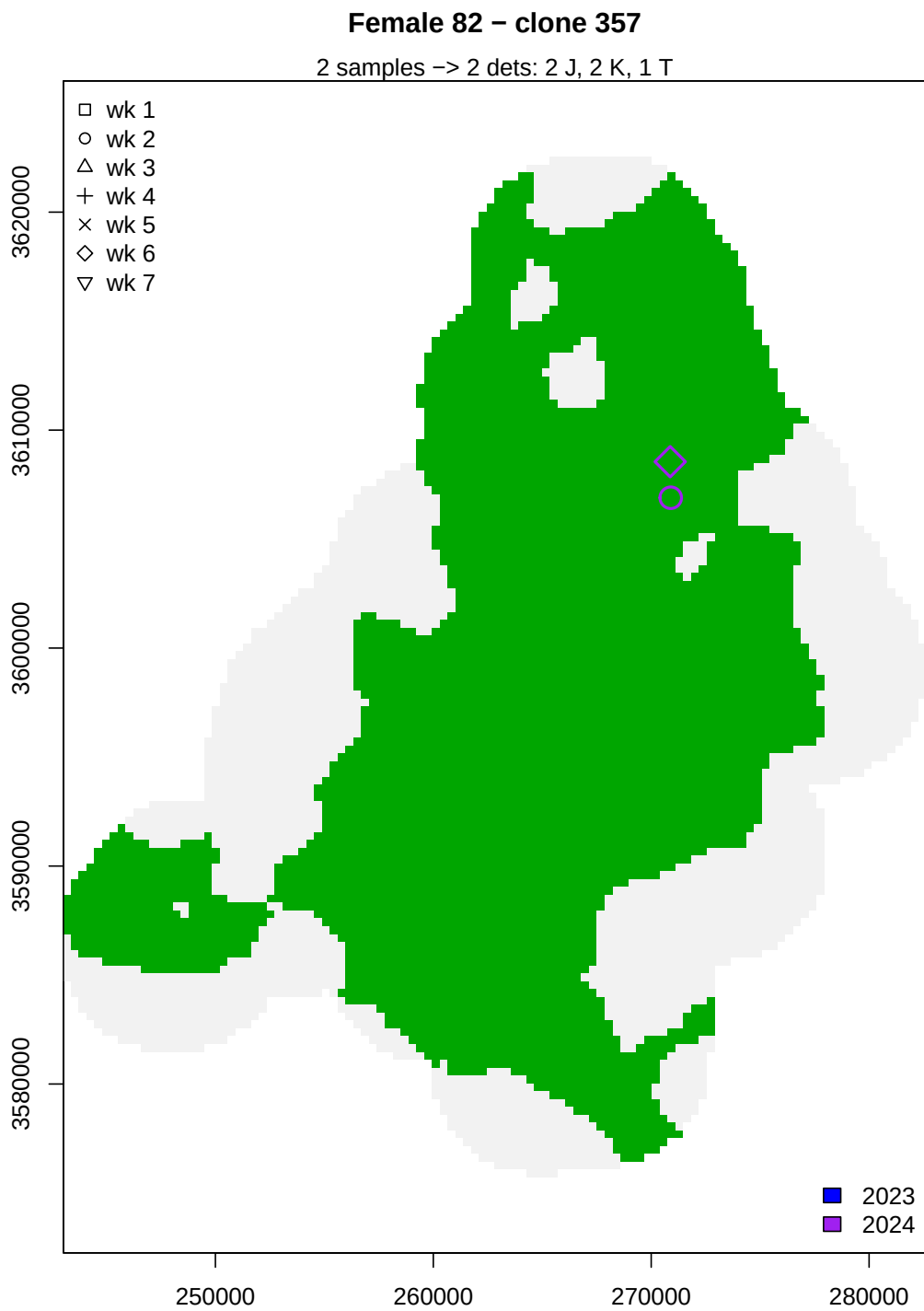
6 samples -> 5 dets: 2 J, 3 K, 2 T



Female 79 – clone 333

2 samples -> 2 dets: 2 J, 2 K, 1 T





Males

Then for males...

```

#Males
caphist <- Mch$y
Mclones <- Mch$clone

sumstats_4d(caphist)

##          ss.totN      ss.totcaps      ss.avgcaps ss.indwsprecap      ss.avgtraps
##          77.000000      254.000000          3.298701          26.000000          2.246753
##          ss.avgyrs
##          1.194805

nind <- dim(caphist)[1]

yr.col <- c("blue","purple")
wk.symbol <- 0:6

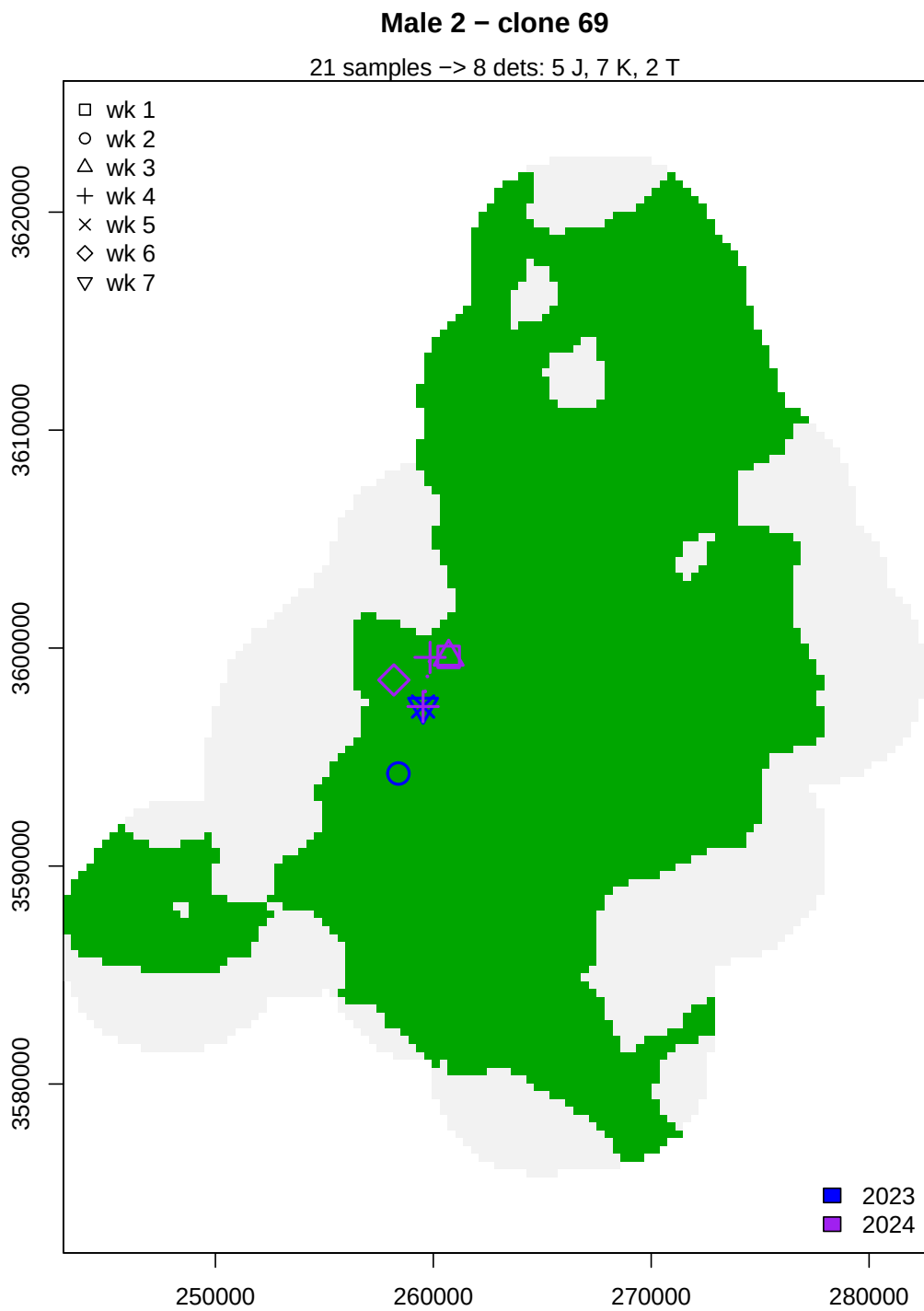
for(i in 1:nind) {
  i.ch <- caphist[i,,]

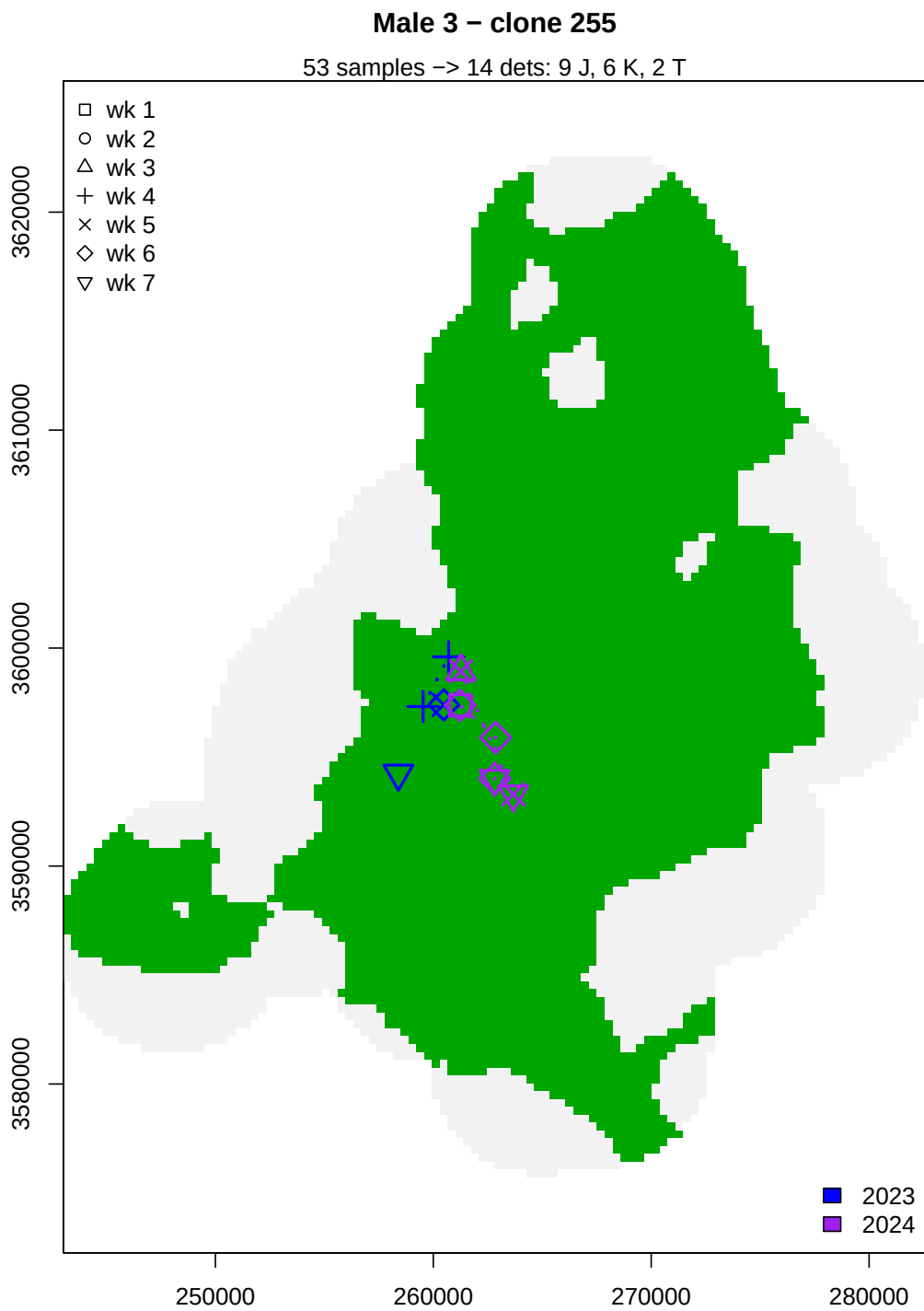
  nTraps <- sum(rowSums(i.ch) > 0)
  if(nTraps > 1) {
    plot.me <- as.data.frame(which(i.ch == 1, arr.ind = TRUE))
    plot(S360, legend = FALSE, main = paste("Male",i,"- clone", Mclones[i]))
    mtext(paste(length(which(smd$finalClone == Mclones[i])), "samples ->",
      nrow(plot.me), "dets:",
      nTraps, "J,",
      length(unique(plot.me$dim2)), "K,",
      length(unique(plot.me$dim3)), "T"))
    points(x=locs[plot.me$dim1,1], locs[plot.me$dim1,2],
      pch = wk.symbol[plot.me$dim2], col = yr.col[plot.me$dim3],
      cex = 2, lwd = 2)

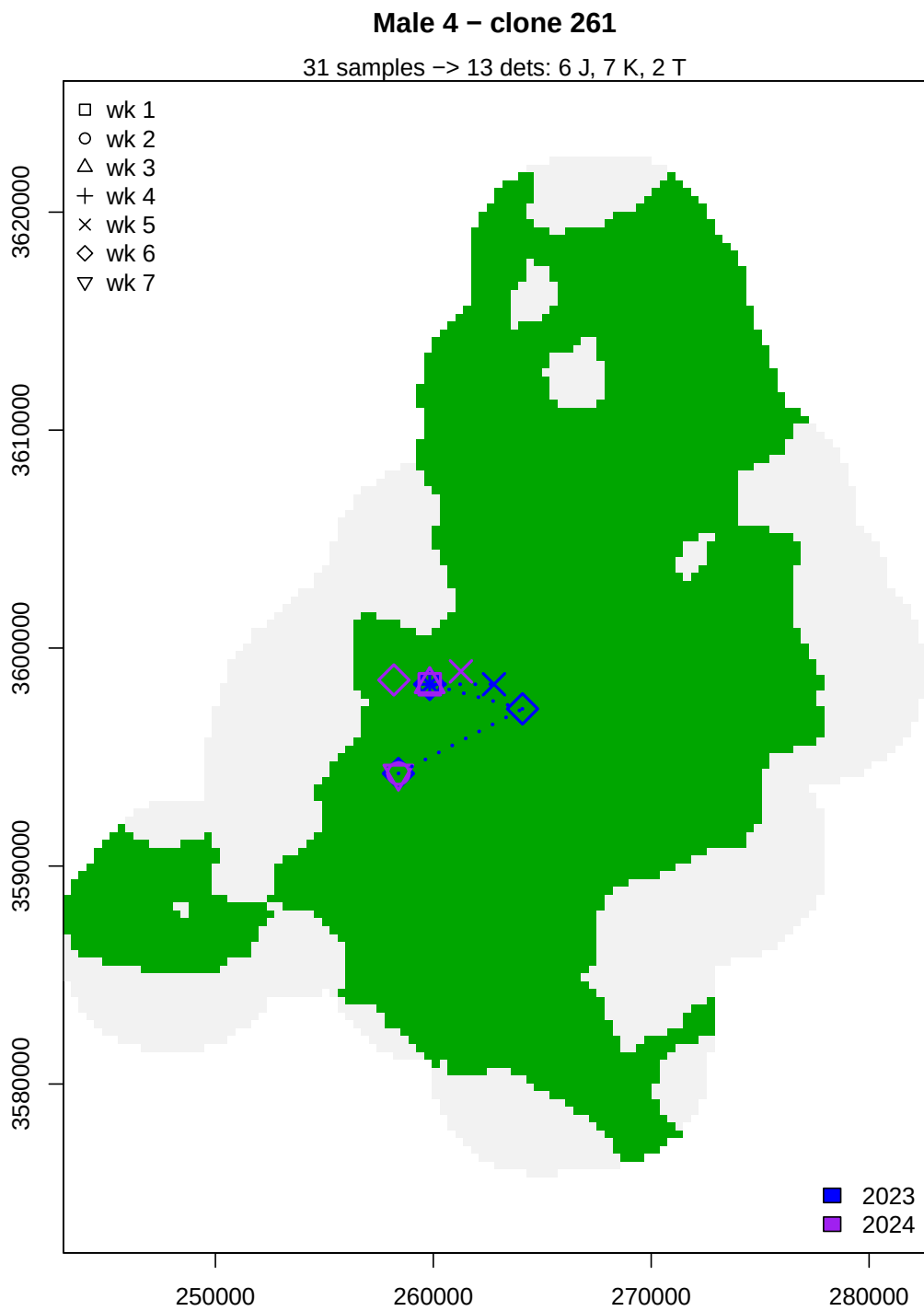
    for(r in 2:nrow(plot.me)) {
      if(plot.me$dim3[r]==plot.me$dim3[r-1] &
        plot.me$dim2[r]==plot.me$dim2[r-1]) {
        lines(x=c(locs[plot.me$dim1[r-1],1], locs[plot.me$dim1[r],1]),
          y=c(locs[plot.me$dim1[r-1],2], locs[plot.me$dim1[r],2]),
          lty="dotted", col = yr.col[plot.me$dim3[r]],
          lwd = 2.5)
      }
    }

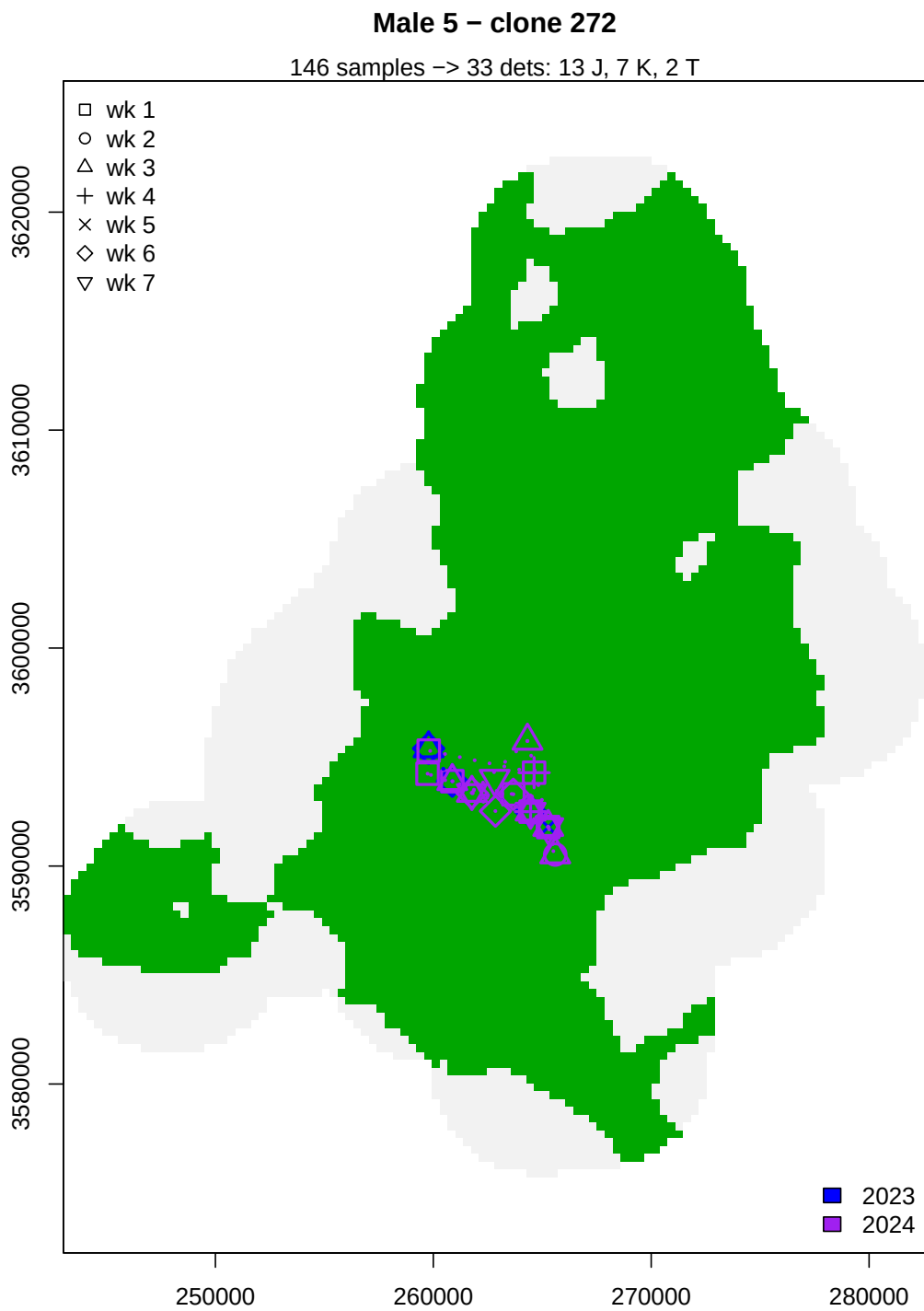
    legend("topleft", bty = "n", pch = 0:6, legend = paste("wk",c(1:7)))
    legend("bottomright", bty = "n", fill = yr.col, legend = c("2023","2024"))
  }
}

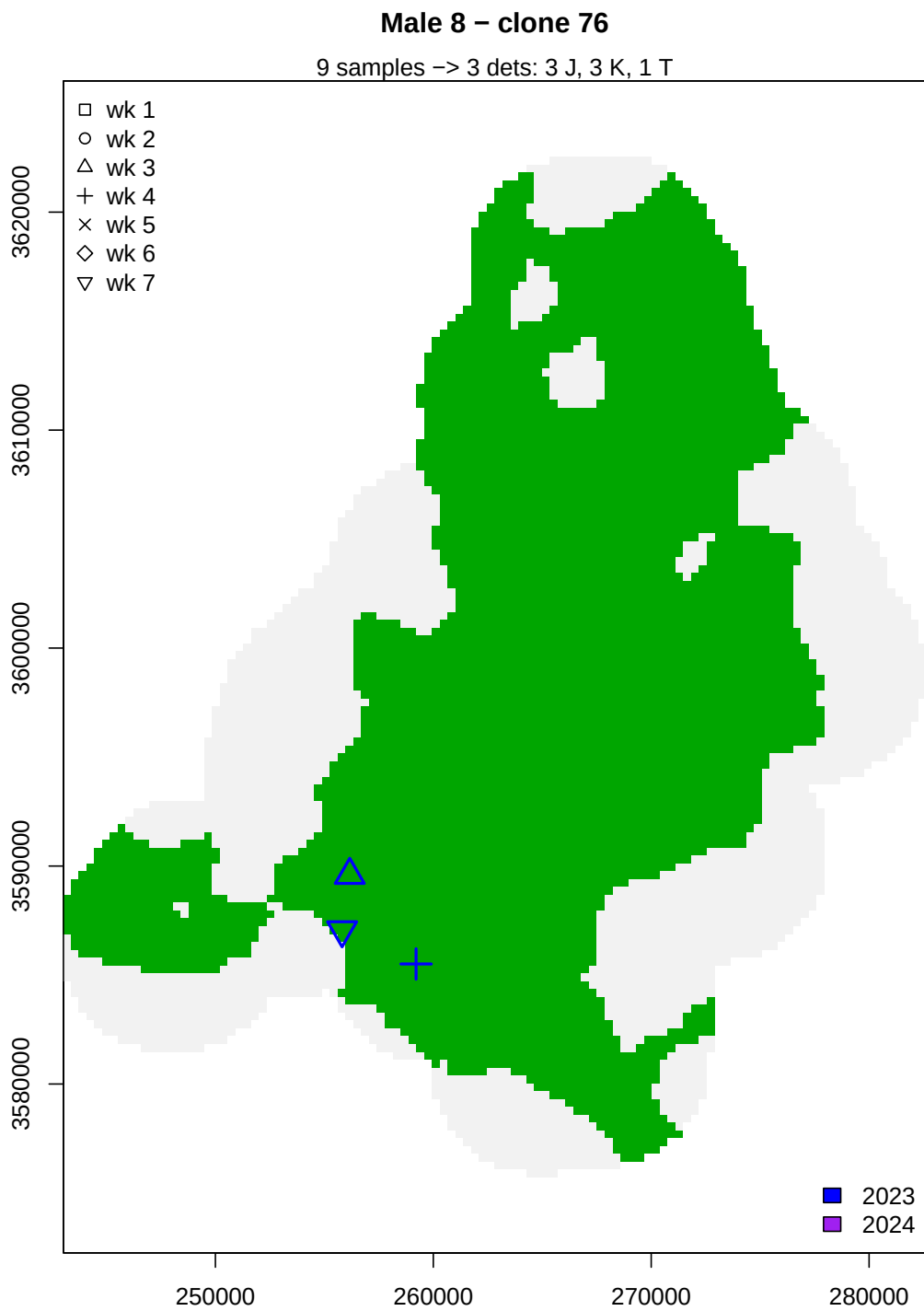
```

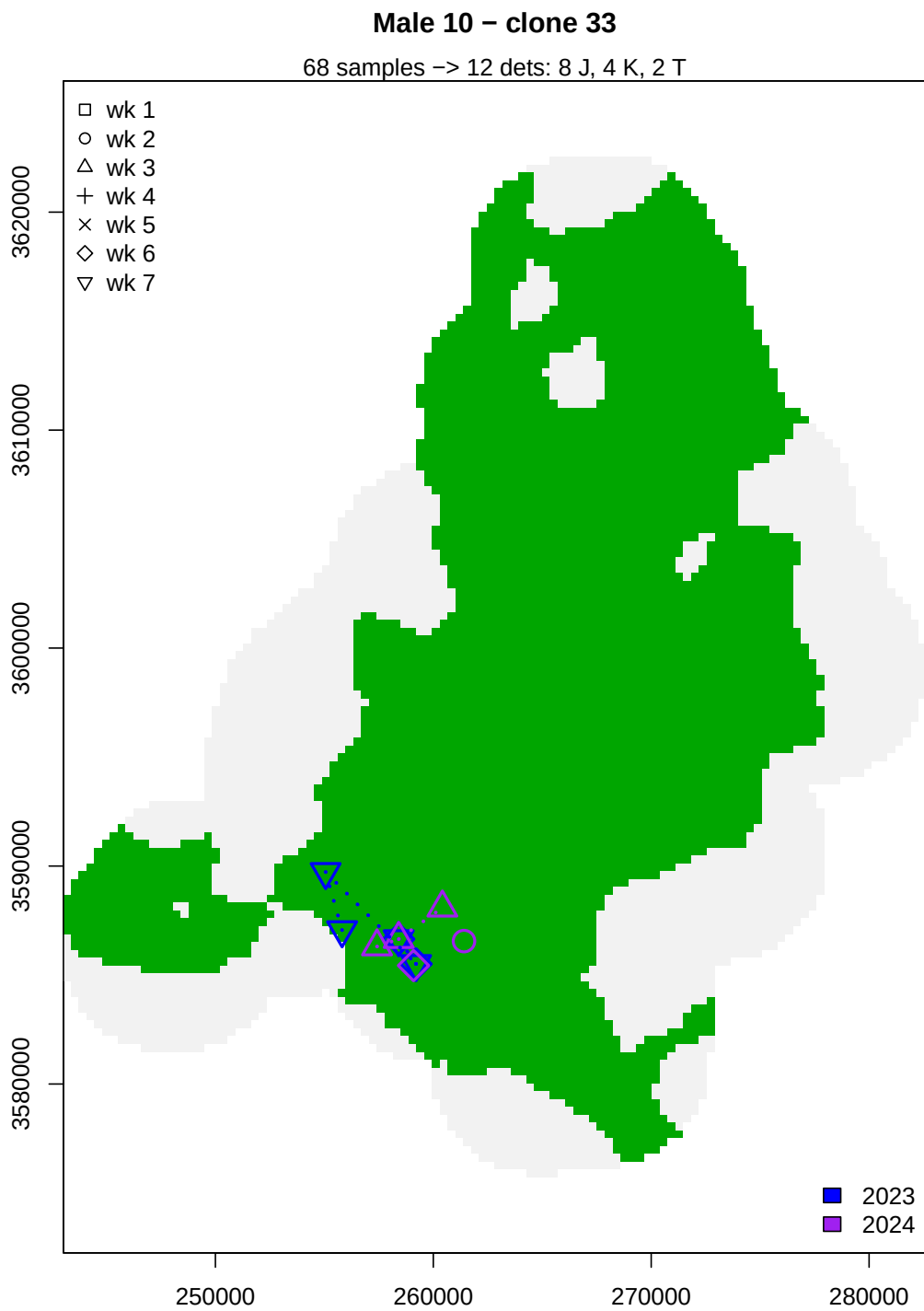






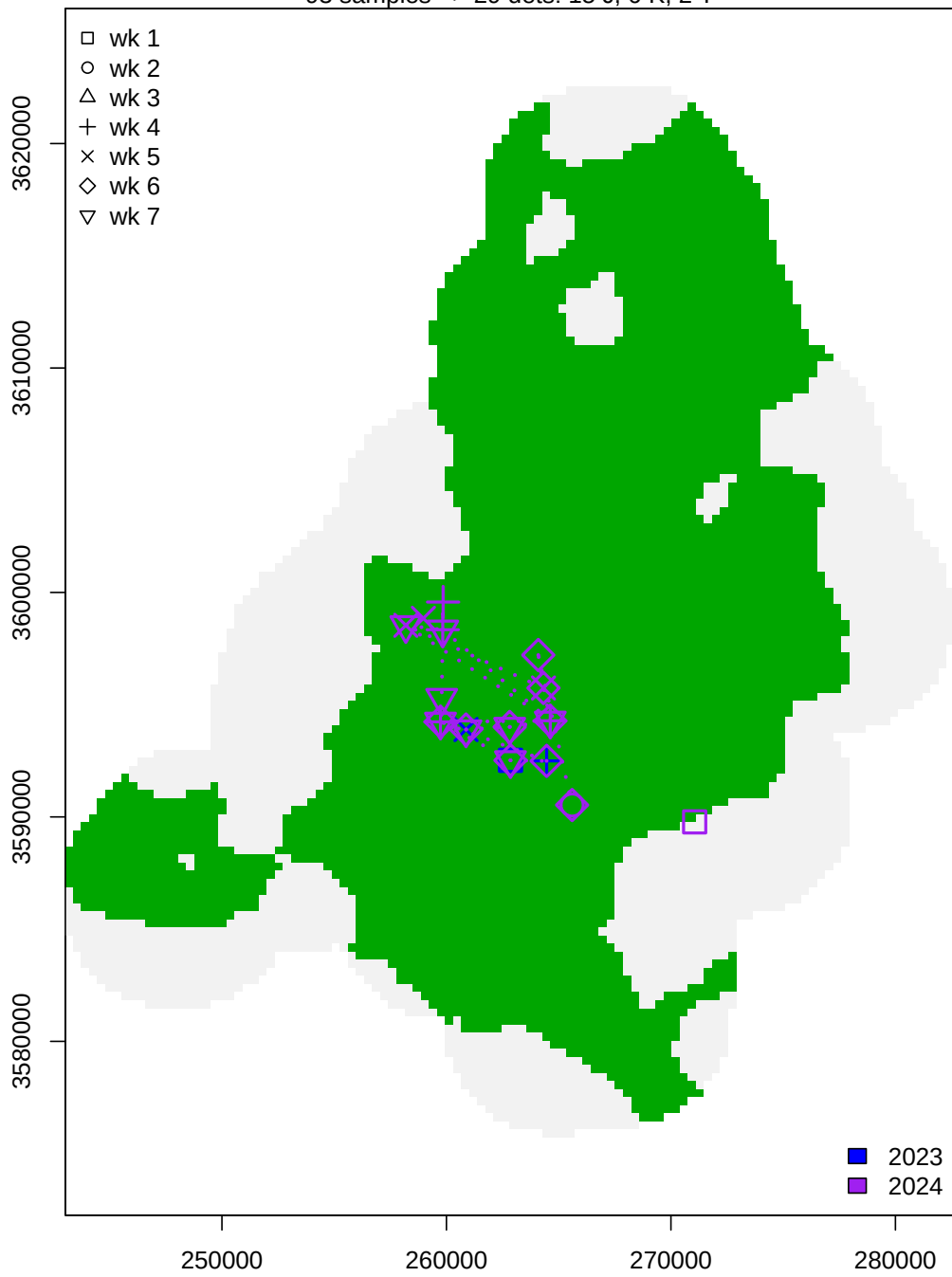


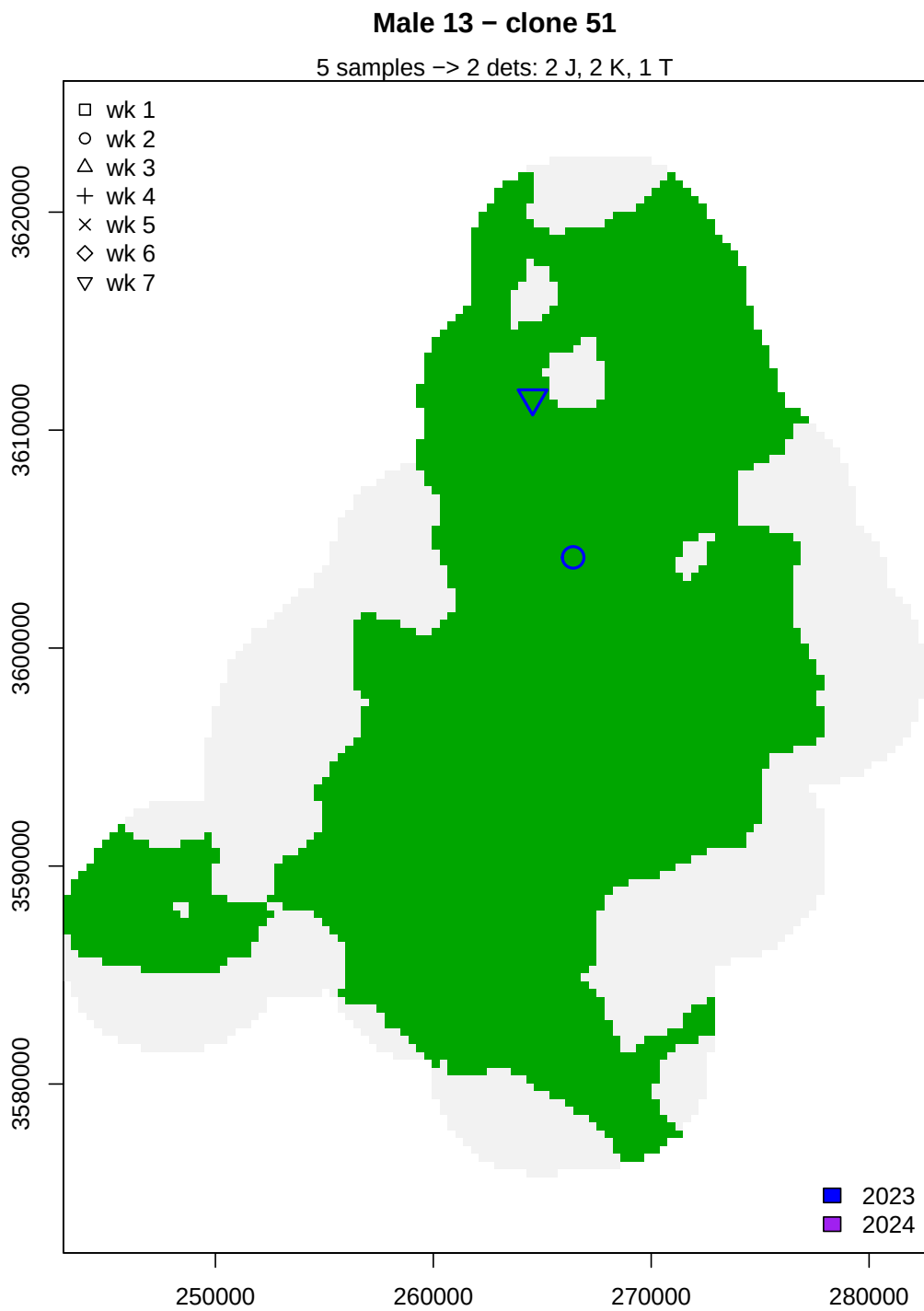


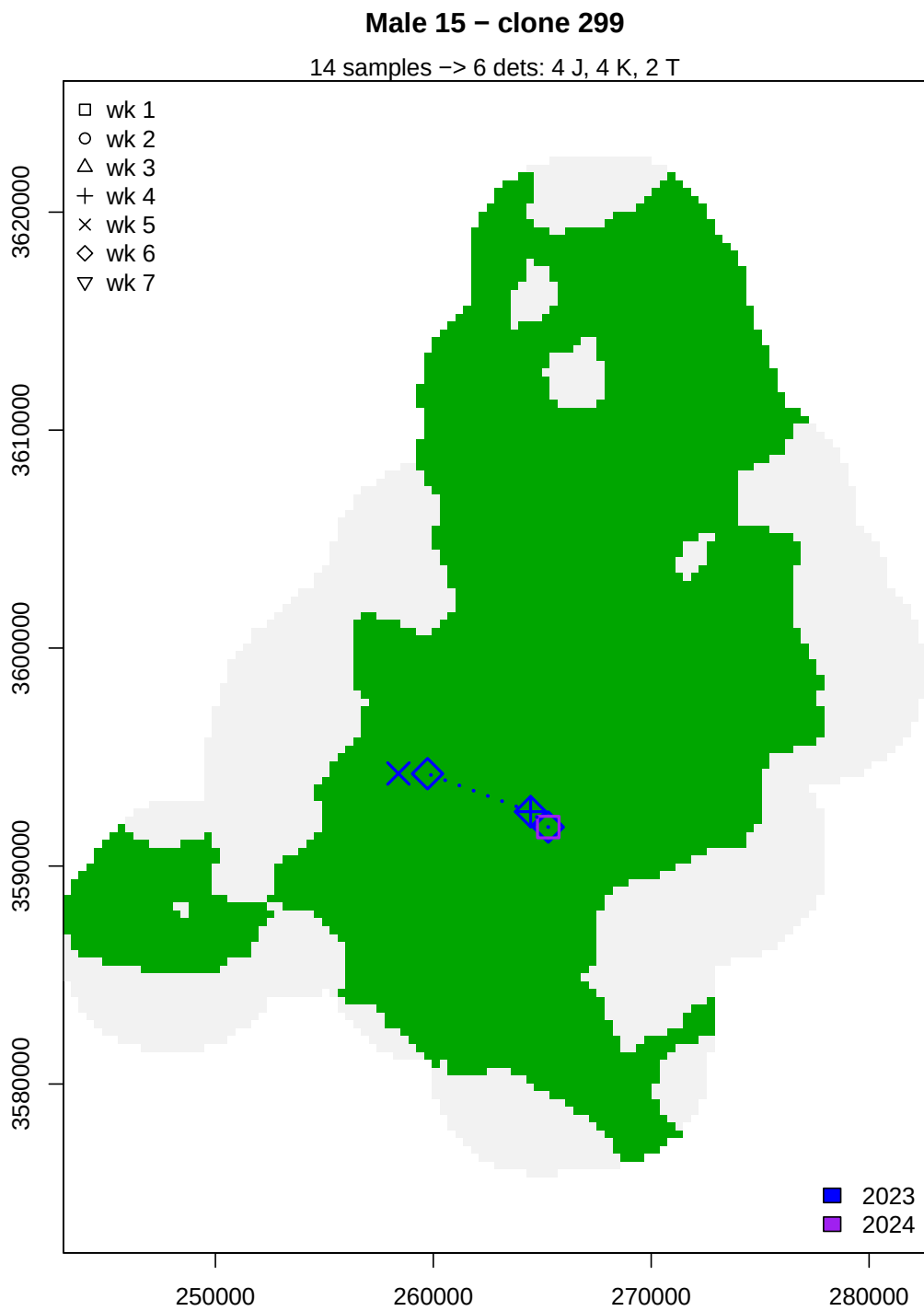


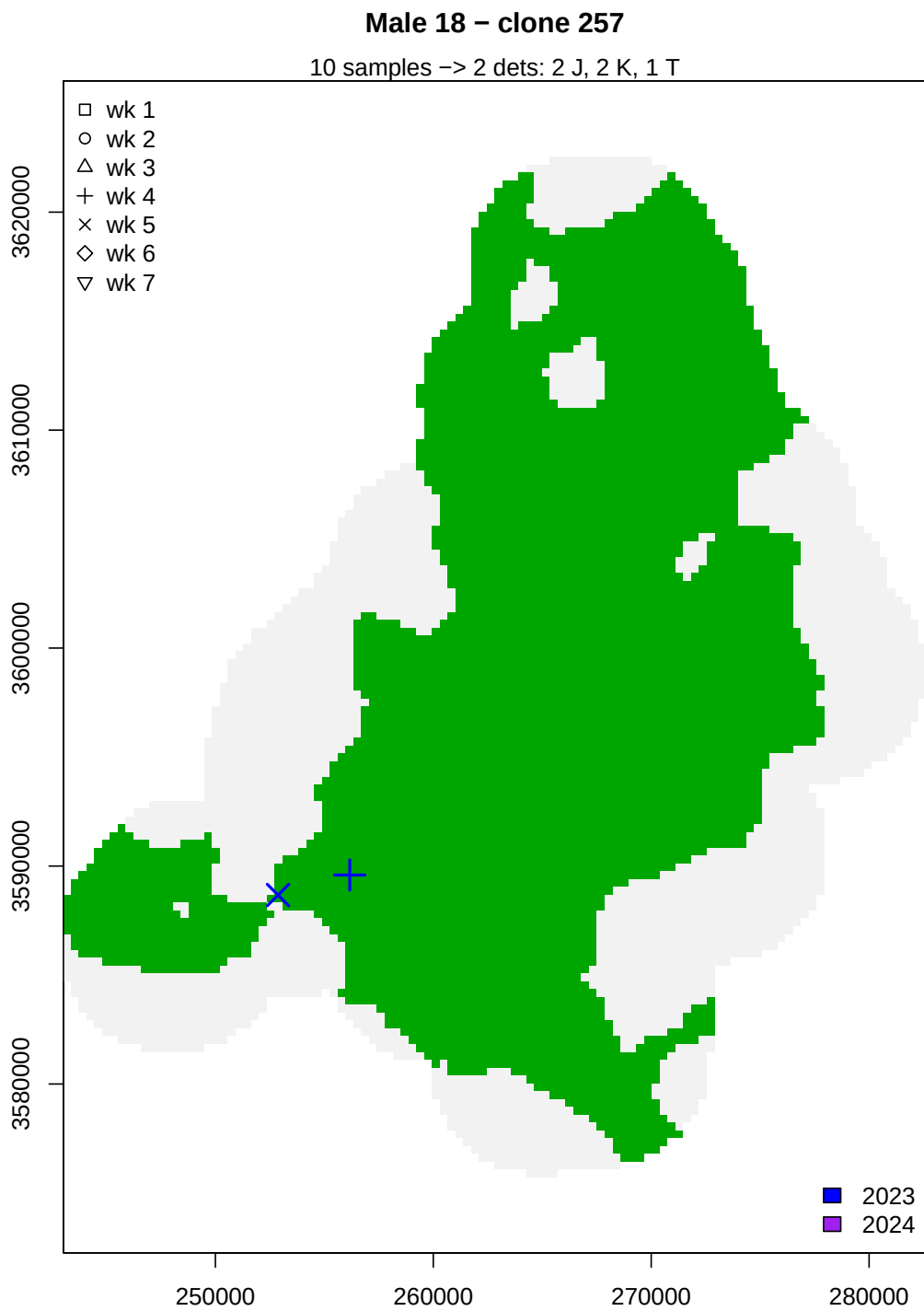
Male 12 – clone 262

93 samples -> 29 dets: 15 J, 6 K, 2 T



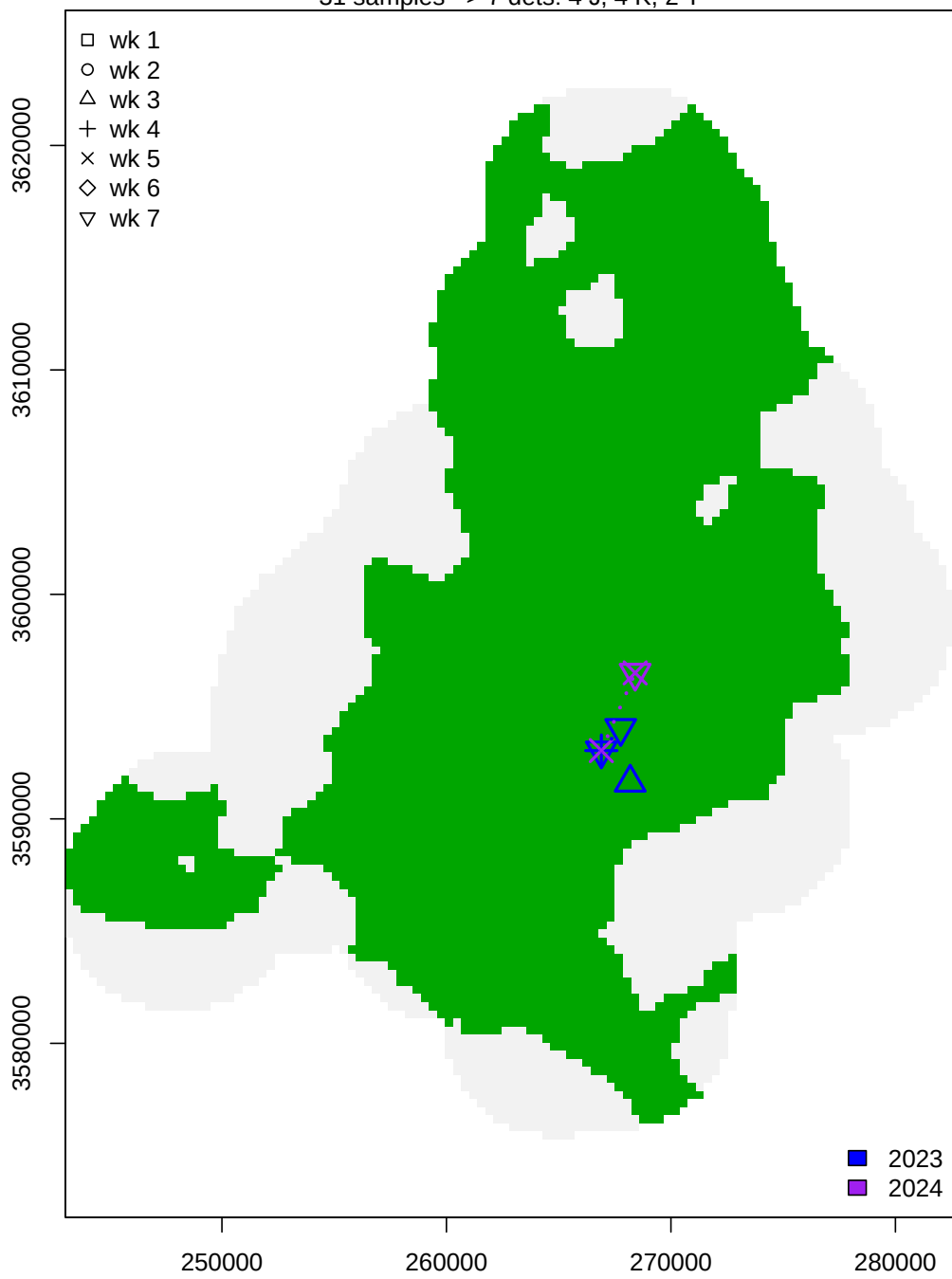






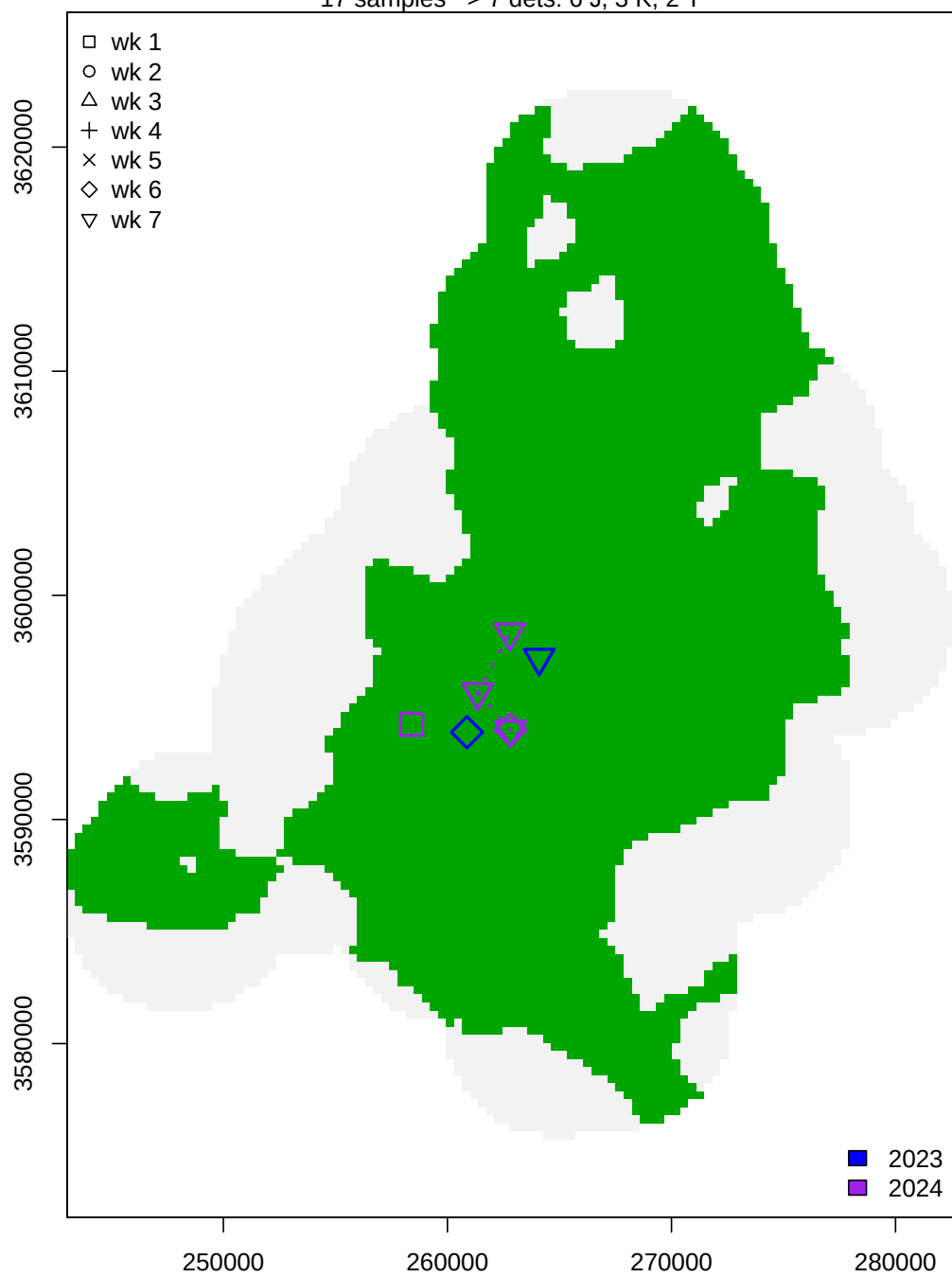
Male 19 – clone 288

31 samples -> 7 dets: 4 J, 4 K, 2 T



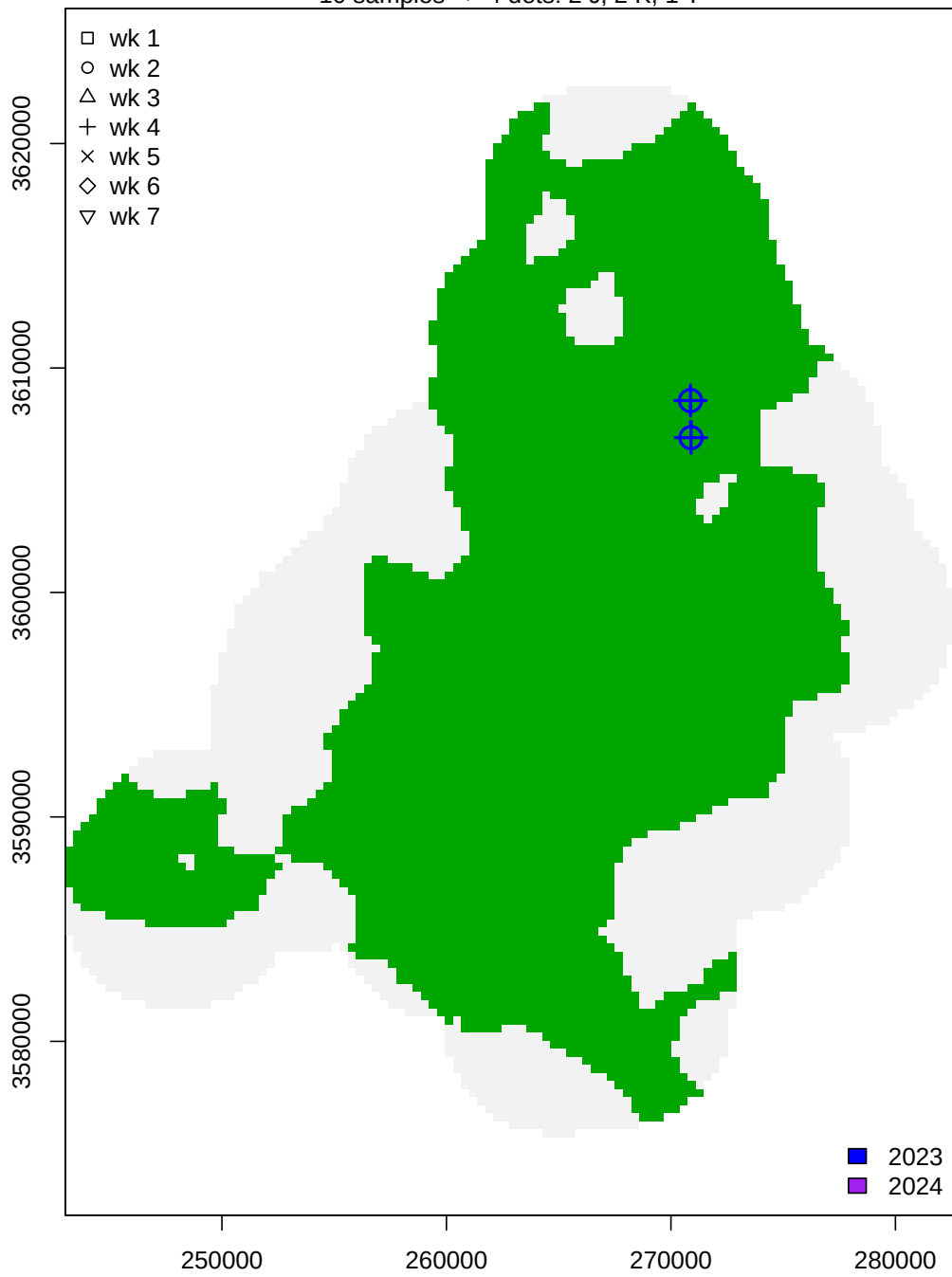
Male 20 – clone 269

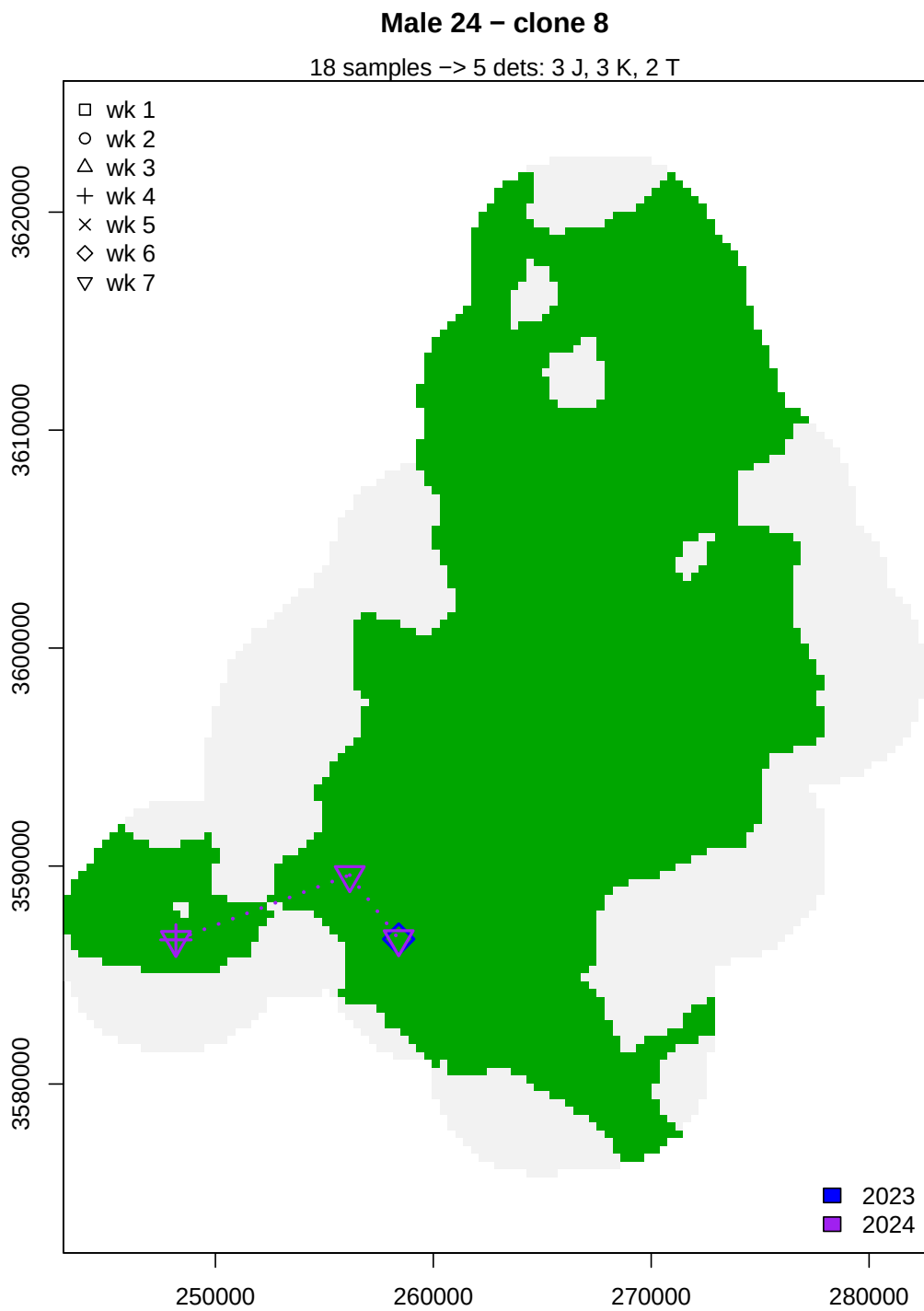
17 samples -> 7 dets: 6 J, 3 K, 2 T



Male 22 – clone 265

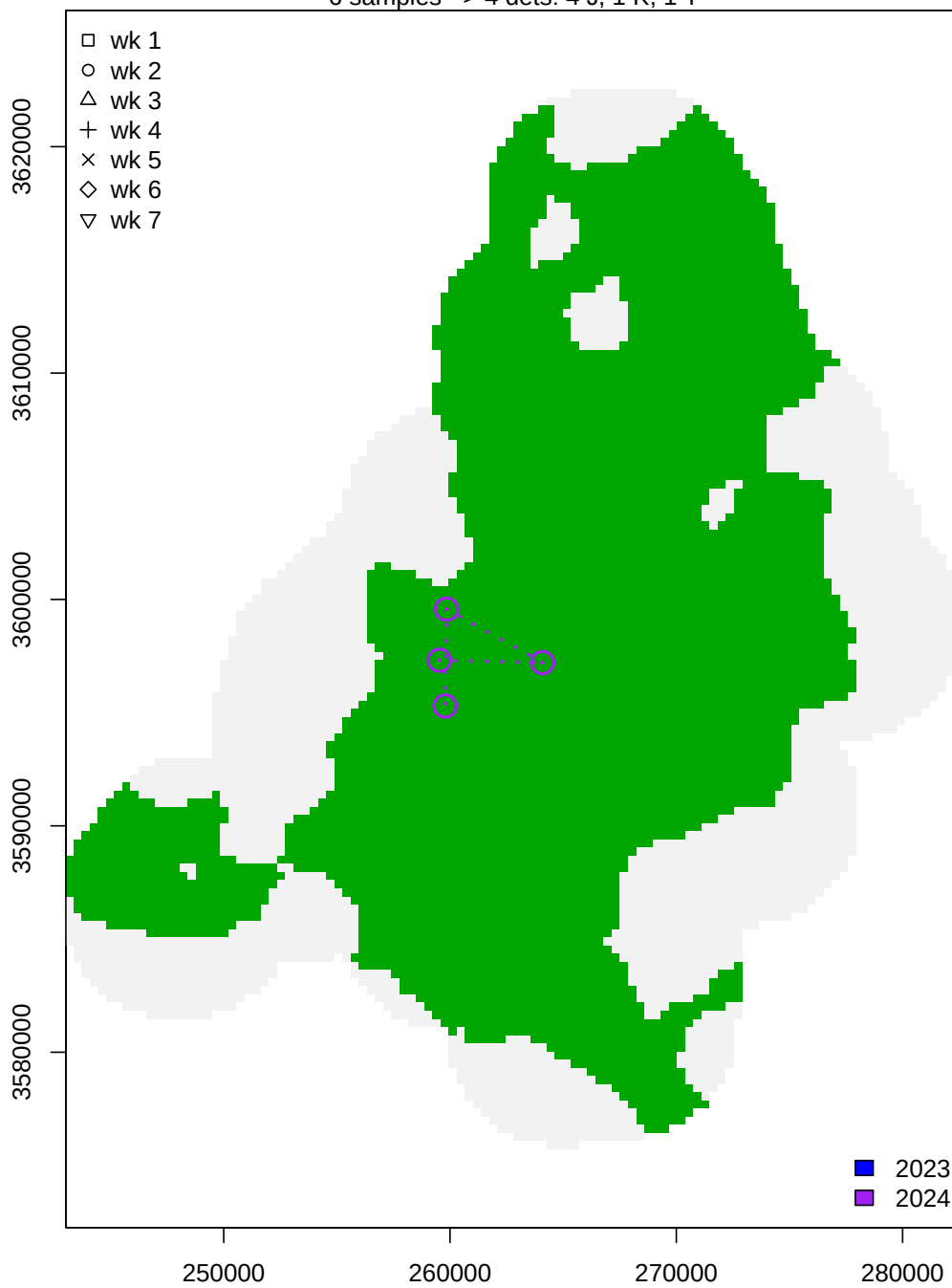
10 samples -> 4 dets: 2 J, 2 K, 1 T

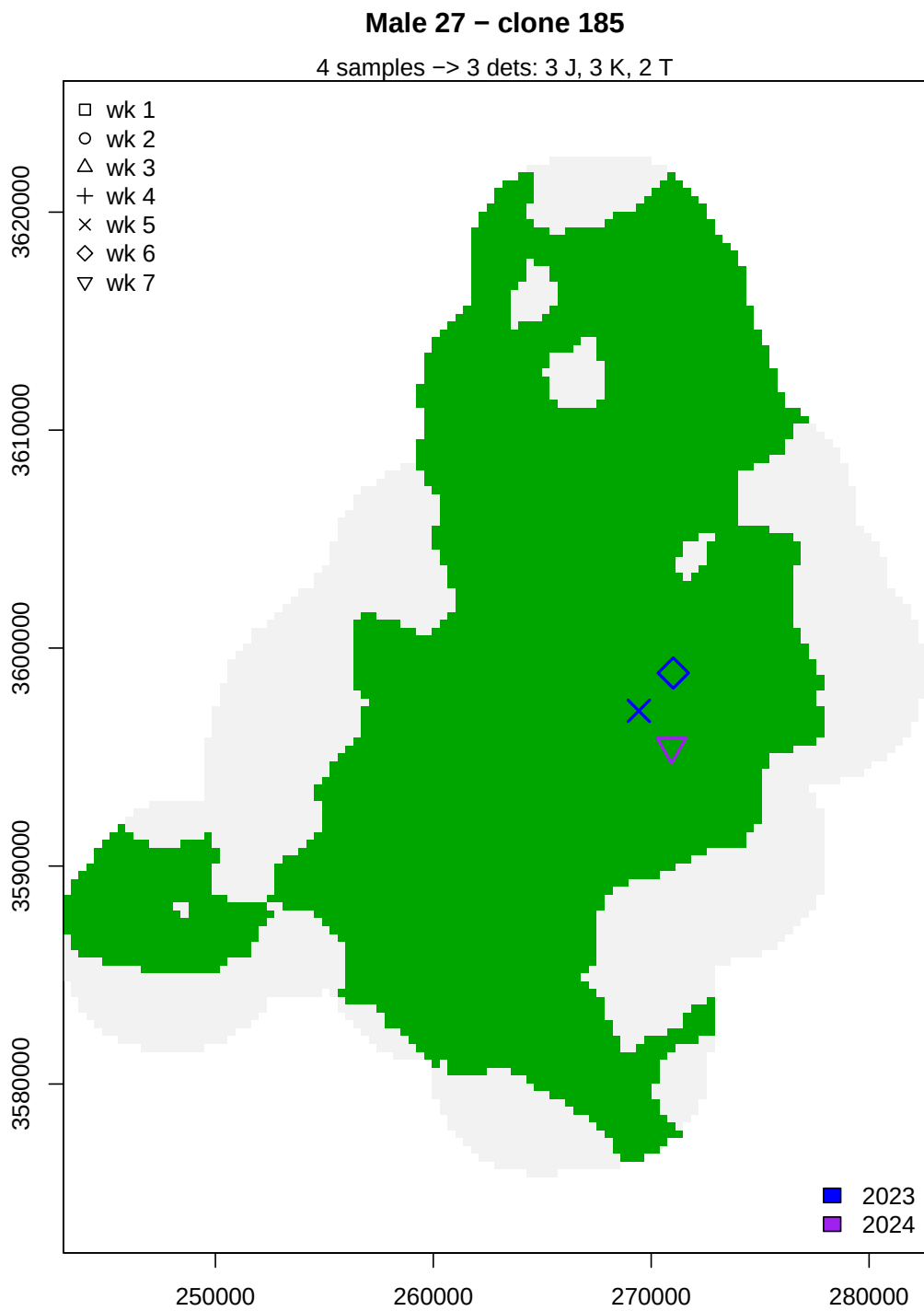


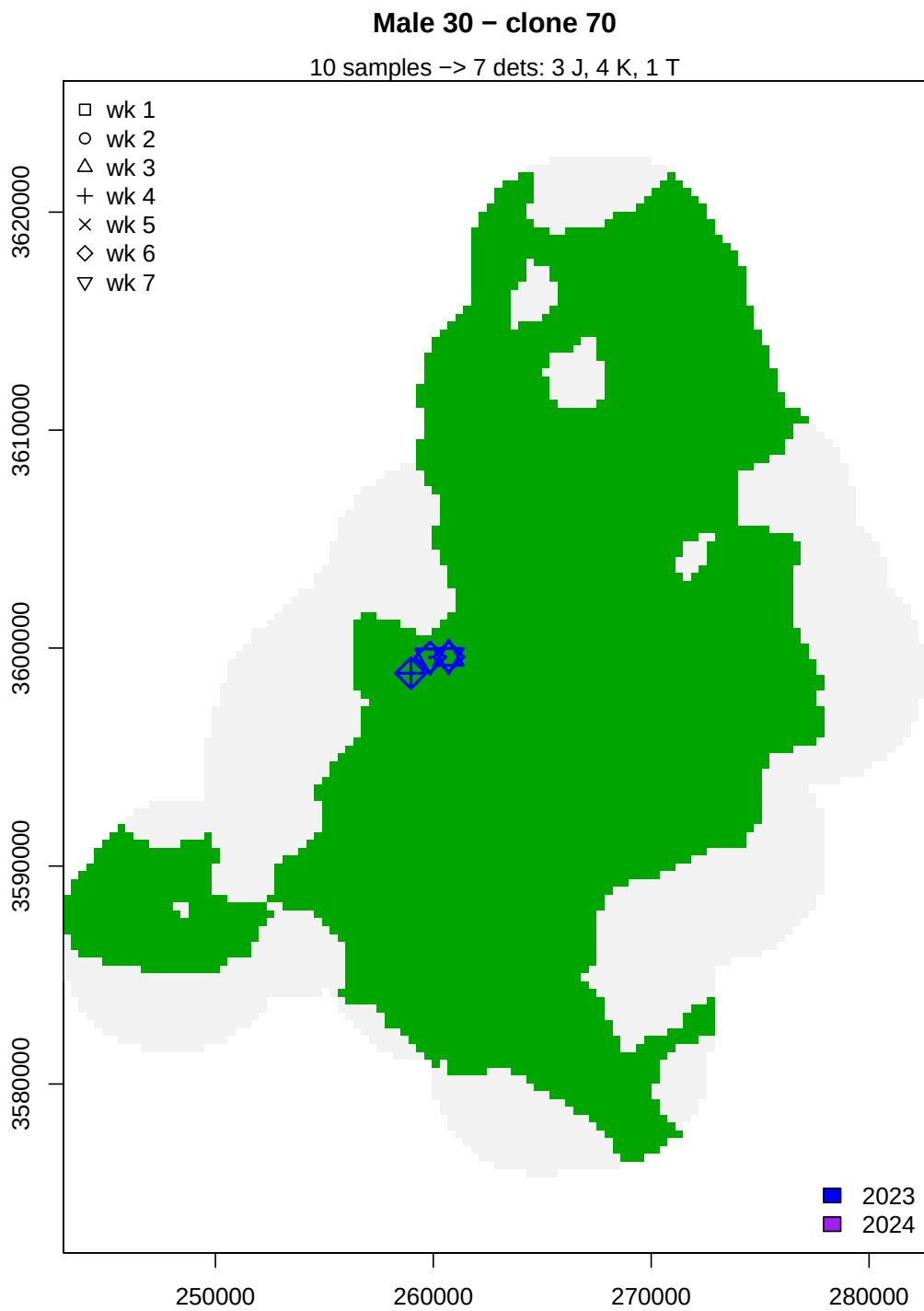


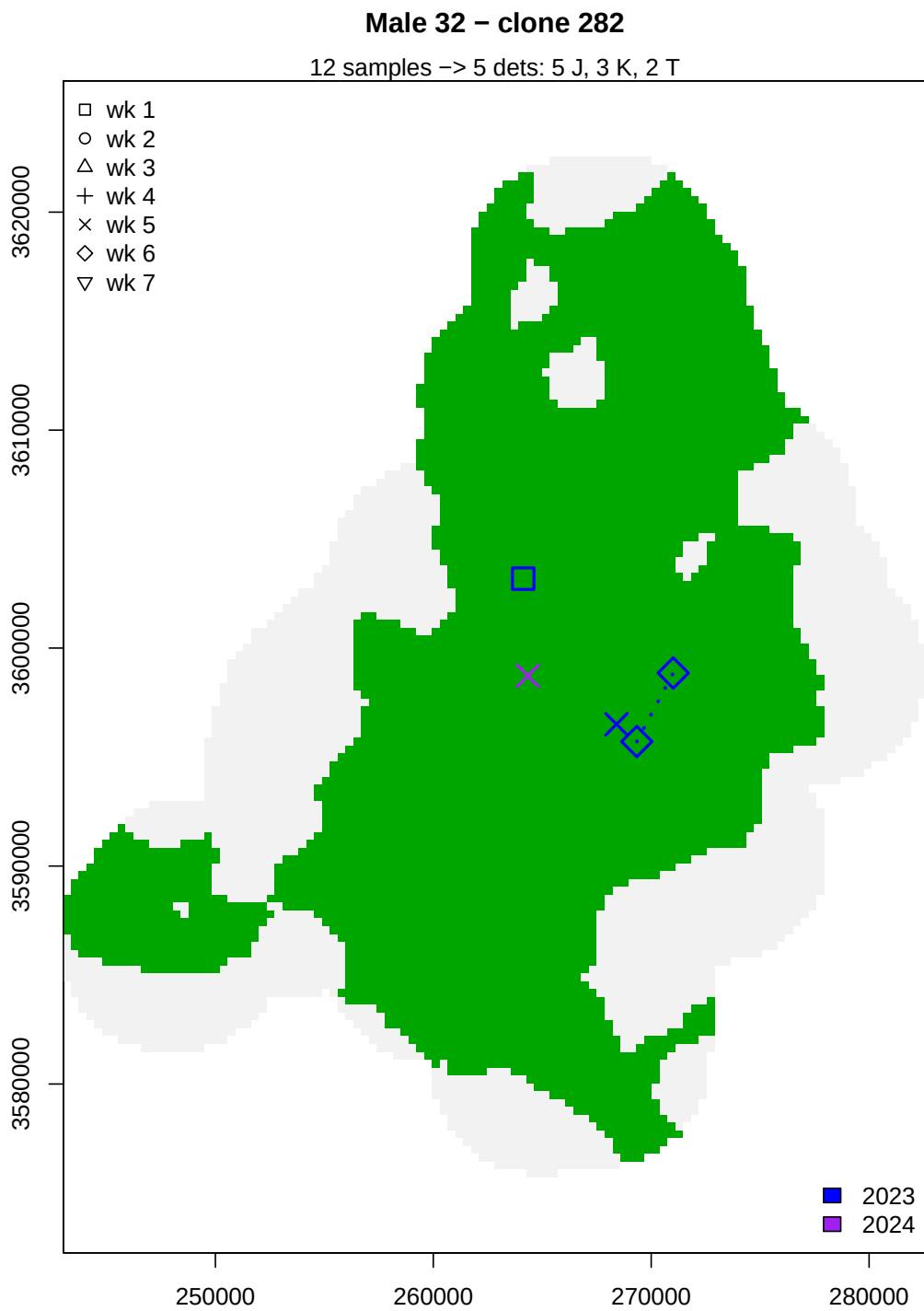
Male 25 – clone 117

6 samples -> 4 dets: 4 J, 1 K, 1 T



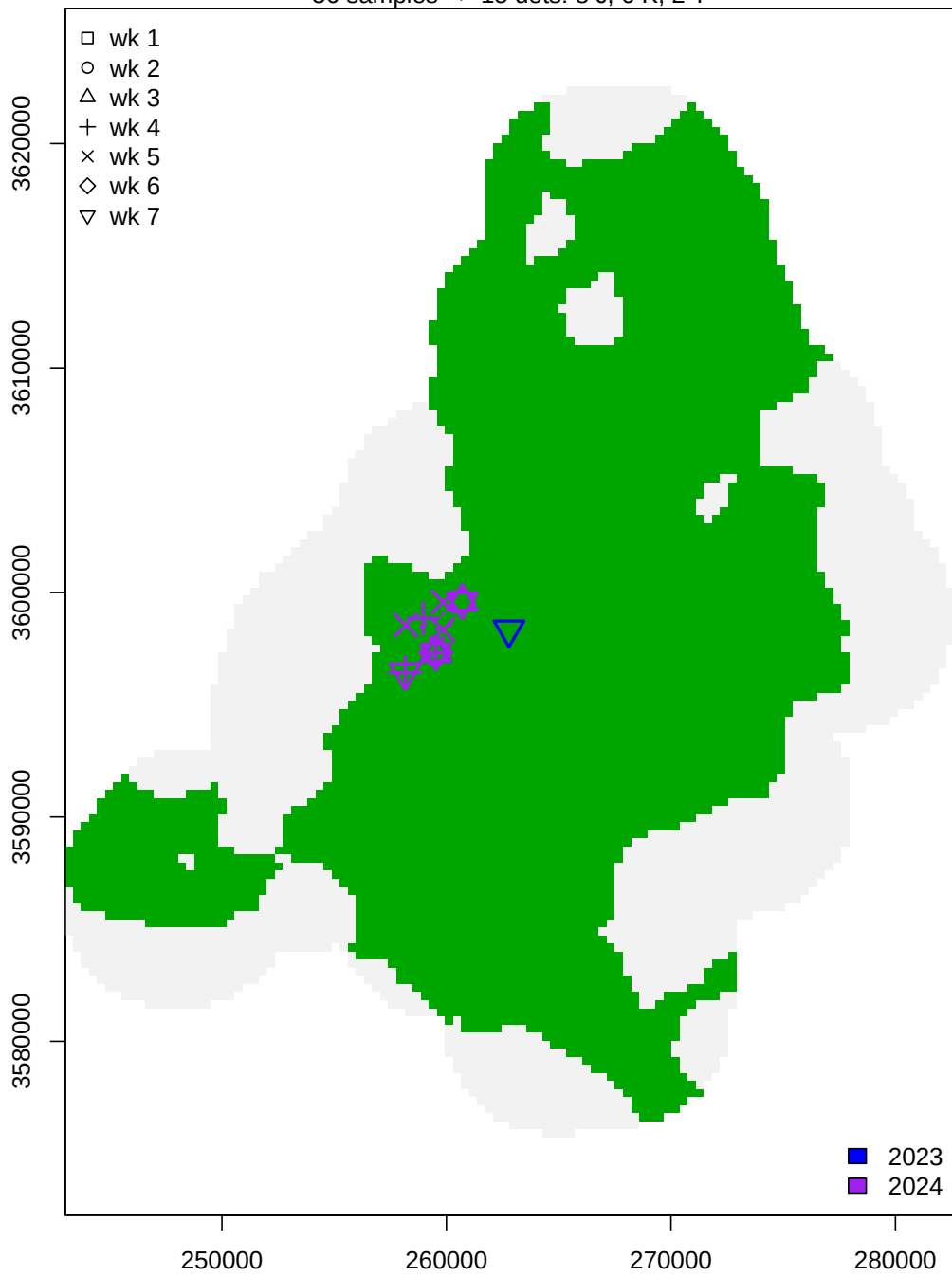






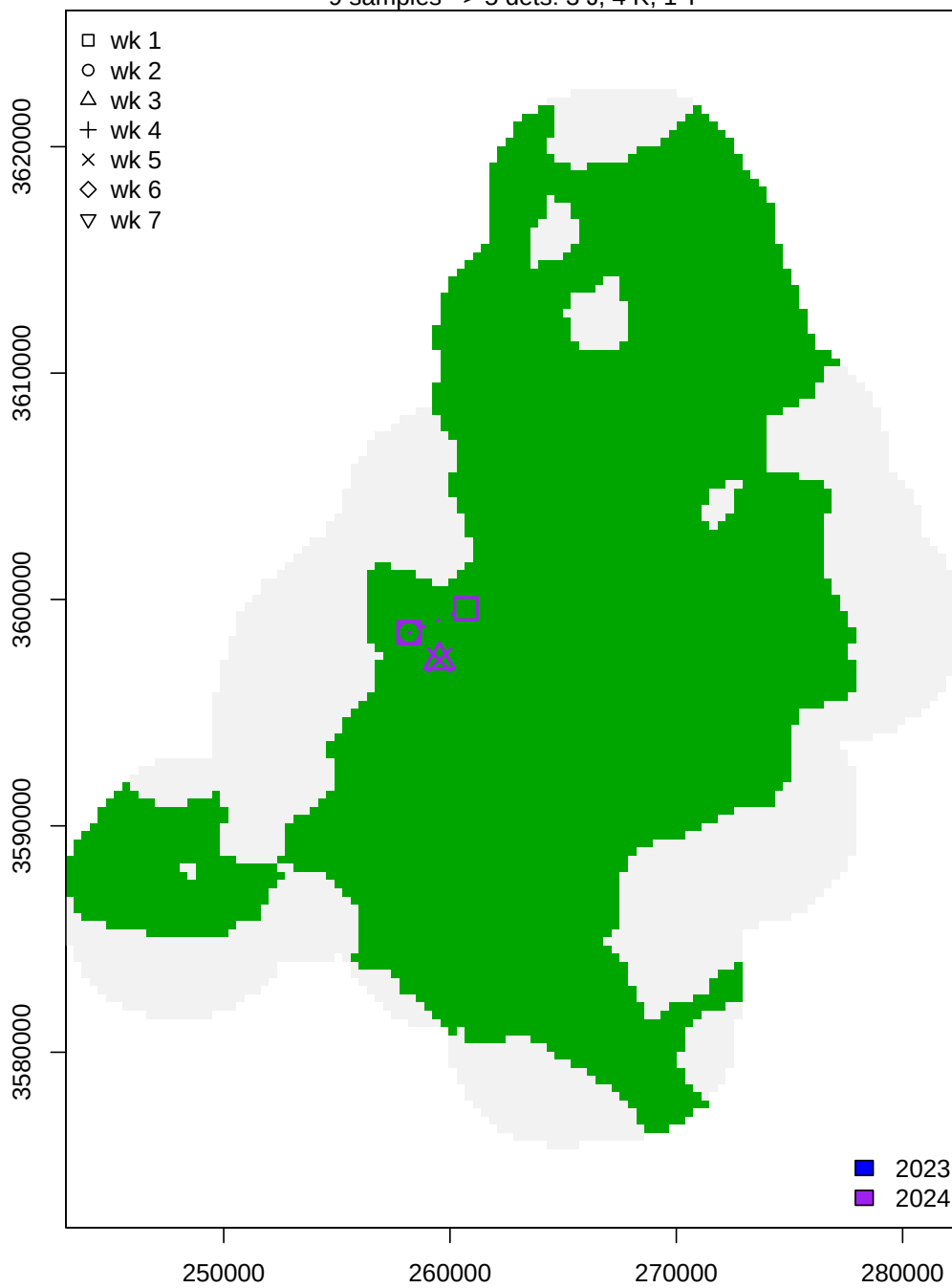
Male 33 – clone 292

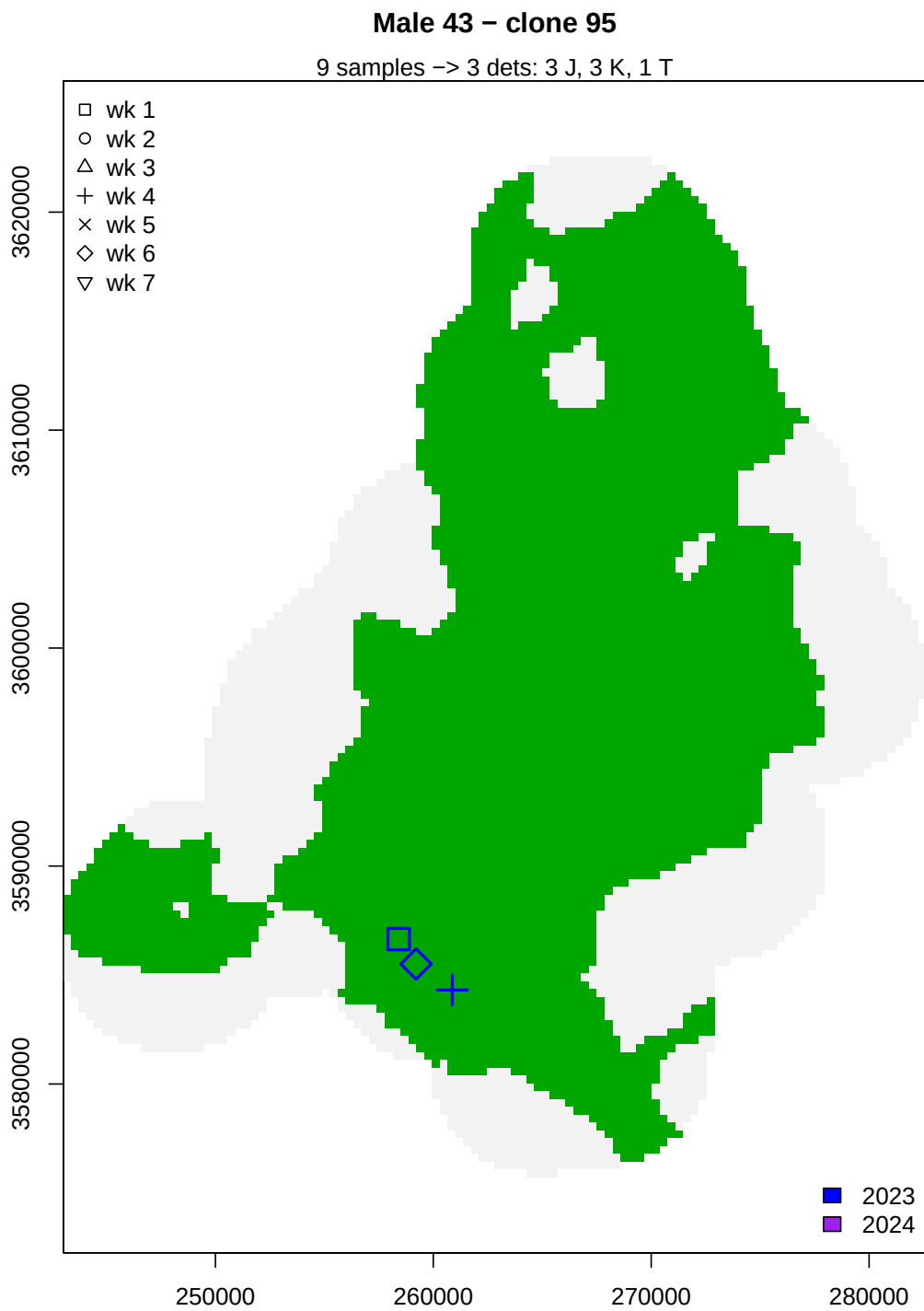
36 samples → 15 dets: 8 J, 6 K, 2 T



Male 39 – clone 219

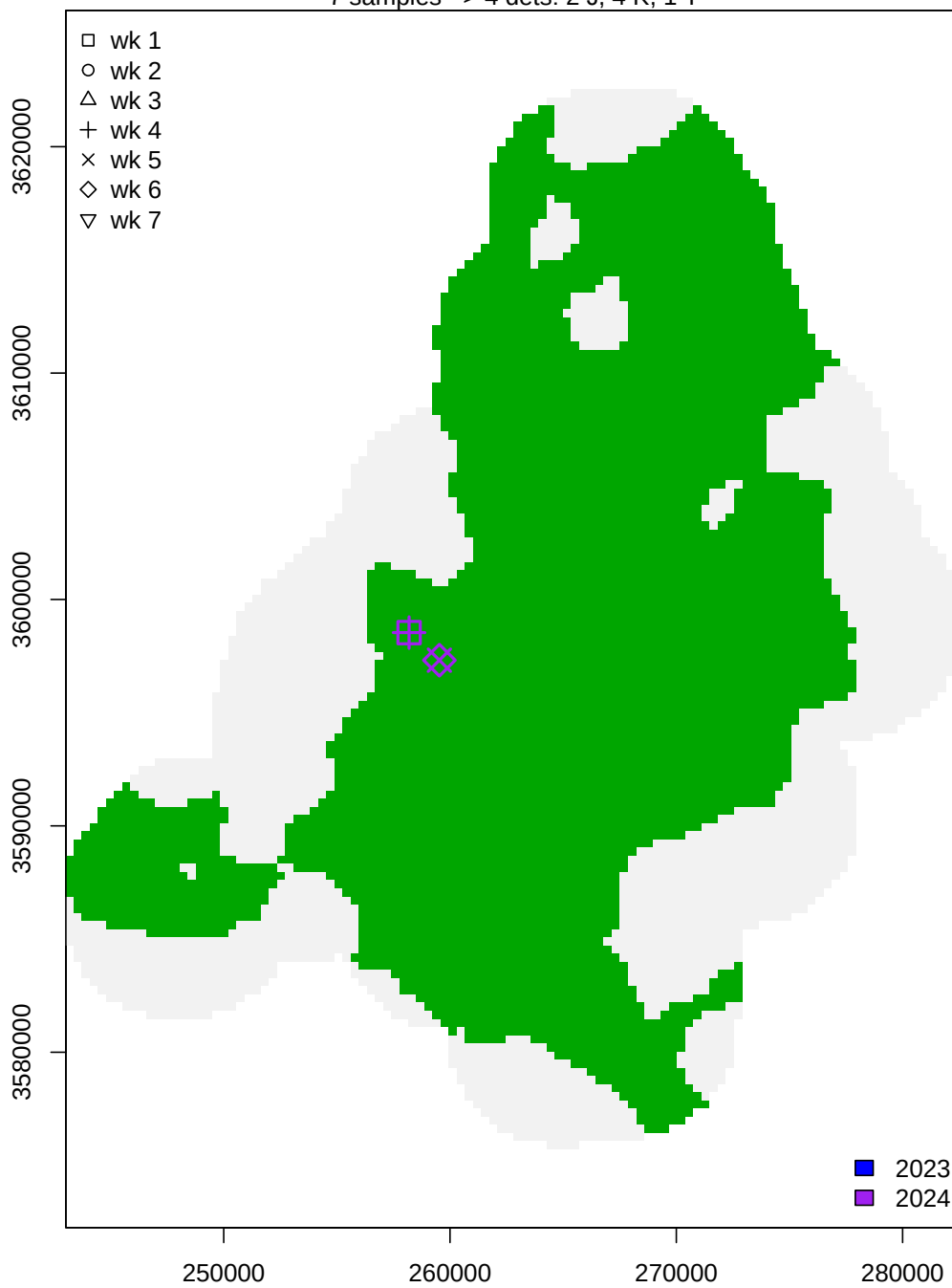
9 samples -> 5 dets: 3 J, 4 K, 1 T





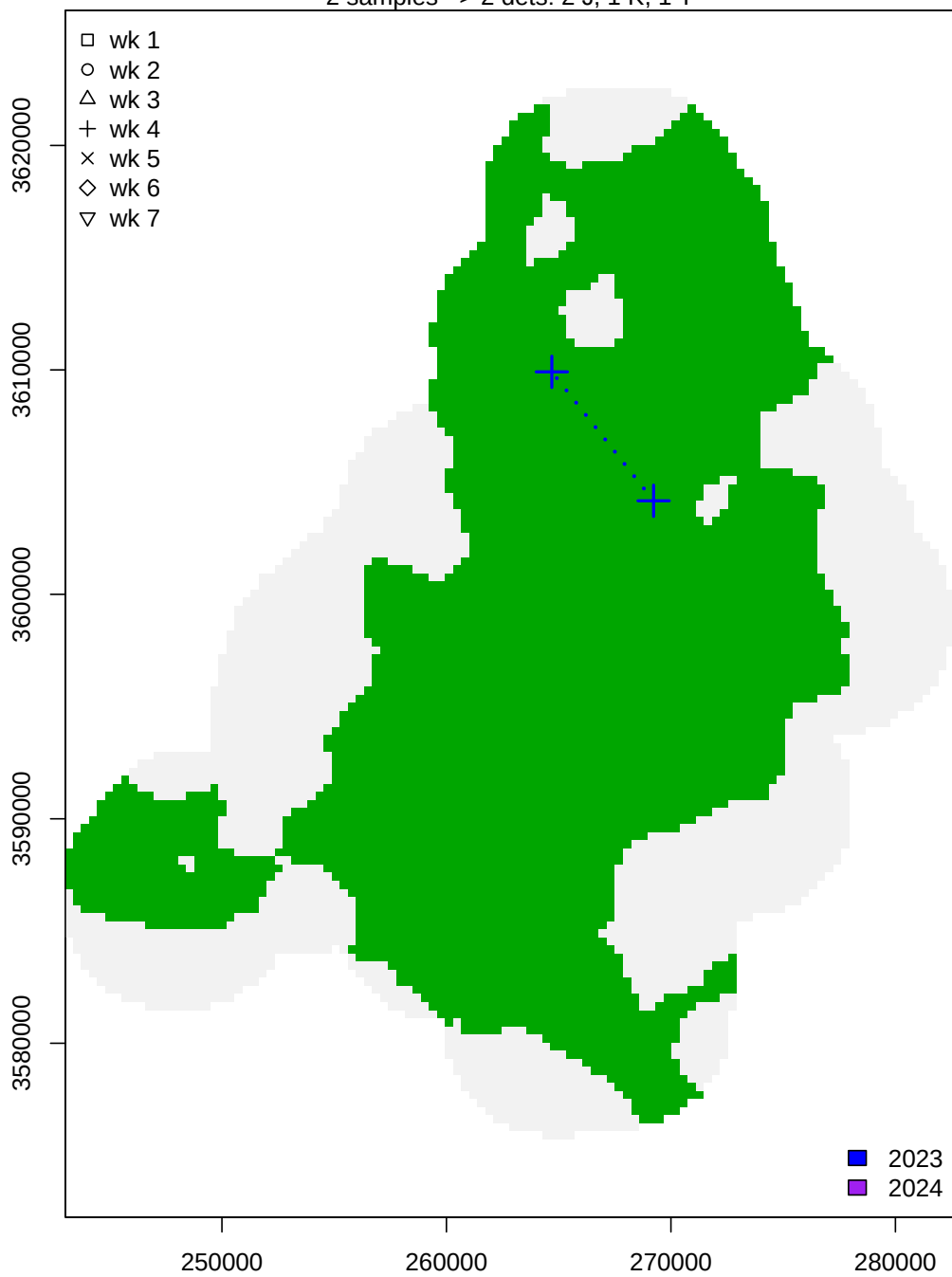
Male 53 – clone 244

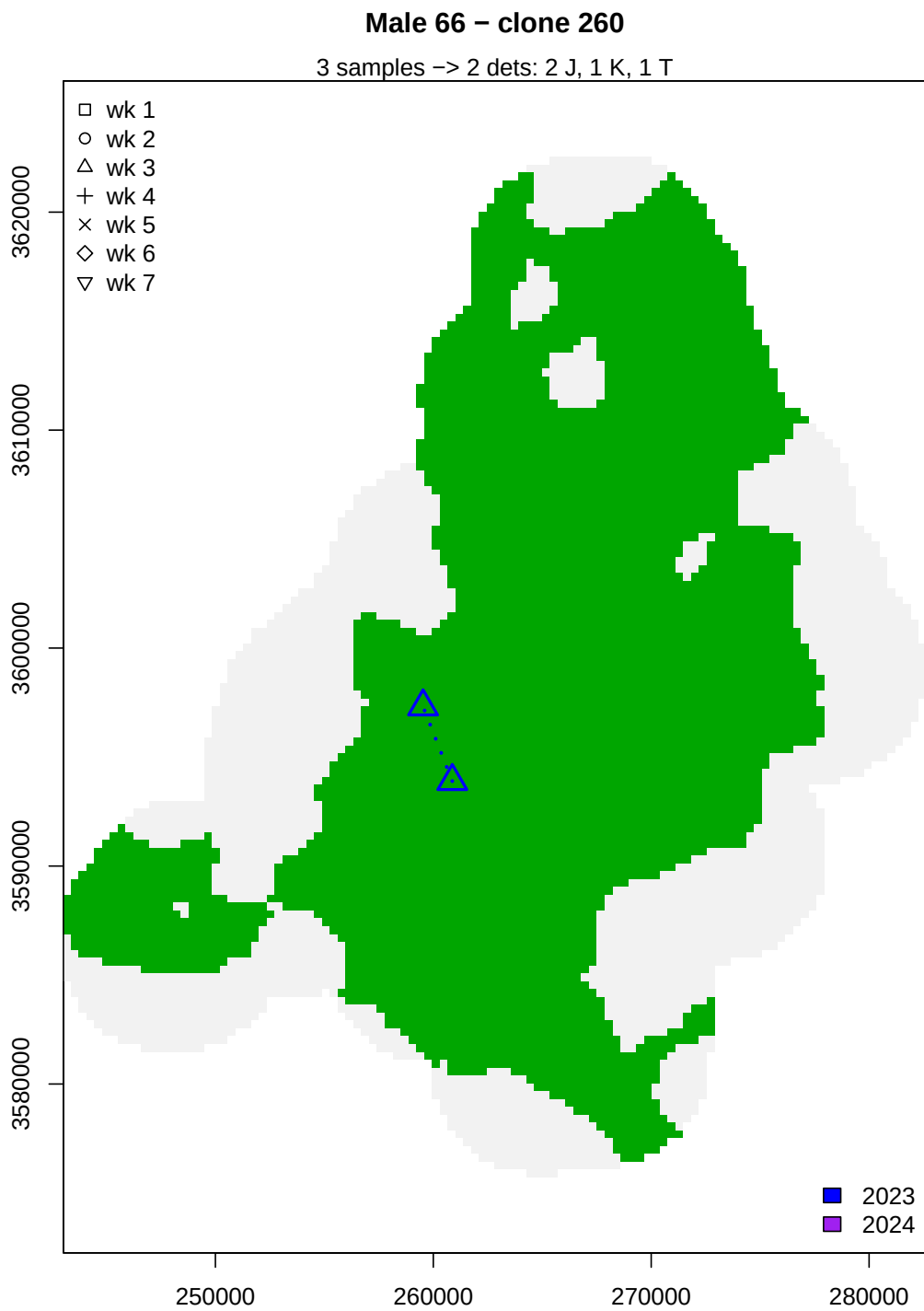
7 samples -> 4 dets: 2 J, 4 K, 1 T

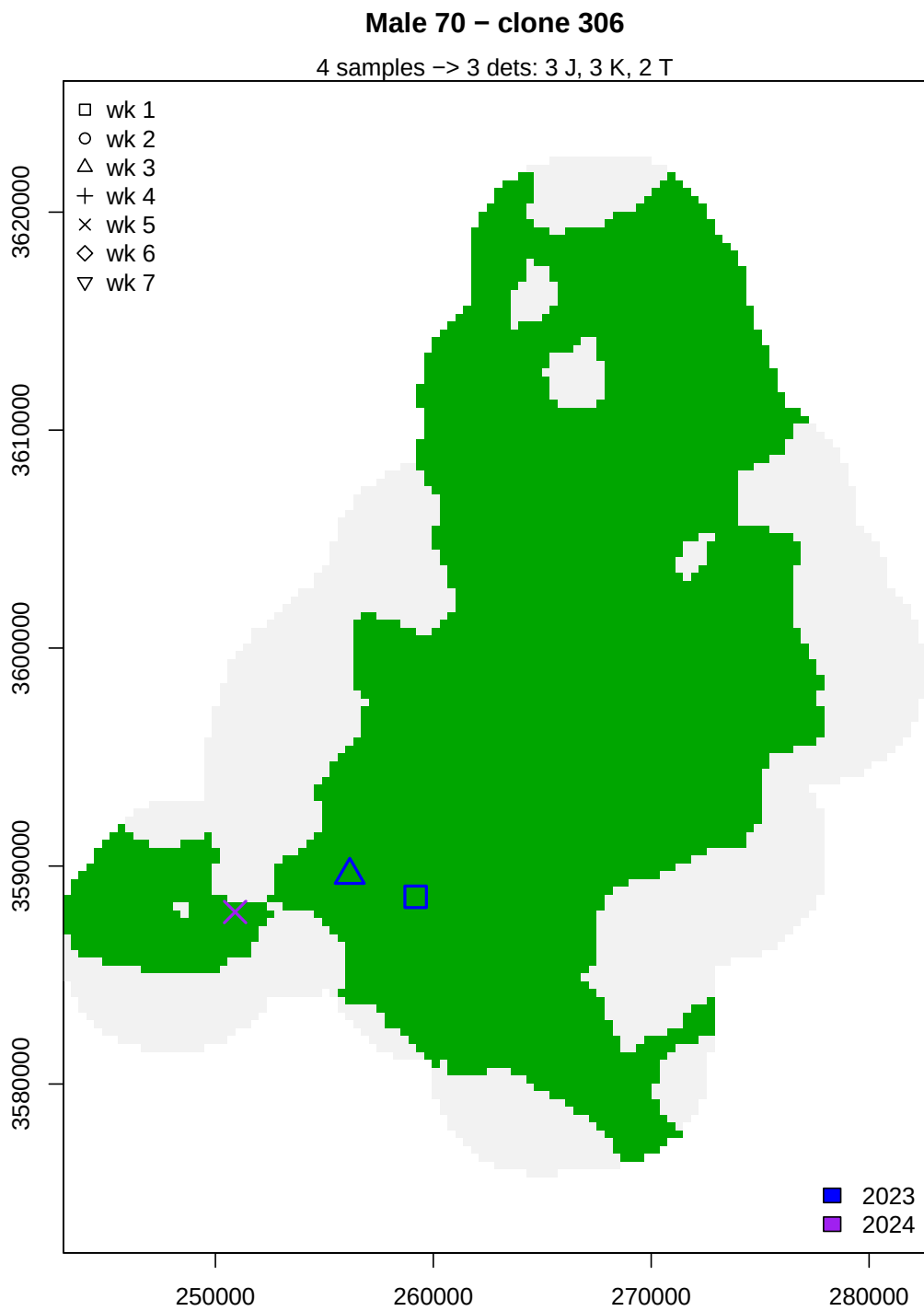


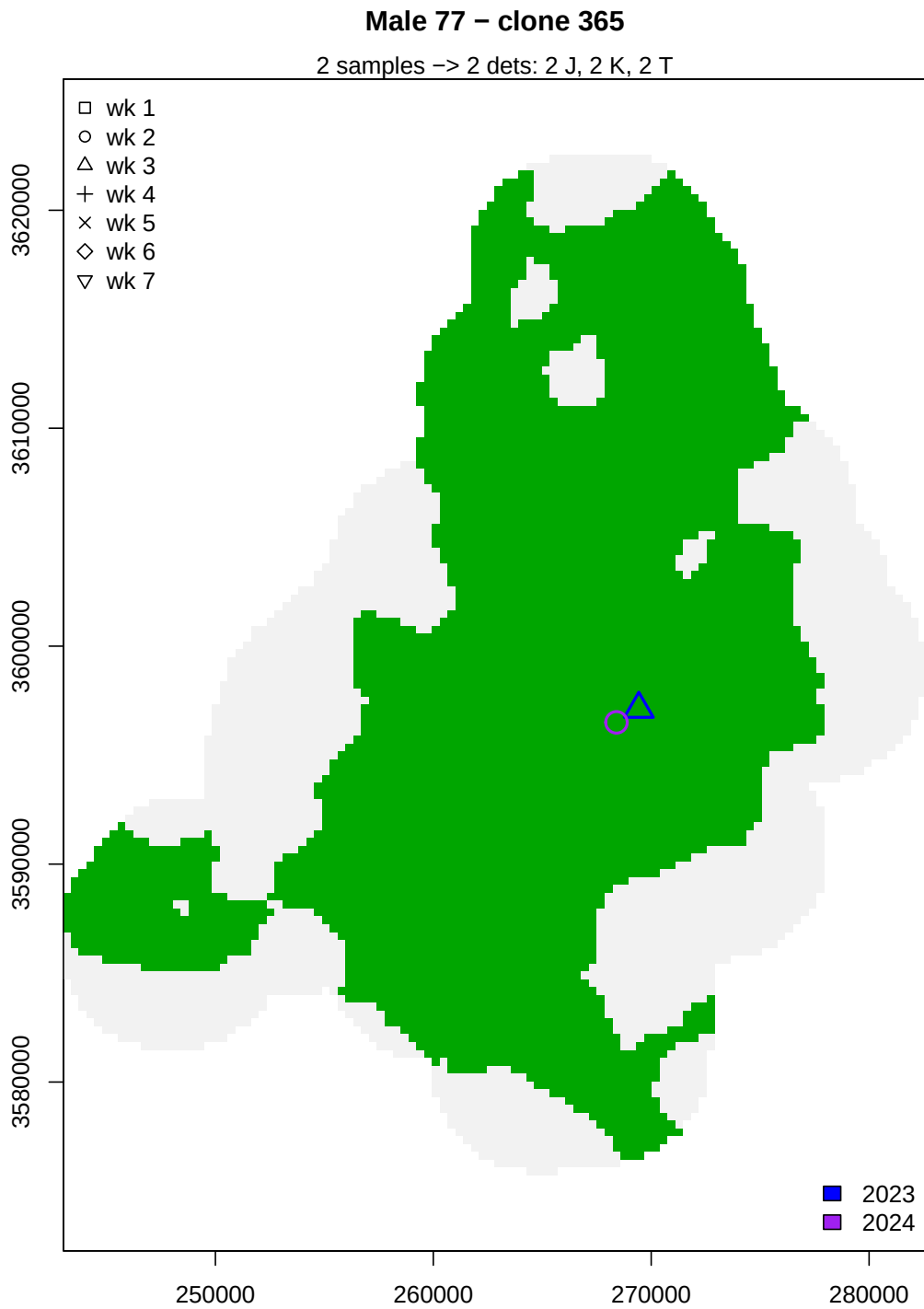
Male 62 – clone 223

2 samples -> 2 dets: 2 J, 1 K, 1 T









Overall, it seems we have little evidence for movement of activity centers between years. By far, the furthest movements recorded in our capture histories come in a single week!

Distances within and between years

Nonetheless, let's look more closely at the question about movement of activity centers. We'll do this in a couple of ways. First, we'll calculate the centroid of trap locations where each individual was detected in each year, then record a distance between these centroids. Second, we'll calculate the maximum distance between detections for each individual across the study period (both years) and compare that to the maximum distance between two detections within a year. From the maps above, we expect to see the largest distances between locations within, rather than among, years.

Females

First, for females...

```
#Load capture history objects
load("../scr/CGA_CH.Rda")

#Load hair snare location data
load("../sampleinfo/snareLocations.Rda")
traps <- locs

#Females first
centroid.yr1 <- centroid.yr2 <- matrix(data = NA, nrow = dim(Fch$y)[1], ncol = 2)
is.spatrecap <- max.within <- max.all <- dist.centroid <- max.yr1 <- max.yr2 <- c()

for(i in 1:dim(Fch$y)[1]) {
  is.spatrecap[i] <- (sum(rowSums(Fch$y[i,,]) > 0) > 1)
  yr1snares <- which(rowSums(Fch$y[i,,1]) > 0)
  yr2snares <- which(rowSums(Fch$y[i,,2]) > 0)

  if(length(yr1snares) > 0) {
    centroid.yr1[i,] <- colMeans(traps[yr1snares,])
    if(length(yr1snares) > 1) {
      max.yr1[i] <- max(dist(traps[yr1snares,], method = "euclidean"))
    } else {
      max.yr1[i] <- 0
    }
  } else {
    max.yr1[i] <- 0
  }

  if(length(yr2snares) > 0) {
    centroid.yr2[i,] <- colMeans(traps[yr2snares,])
    if(length(yr2snares) > 1) {
      max.yr2[i] <- max(dist(traps[yr2snares,], method = "euclidean"))
    } else {
      max.yr2[i] <- 0
    }
  } else {
    max.yr2[i] <- 0
  }

  if(length(yr1snares) > 0 & length(yr2snares) > 0) {
    dist.centroid[i] <- dist(rbind(centroid.yr1[i,], centroid.yr2[i,]),
```

```

        method = "euclidean")
}

max.within[i] <- max(c(max.yr1[i], max.yr2[i]))
if(length(yr1snares) > 0 & length(yr2snares > 0)) {
  max.all[i] <- max(dist(traps[c(yr1snares, yr2snares), ],
                        method = "euclidean"))
} else if(length(yr1snares) > 0 & length(yr2snares) == 0) {
  max.all[i] <- max.yr1[i]
} else if(length(yr1snares) == 0 & length(yr2snares) > 0) {
  max.all[i] <- max.yr2[i]
}

rm(yr1snares, yr2snares)
}

#Focusing on the spatial recaptures
mean(dist.centroid[which(is.spatrecap == TRUE)], na.rm = TRUE)

```

```
## [1] 1367.431
```

```

sr.max.within <- max.within[which(is.spatrecap == TRUE)]
sr.max.all <- max.all[which(is.spatrecap == TRUE)]
mean(sr.max.within)

```

```
## [1] 3138.329
```

```
mean(sr.max.all)
```

```
## [1] 3456.435
```

```
sum(is.spatrecap)
```

```
## [1] 31
```

```
sum(sr.max.within >= sr.max.all)
```

```
## [1] 23
```

The average distance between yearly centroids for females with spatial recaptures was 1367.4 m. Among the 31 females with spatial recaptures, 23 had their greatest distance between two detections *within* a year.

Males

Then for males...

```

#Now for males
centroid.yr1 <- centroid.yr2 <- matrix(data = NA, nrow = dim(Mch$y)[1], ncol = 2)
is.spatrecap <- max.within <- max.all <- dist.centroid <- max.yr1 <- max.yr2 <- c()

for(i in 1:dim(Mch$y)[1]) {
  is.spatrecap[i] <- (sum(rowSums(Mch$y[i,,]) > 0) > 1)
  yr1snares <- which(rowSums(Mch$y[i,,1]) > 0)
  yr2snares <- which(rowSums(Mch$y[i,,2]) > 0)

  if(length(yr1snares) > 0) {
    centroid.yr1[i,] <- colMeans(traps[yr1snares,])
    if(length(yr1snares) > 1) {
      max.yr1[i] <- max(dist(traps[yr1snares,], method = "euclidean"))
    } else {
      max.yr1[i] <- 0
    }
  } else {
    max.yr1[i] <- 0
  }

  if(length(yr2snares) > 0) {
    centroid.yr2[i,] <- colMeans(traps[yr2snares,])
    if(length(yr2snares) > 1) {
      max.yr2[i] <- max(dist(traps[yr2snares,], method = "euclidean"))
    } else {
      max.yr2[i] <- 0
    }
  } else {
    max.yr2[i] <- 0
  }

  if(length(yr1snares) > 0 & length(yr2snares) > 0) {
    dist.centroid[i] <- dist(rbind(centroid.yr1[i,], centroid.yr2[i,]),
                           method = "euclidean")
  }

  max.within[i] <- max(c(max.yr1[i], max.yr2[i]))
  if(length(yr1snares) > 0 & length(yr2snares) > 0) {
    max.all[i] <- max(dist(traps[c(yr1snares, yr2snares),],
                        method = "euclidean"))
  } else if(length(yr1snares) > 0 & length(yr2snares) == 0) {
    max.all[i] <- max.yr1[i]
  } else if(length(yr1snares) == 0 & length(yr2snares) > 0) {
    max.all[i] <- max.yr2[i]
  }

  rm(yr1snares, yr2snares)
}

#Focusing on the spatial recaptures
mean(dist.centroid[which(is.spatrecap == TRUE)], na.rm = TRUE)

```



```
## [1] 2663.472
```

```
sr.max.within <- max.within[which(is.spatrecap == TRUE)]  
sr.max.all <- max.all[which(is.spatrecap == TRUE)]  
mean(sr.max.within)
```

```
## [1] 5161.278
```

```
mean(sr.max.all)
```

```
## [1] 5717.523
```

```
sum(is.spatrecap)
```

```
## [1] 26
```

```
sum(sr.max.within >= sr.max.all)
```

```
## [1] 16
```

The average distance between yearly centroids for males with spatial recaptures was 2663.5 m. Among the 26 males with spatial recaptures, 16 had their greatest distance between two detections *within* a year.

Digging in...

Female clone 7

Looking more closely at the 1st female plotted (clone #7), which had a large within-week movement inferred during week 7 of 2024...

```
#Recreating the object that was used to build the female capture history...
```

```
#Load sample metadata (smd)
```

```
smd <- read.csv("../sampleinfo/sampleMetaData_cloneID.csv", header = TRUE)
```

```
sampsALL <- smd[which(!is.na(smd$finalClone) &  
                      smd$is.SCR == TRUE &  
                      smd$year %in% c(2023,2024)),]
```

```
#Switch name of finalClone column for compatibility with makeCH function  
colnames(sampsALL)[colnames(sampsALL) == "finalClone"] <- "cloneID"
```

```
Fsamps <- sampsALL[which(sampsALL$finalSEX == "F"),]  
length(unique(Fsamps$cloneID[!is.na(Fsamps$cloneID)]))
```

```
## [1] 88
```

```
Fsamps[Fsamps$cloneID == 7,]
```

##		extID	ugaID	is.SCR	is.rep	week	year
## 40		UGA_173101_P015_WD04	414-23	TRUE	FALSE	7	2023
## 102		UGA_173101_P016_WA06	417-24	TRUE	FALSE	3	2024
## 113		UGA_173101_P016_WB05	112-24	TRUE	FALSE	4	2024
## 136		UGA_173101_P016_WD04	59-23	TRUE	FALSE	7	2023
## 179		UGA_173101_P016_WG11	365-23	TRUE	FALSE	6	2023
## 250		UGA_173101_P017_WE10	162-23	TRUE	FALSE	4	2023
## 265		UGA_173101_P017_WG01	217-23	TRUE	FALSE	7	2023
## 273		UGA_173101_P017_WG09	398-24	TRUE	FALSE	7	2024
## 282		UGA_173101_P017_WH06	19-23	TRUE	FALSE	7	2023
## 326		UGA_173101_P018_WD02	420-24	TRUE	FALSE	7	2024
## 349		UGA_173101_P018_WF01	78-23	TRUE	FALSE	4	2023
## 355		UGA_173101_P018_WF07	184-23	TRUE	FALSE	4	2023
## 369		UGA_173101_P019_WA03	26-23	TRUE	FALSE	7	2023
## 393		UGA_173101_P019_WC03	233-23	TRUE	FALSE	7	2023
## 484		UGA_173101_P020_WB10	79-24	TRUE	FALSE	6	2024
## 537		UGA_173101_P020_WG03	242-24	TRUE	FALSE	3	2024
## 539		UGA_173101_P020_WG05	221-24	TRUE	FALSE	4	2024
## 547		UGA_173101_P020_WH01	346-23	TRUE	FALSE	4	2023
## 573		UGA_173101_P021_WB03	181-24	TRUE	FALSE	4	2024
## 635		UGA_173101_P021_WG05	372-24	TRUE	FALSE	3	2024
## 718		UGA_173101_P022_WF04	317-24	TRUE	FALSE	4	2024
## 721		UGA_173101_P022_WF07	458-24	TRUE	FALSE	6	2024
## 748		UGA_173101_P022_WH10	404-23	TRUE	FALSE	7	2023
## 755		UGA_173101_P023_WA05	278-23	TRUE	FALSE	7	2023
## 758		UGA_173101_P023_WA08	491-23	TRUE	FALSE	5	2023
## 770		UGA_173101_P023_WB08	413-24	TRUE	FALSE	3	2024
## 771		UGA_173101_P023_WB09	393-24	TRUE	FALSE	6	2024
## 785		UGA_173101_P023_WC11	352-23	TRUE	FALSE	4	2023
## 788		UGA_173101_P023_WD02	237-24	TRUE	FALSE	4	2024
## 813		UGA_173101_P023_WF03	115-24	TRUE	FALSE	4	2024
## 878	GTseek_024_384i5_051_P098.002_767-23	767-23	TRUE	FALSE	7	2023	
## 886	GTseek_024_384i5_059_P098.002_775-23	775-23	TRUE	FALSE	4	2023	
## 918	GTseek_024_384i5_091_P098.002_785-23	785-23	TRUE	FALSE	2	2023	
## 924	GTseek_024_384i5_097_P098.001_634-23	634-23	TRUE	FALSE	7	2023	
## 981	GTseek_024_384i5_155_P098.002_806-23	806-23	TRUE	FALSE	7	2023	
## 1012	GTseek_024_384i5_186_P098.004_648-24	648-24	TRUE	FALSE	7	2024	
## 1085	GTseek_024_384i5_259_P098.001_695-23	695-23	TRUE	FALSE	7	2023	
## 1109	GTseek_024_384i5_283_P098.002_867-23	867-23	TRUE	FALSE	4	2023	
## 1110	GTseek_024_384i5_284_P098.004_683-24	683-24	TRUE	FALSE	4	2024	
## 1152	GTseek_024_384i5_326_P098.003_537-24	537-24	TRUE	FALSE	3	2024	
## 1186	GTseek_024_384i5_360_P098.003_554-24	554-24	TRUE	FALSE	4	2024	
## 1237	GTseek_025_384i5_027_P098.006_996-24	996-24	TRUE	FALSE	6	2024	
## 1439	GTseek_025_384i5_229_P098.005_871-24	871-24	TRUE	FALSE	4	2024	
## 1525	GTseek_025_384i5_315_P098.006_3009-24	3009-24	TRUE	FALSE	7	2024	
## 1543	GTseek_025_384i5_333_P098.005_954-24	954-24	TRUE	FALSE	4	2024	
## 1559	GTseek_025_384i5_349_P098.006_3025-24	3025-24	TRUE	FALSE	7	2024	
## 1638	GTseek_026_384i5_055_P098.010_2768-24	2768-24	TRUE	FALSE	4	2024	
## 1649	GTseek_026_384i5_071_P098.009_1600-24	1600-24	TRUE	FALSE	4	2024	
## 1713	GTseek_026_384i5_159_P098.010_2821-24	2821-24	TRUE	FALSE	4	2024	
## 1776	GTseek_026_384i5_285_P098.010_2880-24	2880-24	TRUE	FALSE	4	2024	

##	1856	GTseek_028_384i5_031_P098.UGA2_351-23	351-23	TRUE	FALSE	7	2023
##	1890	GTseek_028_384i5_065_P098.UGA1_79-23	79-23	TRUE	FALSE	6	2023
##	1949	GTseek_028_384i5_124_P098.UGA4_400-24	400-24	TRUE	FALSE	4	2024
##	2121	GTseek_028_384i5_296_P098.UGA3_276-24	276-24	TRUE	FALSE	6	2024
##		snare	utmSnare				
##	40	HS 042	259530x3597317				
##	102	HS 045	258191x3598537				
##	113	HS 042	259530x3597317				
##	136	HS 042	259530x3597317				
##	179	HS 042	259530x3597317				
##	250	HS 042	259530x3597317				
##	265	HS 042	259530x3597317				
##	273	HS 066	268064x3590555				
##	282	HS 042	259530x3597317				
##	326	HS 042	259530x3597317				
##	349	HS 042	259530x3597317				
##	355	HS 042	259530x3597317				
##	369	HS 042	259530x3597317				
##	393	HS 042	259530x3597317				
##	484	HS 042	259530x3597317				
##	537	HS 045	258191x3598537				
##	539	HS 042	259530x3597317				
##	547	HS 042	259530x3597317				
##	573	HS 042	259530x3597317				
##	635	HS 045	258191x3598537				
##	718	HS 041	258115x3597270				
##	721	HS 042	259530x3597317				
##	748	HS 045	258191x3598537				
##	755	HS 045	258191x3598537				
##	758	HS 042	259530x3597317				
##	770	HS 045	258191x3598537				
##	771	HS 042	259530x3597317				
##	785	HS 042	259530x3597317				
##	788	HS 041	258115x3597270				
##	813	HS 041	258115x3597270				
##	878	HS 045	258191x3598537				
##	886	HS 042	259530x3597317				
##	918	HS 042	259530x3597317				
##	924	HS 045	258191x3598537				
##	981	HS 045	258191x3598537				
##	1012	HS 045	258191x3598537				
##	1085	HS 042	259530x3597317				
##	1109	HS 042	259530x3597317				
##	1110	HS 041	258115x3597270				
##	1152	HS 045	258191x3598537				
##	1186	HS 042	259530x3597317				
##	1237	HS 042	259530x3597317				
##	1439	HS 042	259530x3597317				
##	1525	HS 042	259530x3597317				
##	1543	HS 041	258115x3597270				
##	1559	HS 042	259530x3597317				
##	1638	HS 042	259530x3597317				
##	1649	HS 041	258115x3597270				
##	1713	HS 042	259530x3597317				

```

## 1776 HS 042 259530x3597317
## 1856 HS 045 258191x3598537
## 1890 HS 042 259530x3597317
## 1949 HS 042 259530x3597317
## 2121 HS 042 259530x3597317
##
##                                     fqfile
## 40   RAPiD-Genomics_F534_UGA_173101_P015_WD04_i5-502_i7-91_S14778_L003_R1_001.fastq
## 102  RAPiD-Genomics_F534_UGA_173101_P016_WA06_i5-504_i7-102_S14840_L003_R1_001.fastq
## 113  RAPiD-Genomics_F534_UGA_173101_P016_WB05_i5-504_i7-113_S14851_L003_R1_001.fastq
## 136  RAPiD-Genomics_F534_UGA_173101_P016_WD04_i5-504_i7-136_S14874_L003_R1_001.fastq
## 179  RAPiD-Genomics_F534_UGA_173101_P016_WG11_i5-504_i7-179_S14916_L003_R1_001.fastq
## 250  RAPiD-Genomics_F534_UGA_173101_P017_WE10_i5-512_i7-154_S14987_L003_R1_001.fastq
## 265  RAPiD-Genomics_F534_UGA_173101_P017_WG01_i5-512_i7-169_S15002_L003_R1_001.fastq
## 273  RAPiD-Genomics_F534_UGA_173101_P017_WG09_i5-512_i7-177_S15010_L003_R1_001.fastq
## 282  RAPiD-Genomics_F534_UGA_173101_P017_WH06_i5-512_i7-186_S15019_L003_R1_001.fastq
## 326  RAPiD-Genomics_F534_UGA_173101_P018_WD02_i5-537_i7-134_S15063_L003_R1_001.fastq
## 349  RAPiD-Genomics_F534_UGA_173101_P018_WF01_i5-537_i7-157_S15088_L003_R1_001.fastq
## 355  RAPiD-Genomics_F534_UGA_173101_P018_WF07_i5-537_i7-163_S15094_L003_R1_001.fastq
## 369  RAPiD-Genomics_F534_UGA_173101_P019_WA03_i5-1102_i7-82_S15108_L003_R1_001.fastq
## 393  RAPiD-Genomics_F534_UGA_173101_P019_WC03_i5-1102_i7-40_S15132_L003_R1_001.fastq
## 484   RAPiD-Genomics_F566_UGA_173101_P020_WB10_i5-505_i7-55_S3912_L001_R1_001.fastq
## 537   RAPiD-Genomics_F566_UGA_173101_P020_WG03_i5-505_i7-22_S3965_L001_R1_001.fastq
## 539   RAPiD-Genomics_F566_UGA_173101_P020_WG05_i5-505_i7-69_S3967_L001_R1_001.fastq
## 547   RAPiD-Genomics_F566_UGA_173101_P020_WH01_i5-505_i7-11_S3975_L001_R1_001.fastq
## 573   RAPiD-Genomics_F566_UGA_173101_P021_WB03_i5-501_i7-2_S15_L003_R1_001.fastq
## 635   RAPiD-Genomics_F566_UGA_173101_P021_WG05_i5-501_i7-69_S77_L003_R1_001.fastq
## 718  RAPiD-Genomics_F566_UGA_173101_P022_WF04_i5-1098_i7-45_S7646_L003_R1_001.fastq
## 721  RAPiD-Genomics_F566_UGA_173101_P022_WF07_i5-1098_i7-33_S7649_L003_R1_001.fastq
## 748  RAPiD-Genomics_F566_UGA_173101_P022_WH10_i5-1098_i7-34_S7676_L003_R1_001.fastq
## 755  RAPiD-Genomics_F566_UGA_173101_P023_WA05_i5-1099_i7-38_S7683_L003_R1_001.fastq
## 758  RAPiD-Genomics_F566_UGA_173101_P023_WA08_i5-1099_i7-16_S7686_L003_R1_001.fastq
## 770  RAPiD-Genomics_F566_UGA_173101_P023_WB08_i5-1099_i7-57_S7698_L003_R1_001.fastq
## 771  RAPiD-Genomics_F566_UGA_173101_P023_WB09_i5-1099_i7-63_S7699_L003_R1_001.fastq
## 785  RAPiD-Genomics_F566_UGA_173101_P023_WC11_i5-1099_i7-70_S7713_L003_R1_001.fastq
## 788  RAPiD-Genomics_F566_UGA_173101_P023_WD02_i5-1099_i7-72_S7716_L003_R1_001.fastq
## 813  RAPiD-Genomics_F566_UGA_173101_P023_WF03_i5-1099_i7-79_S7741_L003_R1_001.fastq
## 878                                     <NA>
## 886                                     <NA>
## 918                                     <NA>
## 924                                     <NA>
## 981                                     <NA>
## 1012                                    <NA>
## 1085                                    <NA>
## 1109                                    <NA>
## 1110                                    <NA>
## 1152                                    <NA>
## 1186                                    <NA>
## 1237                                    <NA>
## 1439                                    <NA>
## 1525                                    <NA>
## 1543                                    <NA>
## 1559                                    <NA>
## 1638                                    <NA>
## 1649                                    <NA>

```

## 1713								<NA>
## 1776								<NA>
## 1856								<NA>
## 1890								<NA>
## 1949								<NA>
## 2121								<NA>
##	knownSEX	sampSOURCE	region	nSNPgt	vcfID	FILT	GTservice	
## 40		Hair Snare	CGA	1049	UGA_173101_P015_WD04	FALSE	LGC	
## 102		Hair Snare	CGA	1099	UGA_173101_P016_WA06	FALSE	LGC	
## 113		Hair Snare	CGA	1195	UGA_173101_P016_WB05	FALSE	LGC	
## 136		Hair Snare	CGA	857	UGA_173101_P016_WD04	FALSE	LGC	
## 179		Hair Snare	CGA	773	UGA_173101_P016_WG11	FALSE	LGC	
## 250		Hair Snare	CGA	747	UGA_173101_P017_WE10	FALSE	LGC	
## 265		Hair Snare	CGA	650	UGA_173101_P017_WG01	FALSE	LGC	
## 273		Hair Snare	CGA	461	UGA_173101_P017_WG09	FALSE	LGC	
## 282		Hair Snare	CGA	1160	UGA_173101_P017_WH06	FALSE	LGC	
## 326		Hair Snare	CGA	1186	UGA_173101_P018_WD02	FALSE	LGC	
## 349		Hair Snare	CGA	1176	UGA_173101_P018_WF01	FALSE	LGC	
## 355		Hair Snare	CGA	1155	UGA_173101_P018_WF07	FALSE	LGC	
## 369		Hair Snare	CGA	1220	UGA_173101_P019_WA03	FALSE	LGC	
## 393		Hair Snare	CGA	1219	UGA_173101_P019_WC03	FALSE	LGC	
## 484		Hair Snare	CGA	1209	UGA_173101_P020_WB10	FALSE	LGC	
## 537		Hair Snare	CGA	1200	UGA_173101_P020_WG03	FALSE	LGC	
## 539		Hair Snare	CGA	1136	UGA_173101_P020_WG05	FALSE	LGC	
## 547		Hair Snare	CGA	1218	UGA_173101_P020_WH01	FALSE	LGC	
## 573		Hair Snare	CGA	1213	UGA_173101_P021_WB03	FALSE	LGC	
## 635		Hair Snare	CGA	1190	UGA_173101_P021_WG05	FALSE	LGC	
## 718		Hair Snare	CGA	1171	UGA_173101_P022_WF04	FALSE	LGC	
## 721		Hair Snare	CGA	1017	UGA_173101_P022_WF07	FALSE	LGC	
## 748		Hair Snare	CGA	1216	UGA_173101_P022_WH10	FALSE	LGC	
## 755		Hair Snare	CGA	1199	UGA_173101_P023_WA05	FALSE	LGC	
## 758		Hair Snare	CGA	1196	UGA_173101_P023_WA08	FALSE	LGC	
## 770		Hair Snare	CGA	1204	UGA_173101_P023_WB08	FALSE	LGC	
## 771		Hair Snare	CGA	1201	UGA_173101_P023_WB09	FALSE	LGC	
## 785		Hair Snare	CGA	1116	UGA_173101_P023_WC11	FALSE	LGC	
## 788		Hair Snare	CGA	1035	UGA_173101_P023_WD02	FALSE	LGC	
## 813		Hair Snare	CGA	1194	UGA_173101_P023_WF03	FALSE	LGC	
## 878		Hair Sample	CGA	123	GTS_767-23	FALSE	GTSeek	
## 886		Hair Sample	CGA	123	GTS_775-23	FALSE	GTSeek	
## 918		Hair Sample	CGA	123	GTS_785-23	FALSE	GTSeek	
## 924		Hair Sample	CGA	115	GTS_634-23	FALSE	GTSeek	
## 981		Hair Sample	CGA	123	GTS_806-23	FALSE	GTSeek	
## 1012		Hair Sample	CGA	120	GTS_648-24	FALSE	GTSeek	
## 1085		Hair Sample	CGA	117	GTS_695-23	FALSE	GTSeek	
## 1109		Hair Sample	CGA	124	GTS_867-23	FALSE	GTSeek	
## 1110		Hair Sample	CGA	123	GTS_683-24	FALSE	GTSeek	
## 1152		Hair Sample	CGA	123	GTS_537-24	FALSE	GTSeek	
## 1186		Hair Sample	CGA	122	GTS_554-24	FALSE	GTSeek	
## 1237		Hair Sample	CGA	125	GTS_996-24	FALSE	GTSeek	
## 1439		Hair Sample	CGA	125	GTS_871-24	FALSE	GTSeek	
## 1525		Hair Sample	CGA	125	GTS_3009-24	FALSE	GTSeek	
## 1543		Hair Sample	CGA	124	GTS_954-24	FALSE	GTSeek	
## 1559		Hair Sample	CGA	126	GTS_3025-24	FALSE	GTSeek	
## 1638		Hair Sample	CGA	123	GTS_2768-24	FALSE	GTSeek	

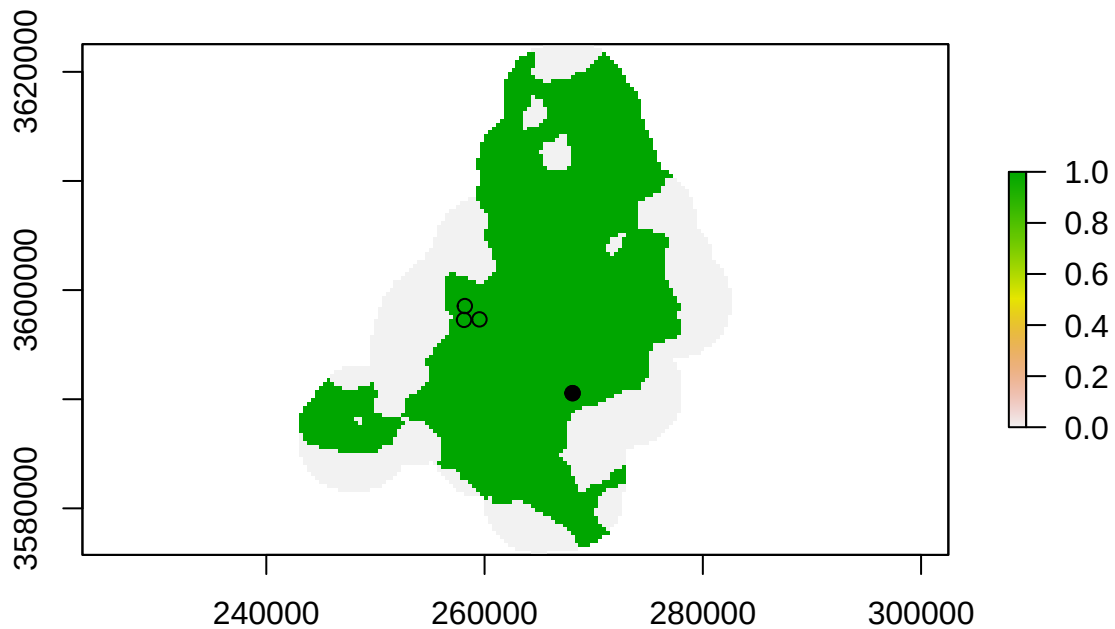
## 1649		Hair Sample	CGA	124	GTS_1600-24	FALSE	GTSeek
## 1713		Hair Sample	CGA	124	GTS_2821-24	FALSE	GTSeek
## 1776		Hair Sample	CGA	125	GTS_2880-24	FALSE	GTSeek
## 1856		Hair Sample	CGA	123	GTS_351-23	FALSE	GTSeek
## 1890		Hair Sample	CGA	123	GTS_79-23	FALSE	GTSeek
## 1949		Hair Sample	CGA	125	GTS_400-24	FALSE	GTSeek
## 2121		Hair Sample	CGA	124	GTS_276-24	FALSE	GTSeek
##	sexGT	COLclone	cloneID	finalSEX			
## 40	F	7	7	F			
## 102	F	7	7	F			
## 113	F	7	7	F			
## 136	F	7	7	F			
## 179	F	7	7	F			
## 250	F	7	7	F			
## 265	F	7	7	F			
## 273	F	7	7	F			
## 282	F	7	7	F			
## 326	F	7	7	F			
## 349	F	7	7	F			
## 355	F	7	7	F			
## 369	F	7	7	F			
## 393	F	7	7	F			
## 484	F	7	7	F			
## 537	F	7	7	F			
## 539	F	7	7	F			
## 547	F	7	7	F			
## 573	F	7	7	F			
## 635	F	7	7	F			
## 718	F	7	7	F			
## 721	undet	7	7	F			
## 748	F	7	7	F			
## 755	F	7	7	F			
## 758	F	7	7	F			
## 770	F	7	7	F			
## 771	F	7	7	F			
## 785	F	7	7	F			
## 788	F	7	7	F			
## 813	F	7	7	F			
## 878	F	273	7	F			
## 886	F	276	7	F			
## 918	F	276	7	F			
## 924	F	273	7	F			
## 981	F	276	7	F			
## 1012	F	276	7	F			
## 1085	F	273	7	F			
## 1109	F	276	7	F			
## 1110	F	276	7	F			
## 1152	F	276	7	F			
## 1186	F	276	7	F			
## 1237	F	273	7	F			
## 1439	F	276	7	F			
## 1525	F	276	7	F			
## 1543	F	273	7	F			
## 1559	F	276	7	F			

```
## 1638      F      273      7      F
## 1649      F      273      7      F
## 1713      F      273      7      F
## 1776      F      273      7      F
## 1856      F      276      7      F
## 1890      F      276      7      F
## 1949      F      276      7      F
## 2121      F      276      7      F
```

```
table(Fsamps$snare[which(Fsamps$cloneID == 7)])
```

```
##
## HS 041 HS 042 HS 045 HS 066
##      6      35      12      1
```

```
plot(S360)
points(locs[rownames(locs) %in% Fsamps$utmSnare[Fsamps$cloneID == 7],])
points(locs[rownames(locs) %in%
           Fsamps$utmSnare[which(Fsamps$cloneID == 7 &
                                Fsamps$snare == "HS 066")],],
       pch = 19)
```



```
Fsamps[Fsamps$cloneID == 7 & Fsamps$snare == "HS 066",]
```

```
##               extID  ugaID is.SCR is.rep week year  snare      utmSnare
## 273 UGA_173101_P017_WG09 398-24   TRUE  FALSE    7 2024 HS 066 268064x3590555
##
##                                     fqfile
## 273 RAPiD-Genomics_F534_UGA_173101_P017_WG09_i5-512_i7-177_S15010_L003_R1_001.fastq
##      knownSEX sampSOURCE region nSNPggt          vcfID  FILT GTservice
## 273      Hair Snare      CGA      461 UGA_173101_P017_WG09 FALSE      LGC
##      sexGT COLclone cloneID finalSEX
## 273      F      7      7      F
```

The distant sample in this case was identified as a recapture by COLONY, not PLINK only, but the sample did have fewer SNPs genotyped. Still well enough to make a reliable inference here, especially given the number of samples identified as part of this clone. It seems unlikely that this is a false-positive recapture.

Female clone 43

Looking more closely at the 9th female plotted (clone #43), which had a large within-week movement inferred during week 4 of 2023...

```
#Recreating the object that was used to build the female capture history...
```

```
#Load sample metadata (smd)
```

```
smd <- read.csv("../sampleinfo/sampleMetaData_cloneID.csv", header = TRUE)
```

```
sampsALL <- smd[which(!is.na(smd$finalClone) &
                      smd$is.SCR == TRUE &
                      smd$year %in% c(2023,2024)),]
```

```
#Switch name of finalClone column for compatibility with makeCH function
```

```
colnames(sampsALL)[colnames(sampsALL) == "finalClone"] <- "cloneID"
```

```
Fsamps <- sampsALL[which(sampsALL$finalSEX == "F"),]
length(unique(Fsamps$cloneID[!is.na(Fsamps$cloneID)]))
```

```
## [1] 88
```

```
Fsamps[Fsamps$cloneID == 43,]
```

```
##               extID  ugaID is.SCR is.rep week year
## 107          UGA_173101_P016_WA11 372-23   TRUE  FALSE    3 2023
## 168          UGA_173101_P016_WF12 370-23   TRUE  FALSE    2 2023
## 196          UGA_173101_P017_WA04 339-23   TRUE  FALSE    2 2023
## 222          UGA_173101_P017_WC06  21-23   TRUE   TRUE    2 2023
## 337          UGA_173101_P018_WE01 103-23   TRUE  FALSE    6 2023
## 359          UGA_173101_P018_WF11 476-23   TRUE  FALSE    6 2023
## 404          UGA_173101_P019_WD02 299-23   TRUE  FALSE    4 2023
## 753          UGA_173101_P023_WA03  21-23   TRUE   TRUE    2 2023
## 798          UGA_173101_P023_WD12 197-23   TRUE  FALSE    6 2023
## 996  GTseek_024_384i5_170_P098.003_996-23 996-23   TRUE  FALSE    2 2023
## 1031 GTseek_024_384i5_205_P098.001_678-23 678-23   TRUE  FALSE    2 2023
## 1165 GTseek_024_384i5_339_P098.002_895-23 895-23   TRUE  FALSE    6 2023
## 1342 GTseek_025_384i5_132_P098.007_886-23 886-23   TRUE  FALSE    6 2023
## 1404 GTseek_025_384i5_194_P098.007_987-23 987-23   TRUE  FALSE    6 2023
```



```

## 1924 GTseek_028_384i5_099_P098.UGA1_107-23 107-23 TRUE FALSE 4 2023
## 1958 GTseek_028_384i5_133_P098.UGA1_143-23 143-23 TRUE FALSE 3 2023
## 2014 GTseek_028_384i5_189_P098.UGA2_459-23 459-23 TRUE FALSE 2 2023
##      snare      utmSnare
## 107 HS 068 268186x3591620
## 168 HS 068 268186x3591620
## 196 HS 068 268186x3591620
## 222 HS 068 268186x3591620
## 337 HS 068 268186x3591620
## 359 HS 066 268064x3590555
## 404 HS 068 268186x3591620
## 753 HS 068 268186x3591620
## 798 HS 066 268064x3590555
## 996 HS 068 268186x3591620
## 1031 HS 068 268186x3591620
## 1165 HS 066 268064x3590555
## 1342 HS 066 268064x3590555
## 1404 HS 066 268064x3590555
## 1924 HS 126 270872x3608550
## 1958 HS 068 268186x3591620
## 2014 HS 068 268186x3591620
##
##                                     fqfile
## 107 RAPiD-Genomics_F534_UGA_173101_P016_WA11_i5-504_i7-107_S14845_L003_R1_001.fastq
## 168 RAPiD-Genomics_F534_UGA_173101_P016_WF12_i5-504_i7-168_S14905_L003_R1_001.fastq
## 196 RAPiD-Genomics_F534_UGA_173101_P017_WA04_i5-512_i7-100_S14933_L003_R1_001.fastq
## 222 RAPiD-Genomics_F534_UGA_173101_P017_WC06_i5-512_i7-126_S14959_L003_R1_001.fastq
## 337 RAPiD-Genomics_F534_UGA_173101_P018_WE01_i5-537_i7-145_S15074_L003_R1_001.fastq
## 359 RAPiD-Genomics_F534_UGA_173101_P018_WF11_i5-537_i7-167_S15098_L003_R1_001.fastq
## 404 RAPiD-Genomics_F534_UGA_173101_P019_WD02_i5-1102_i7-72_S15143_L003_R1_001.fastq
## 753 RAPiD-Genomics_F566_UGA_173101_P023_WA03_i5-1099_i7-82_S7681_L003_R1_001.fastq
## 798 RAPiD-Genomics_F566_UGA_173101_P023_WD12_i5-1099_i7-9_S7726_L003_R1_001.fastq
## 996
## 1031
## 1165
## 1342
## 1404
## 1924
## 1958
## 2014
##
##      knownSEX  sampSOURCE region nSNPgts vcfID  FILT GTservice
## 107      Hair Snare      CGA    1220 UGA_173101_P016_WA11 FALSE      LGC
## 168      Hair Snare      CGA    1136 UGA_173101_P016_WF12 FALSE      LGC
## 196      Hair Snare      CGA    1218 UGA_173101_P017_WA04 FALSE      LGC
## 222      Hair Snare      CGA     868 UGA_173101_P017_WC06 FALSE      LGC
## 337      Hair Snare      CGA    1220 UGA_173101_P018_WE01 FALSE      LGC
## 359      Hair Snare      CGA    1212 UGA_173101_P018_WF11 FALSE      LGC
## 404      Hair Snare      CGA     809 UGA_173101_P019_WD02 FALSE      LGC
## 753      Hair Snare      CGA     919 UGA_173101_P023_WA03 FALSE      LGC
## 798      Hair Snare      CGA     640 UGA_173101_P023_WD12 FALSE      LGC
## 996      Hair Sample      CGA     121      GTS_996-23 FALSE    GTSeek
## 1031     Hair Sample      CGA     120      GTS_678-23 FALSE    GTSeek
## 1165     Hair Sample      CGA     121      GTS_895-23 FALSE    GTSeek
## 1342     Hair Sample      CGA     121      GTS_886-23 FALSE    GTSeek
## 1404     Hair Sample      CGA     122      GTS_987-23 FALSE    GTSeek

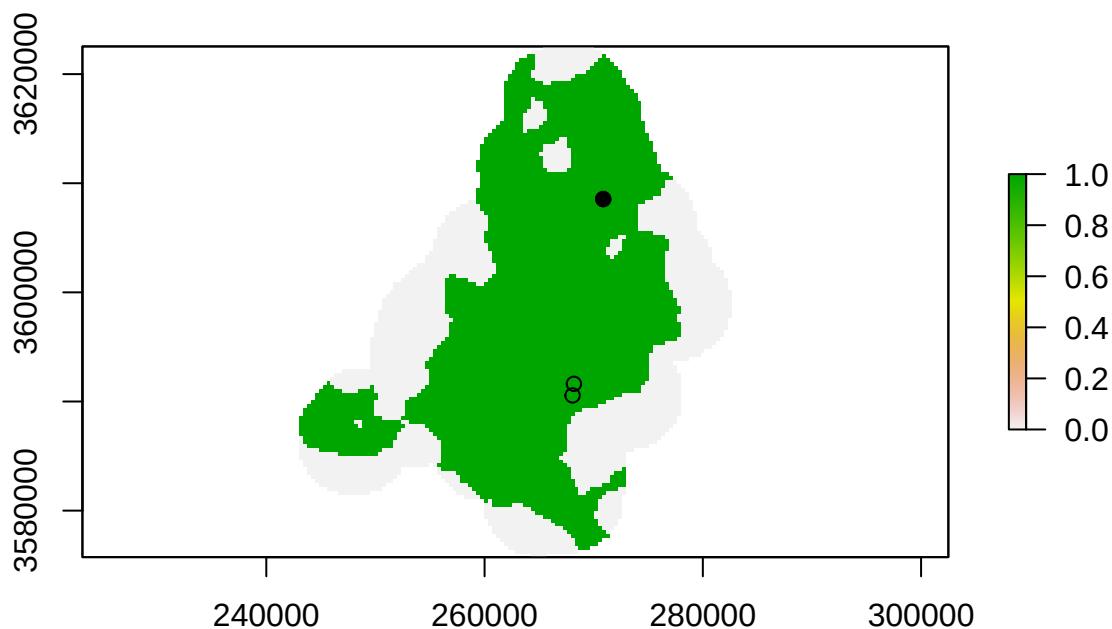
```

```
## 1924      Hair Sample   CGA   123      GTS_107-23 FALSE   GTSeek
## 1958      Hair Sample   CGA   123      GTS_143-23 FALSE   GTSeek
## 2014      Hair Sample   CGA   123      GTS_459-23 FALSE   GTSeek
##      sexGT COLclone  cloneID finalSEX
## 107      F      43      43      F
## 168      F      43      43      F
## 196      F      43      43      F
## 222      F      43      43      F
## 337      F      43      43      F
## 359      F      43      43      F
## 404      F      43      43      F
## 753      F      43      43      F
## 798      F      43      43      F
## 996      F     296      43      F
## 1031     F     296      43      F
## 1165     F     296      43      F
## 1342     F     296      43      F
## 1404     F     296      43      F
## 1924     F     296      43      F
## 1958     F     296      43      F
## 2014     F     296      43      F
```

```
table(Fsamps$snare[which(Fsamps$cloneID == 43)])
```

```
##
## HS 066 HS 068 HS 126
##      5      11      1
```

```
plot(S360)
points(locs[rownames(locs) %in% Fsamps$utmSnare[Fsamps$cloneID == 43],])
points(locs[rownames(locs) %in%
           Fsamps$utmSnare[which(Fsamps$cloneID == 43 &
                                Fsamps$snare == "HS 126")],],
       pch = 19)
```



```
Fsamps[Fsamps$cloneID == 43 & Fsamps$snare == "HS 126",]
```

```
##                               extID  ugaID is.SCR is.rep week year
## 1924 GTseek_028_384i5_099_P098.UGA1_107-23 107-23  TRUE  FALSE   4 2023
##      snare      utmSnare fqfile knownSEX  sampSOURCE region nSNPgt   vcfID
## 1924 HS 126 270872x3608550    <NA>      Hair Sample   CGA    123 GTS_107-23
##      FILT GTservice sexGT COLclone cloneID finalSEX
## 1924 FALSE   GTSeek   F      296     43         F
```

```
#Check kinship of this original COLONY clone
```

```
kingrel <- read.table("recapPLINK.kin0", header = FALSE)
colnames(kingrel) <- c("id1", "id2", "nsnp", "hehet", "ibs0", "kinship")
kingrel$kinship[which(kingrel$id1 %in%
                      Fsamps$vcfID[which(Fsamps$COLclone == 296)] &
                      kingrel$id2 %in%
                      Fsamps$vcfID[which(Fsamps$COLclone == 296)])]
```

```
## [1] 0.488636 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000
## [9] 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000
## [17] 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000 0.500000
## [25] 0.500000 0.500000 0.500000 0.500000
```

```
#Samples assigned to the same original clone by COLONY
```

```
Fsamps[which(Fsamps$COLclone == 296),]
```

##		extID	ugaID	is.SCR	is.rep	week	year		
##	996	GTseek_024_384i5_170_P098.003_996-23	996-23	TRUE	FALSE	2	2023		
##	1031	GTseek_024_384i5_205_P098.001_678-23	678-23	TRUE	FALSE	2	2023		
##	1165	GTseek_024_384i5_339_P098.002_895-23	895-23	TRUE	FALSE	6	2023		
##	1342	GTseek_025_384i5_132_P098.007_886-23	886-23	TRUE	FALSE	6	2023		
##	1404	GTseek_025_384i5_194_P098.007_987-23	987-23	TRUE	FALSE	6	2023		
##	1924	GTseek_028_384i5_099_P098.UGA1_107-23	107-23	TRUE	FALSE	4	2023		
##	1958	GTseek_028_384i5_133_P098.UGA1_143-23	143-23	TRUE	FALSE	3	2023		
##	2014	GTseek_028_384i5_189_P098.UGA2_459-23	459-23	TRUE	FALSE	2	2023		
##	snare	utmSnare	fqfile	knownSEX	sampSOURCE	region	nSNPgt	vcfID	
##	996	HS 068 268186x3591620	<NA>		Hair Sample	CGA	121	GTS_996-23	
##	1031	HS 068 268186x3591620	<NA>		Hair Sample	CGA	120	GTS_678-23	
##	1165	HS 066 268064x3590555	<NA>		Hair Sample	CGA	121	GTS_895-23	
##	1342	HS 066 268064x3590555	<NA>		Hair Sample	CGA	121	GTS_886-23	
##	1404	HS 066 268064x3590555	<NA>		Hair Sample	CGA	122	GTS_987-23	
##	1924	HS 126 270872x3608550	<NA>		Hair Sample	CGA	123	GTS_107-23	
##	1958	HS 068 268186x3591620	<NA>		Hair Sample	CGA	123	GTS_143-23	
##	2014	HS 068 268186x3591620	<NA>		Hair Sample	CGA	123	GTS_459-23	
##	FILT	GTservice	sexGT	COLclone	cloneID	finalSEX			
##	996	FALSE	GTSeek	F	296	43	F		
##	1031	FALSE	GTSeek	F	296	43	F		
##	1165	FALSE	GTSeek	F	296	43	F		
##	1342	FALSE	GTSeek	F	296	43	F		
##	1404	FALSE	GTSeek	F	296	43	F		
##	1924	FALSE	GTSeek	F	296	43	F		
##	1958	FALSE	GTSeek	F	296	43	F		
##	2014	FALSE	GTSeek	F	296	43	F		

```
Fsamps[which(Fsamps$COLclone == 43),]
```

##		extID	ugaID	is.SCR	is.rep	week	year	snare	utmSnare
##	107	UGA_173101_P016_WA11	372-23	TRUE	FALSE	3	2023	HS 068	268186x3591620
##	168	UGA_173101_P016_WF12	370-23	TRUE	FALSE	2	2023	HS 068	268186x3591620
##	196	UGA_173101_P017_WA04	339-23	TRUE	FALSE	2	2023	HS 068	268186x3591620
##	222	UGA_173101_P017_WC06	21-23	TRUE	TRUE	2	2023	HS 068	268186x3591620
##	337	UGA_173101_P018_WE01	103-23	TRUE	FALSE	6	2023	HS 068	268186x3591620
##	359	UGA_173101_P018_WF11	476-23	TRUE	FALSE	6	2023	HS 066	268064x3590555
##	404	UGA_173101_P019_WD02	299-23	TRUE	FALSE	4	2023	HS 068	268186x3591620
##	753	UGA_173101_P023_WA03	21-23	TRUE	TRUE	2	2023	HS 068	268186x3591620
##	798	UGA_173101_P023_WD12	197-23	TRUE	FALSE	6	2023	HS 066	268064x3590555
##									fqfile
##	107	RAPiD-Genomics_F534_UGA_173101_P016_WA11_i5-504_i7-107_S14845_L003_R1_001.fastq							
##	168	RAPiD-Genomics_F534_UGA_173101_P016_WF12_i5-504_i7-168_S14905_L003_R1_001.fastq							
##	196	RAPiD-Genomics_F534_UGA_173101_P017_WA04_i5-512_i7-100_S14933_L003_R1_001.fastq							
##	222	RAPiD-Genomics_F534_UGA_173101_P017_WC06_i5-512_i7-126_S14959_L003_R1_001.fastq							
##	337	RAPiD-Genomics_F534_UGA_173101_P018_WE01_i5-537_i7-145_S15074_L003_R1_001.fastq							
##	359	RAPiD-Genomics_F534_UGA_173101_P018_WF11_i5-537_i7-167_S15098_L003_R1_001.fastq							
##	404	RAPiD-Genomics_F534_UGA_173101_P019_WD02_i5-1102_i7-72_S15143_L003_R1_001.fastq							
##	753	RAPiD-Genomics_F566_UGA_173101_P023_WA03_i5-1099_i7-82_S7681_L003_R1_001.fastq							
##	798	RAPiD-Genomics_F566_UGA_173101_P023_WD12_i5-1099_i7-9_S7726_L003_R1_001.fastq							
##	knownSEX	sampSOURCE	region	nSNPgt	vcfID	FILT	GTservice		
##	107	Hair Snare	CGA	1220	UGA_173101_P016_WA11	FALSE	LGC		
##	168	Hair Snare	CGA	1136	UGA_173101_P016_WF12	FALSE	LGC		
##	196	Hair Snare	CGA	1218	UGA_173101_P017_WA04	FALSE	LGC		

```
## 222      Hair Snare   CGA    868 UGA_173101_P017_WC06 FALSE      LGC
## 337      Hair Snare   CGA   1220 UGA_173101_P018_WE01 FALSE      LGC
## 359      Hair Snare   CGA   1212 UGA_173101_P018_WF11 FALSE      LGC
## 404      Hair Snare   CGA    809 UGA_173101_P019_WD02 FALSE      LGC
## 753      Hair Snare   CGA    919 UGA_173101_P023_WA03 FALSE      LGC
## 798      Hair Snare   CGA    640 UGA_173101_P023_WD12 FALSE      LGC
##      sexGT COLclone cloneID finalSEX
## 107      F      43      43      F
## 168      F      43      43      F
## 196      F      43      43      F
## 222      F      43      43      F
## 337      F      43      43      F
## 359      F      43      43      F
## 404      F      43      43      F
## 753      F      43      43      F
## 798      F      43      43      F
```

```
#And kinship of the sample in question to other members of the final clone
kingrel$kinship[which(kingrel$id1 == "GTS_107-23" &
                      kingrel$id2 %in%
                      Fsamps$vcfID[which(Fsamps$cloneID ==43)])]
```

```
## [1] 0.489583 0.489130 0.489583 0.468750 0.500000 0.478261 0.486111 0.480769
## [9] 0.454545 0.500000 0.500000 0.500000 0.500000 0.500000
```

```
kingrel$kinship[which(kingrel$id1 %in%
                      Fsamps$vcfID[which(Fsamps$cloneID ==43)] &
                      kingrel$id2 == "GTS_107-23")]
```

```
## [1] 0.5 0.5
```

```
#Kinship is high to all other members of this final clone!
```

```
#Did this clone get split by genotyping service?
table(Fsamps$GTservice[which(Fsamps$COLclone == 43)])
```

```
##
## LGC
## 9
```

```
table(Fsamps$GTservice[which(Fsamps$COLclone == 296)])
```

```
##
## GTseek
## 8
```

```
#Yes!
```

So the recapture inference in this case is from PLINK only, not COLONY, and the sample was genotyped by GTseek, so it has fewer SNP loci available to make the match. The sample is well-genotyped, though. It appears that COLONY split detections of this individual by dataset - LGC and GTseek. GTseek detections

were predominantly from the same hair snares (HS 066 and HS 068) as the detections in the LGC data. Given the uniformly high kinship coefficients among members of these two COLONY clones, it seems unlikely that the detection at HS 126 was a false-positive recapture.
